Rewiring the market time clock in Spanish electricity markets: strategic behaviour and market power with higher granularity

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Contents

Glossary of acronyms

OMIE	Operador del Mercado Ibérico de Energía	PDBC	Programa Diario Base de Casación
REE	Red Eléctrica de España	PDBF	PDBC plus bilateral contracts
CNMC	Comisión Nacional de los Mercados y la Competencia	PIBCI	Programa Intradiario Base de Casación Incremental
ENTSO-E	European Network of TSOs for Electricity	TR	Tiempo Real (real-time technical restrictions)
ESIOS	Sistema de Información del Operador del Sistema	Fase I/II	pre-/post-IDA technical-restriction redispatch
EU	European Union	PO	Procedimientos de Operación
DA	day-ahead auction	aFRR	automatic Frequency Restoration Reserve
IDA	intraday auctions	mFRR	manual Frequency Restoration Reserve
SIDC	Single Intraday Coupling	RR	Replacement Reserve
XBID	cross-border intraday continuous market	BRP	Balance Responsible Party
EPEX	European Power Exchange	BSP	Balance Service Provider
MTU	market time unit (MTU60 hourly; MTU15 15-min)	CCGT	combined-cycle gas turbine
ISP15	imbalance settlement at 15 min (2024-12-11)	RES	renewable energy sources
ID15	intraday at 15 min (2025-03-19)	PV	photovoltaic
DA15	day-ahead at 15 min (2025-10-01)	MIC	minimum-income condition (offer type)
DA60	pre-DA15 hourly day-ahead	HHI	Herfindahl–Hirschman Index
MCP	market clearing price	MW / MWh / GWh	megawatt / megawatt-hour / gigawatt-hour

1 Introduction

[To be written. Concepts to cover, mirroring the "Question and headline" slide:]

- **Setting.** Spanish wholesale electricity is a sequence of auctions cleared from one day before delivery down to real time. In 2025 the trading time unit moved from 60 minutes to 15 minutes in two staggered reforms — intraday in March (ID15), day-ahead in October (DA15).
- **Question.** Does finer time resolution reshape strategic bidding? Does market power go up or down? What is the effect on prices, on the day-ahead vs intraday wedge, and on imbalance penalties?
- **Headline answer.** Granularity made the market more competitive without moving aggregate quantities: prices fell substantially at ID15 and were unchanged at DA15; per-firm residual demand thickened on every Big-4 dominant firm at every reform; dispatchables grade their critical-hour ladders more finely while renewables compress; imbalance penalty per MWh dropped by a third at ID15 with aggregate imbalance volume essentially unchanged.
- **Identification challenge.** An April 2025 blackout triggered a separate "operación reforzada" regulatory regime that inflated ancillary-services costs. Granularity and that regime are calendar-overlapping but mechanistically distinct; the empirical windows are constructed to keep them separable.
- **Contribution.** Bid-level identification on the new MTU15 sequence; a sequential-markets model in which the optimal step offer is solved exactly — the bid ladder places equal-MW tranches on an even price grid along the bidder's monopoly locus, so the measured bid-shape statistics $(\sigma_p, \mathrm{HHI}_p, \hat{\beta})$ have closed-form model counterparts; per-firm Ito-Reguant residual-demand recovery for Spanish data; convex-renewable robustness on the price effect; a strategic-markup proxy that nets out infra-marginal own production from the Cournot first-order condition.
- **Roadmap.** Institutional setting (§2); theory (§3); data (§4); empirical strategy (§5); results (§6); conclusion (§7).

2 Institutional setting: sequential markets, three reforms, and a blackout

2.1 Spanish wholesale architecture: sequential auctions and balancing

Spain runs a sequential auction stack — day-ahead → Fase I restrictions → three intraday auctions → continuous intraday → Fase II → balancing settlement — under uniform-price multi-unit clearing. OMIE auction-cleared programmes are pre-restriction by construction (REE redispatch enters through separate offers), so we can isolate competitive clearing from system-operator interventions. Wholesale concentration is moderate (HHI ∼1 100); ancillary-service concentration is high (HHI >4 000).

[To be written. Concepts to cover:]

- OMIE clears the day-ahead market (PDBC), three European intraday auctions (PIBCI), and the continuous intraday market; REE runs the technical-restrictions process before IDA (*Fase I*) and after IDA (*Fase II*), real-time balancing, and the imbalance settlement.

- Day-ahead and intraday auctions both use *uniform-price multi-unit clearing*; within-hour discrimination is possible only through the bid stack, not through differential clearing rules.
- OMIE auction-cleared programmes are *pre-restriction by construction*: REE's redispatch enters through separate restriction-offer markets, so OMIE-cleared volumes reflect competitive clearing rather than the system operator's restriction calls.
- Wholesale HHI $\sim$1 100 (moderate) vs ancillary-services HHI >4 000 (high): the strategically interesting margin is per-product within each unit's competing tier, and concentration migrates from wholesale to ancillary services.

2.2 The three 15-minute reforms: ISP15, ID15, DA15

Spain layered three reforms in under a year. ISP15 (December 2024) moved REE's imbalance settlement period from 60 to 15 minutes while OMIE contracts stayed hourly; ID15 (March 2025) moved the intraday auctions and the continuous intraday market to 15 minutes; DA15 (October 2025) completed the rollout on the day-ahead market. Wherever an effect could plausibly attach to more than one reform, we identify the channel separately by window.

[To be written. Concepts to cover:]

- **ISP15** (2024-12-11): REE's imbalance settlement reference moves from hour to quarter-hour; OMIE contract grid stays hourly. Within-hour deviations from schedule now resolve at 15-min resolution.
- **ID15** (2025-03-19): intraday auctions and continuous intraday market move to 15 minutes. Each clock-hour of delivery now hosts four 15-min products in IDA where it previously hosted one.
- **DA15** (2025-10-01): day-ahead market follows the same logic, completing the granularity rollout.
- Two earlier 2024 reforms set the stage: the European IDA reform of 14 June 2024 (six MIBEL auctions $\rightarrow$ three SIDC auctions), and the CNMC PO 3/14 modification (March 2024) that moved balancing services and BSP habilitation to 15-min.
- The reforms are *layered* rather than independent — ISP15 changes the settlement reference for every subsequent auction; ID15 lands on the new reference; DA15 lands on both.

2.3 The April 2025 blackout and the reinforced-operation regime

The blackout of 28 April 2025 cascaded from voltage- and reactive-power-control failures, not from market microstructure (ENTSO-E Expert Panel). REE's continuous reinforced-operation regime (*operación reforzada*) that followed is a separate regulatory layer: it tightens voltage-control requirements and expands Fase I up-redispatch payments by EUR $\sim$5M/day, almost entirely on combined-cycle gas. The DA15 BSTS pre-window starts on the blackout date, holding reforzada constant across the cutover by window construction; the ID15 windows sit entirely before the blackout.

[To be written. Concepts to cover:]

- ENTSO-E Expert Panel report (ENTSO-E Expert Panel, 2026) attributes the cascade to voltage- and reactive-power-control failures: fixed-power-factor renewable plants, manually-operated shunt reactors, divergent overvoltage protection, converter-driven instability, missing power-system stabilisers. Market microstructure is *not* in the panel's root-cause tree.

- REE's response was *operación reforzada* (CNMC, 2025b), continuous from 28 April 2025: tightened voltage-control requirements and expanded Fase I up-redispatch to keep CCGT synchronised for voltage support even when not day-ahead competitive. Fase I cost balloons by EUR ∼5 M/day, almost entirely on CCGT.
- **Identification consequence.** Reforzada overlaps DA15 calendar-wise but *not* ID15 (clean 40-day pre-blackout post-window). DA15 windows start on the blackout date so reforzada is held constant across the DA15 cutover by window construction. The headline ajuste-cost rise across 2025 is reforzada-driven, not MTU15-driven.

3 Theory: a sequential-market model of bid ladders with granularity selectors

3.1 Setup: three sequential stages, three agents, and the within-hour state

The model is a sequential-market framework in the Ito and Reguant (2016) tradition — a strategic bidder, a forecast-driven renewable aggregator, and a capacity-bounded arbitrageur — augmented with a granularity selector $Q^m \in \{1, Q\}$ at each stage and an explicit within-hour state (a deterministic ramp plus redistributive news) that hourly products mechanically average away. The mechanism in one line: granularity determines which within-hour states are *contractible*; the strategic bidder's step offer tracks its monopoly locus over the states its product is exposed to, so when a venue becomes per-quarter its competing tier widens and regrades precisely in the hours where the within-hour state varies; equilibrium price levels move only through residual-demand slopes and venue supply, because ladder geometry is level-neutral; and the within-hour wedge dispersion does not collapse at (Q, Q) because per-product arbitrage capacity binds at K_a/Q. Full derivations live in Appendix D; the main body collects the closed forms and the comparative statics that §6 takes to the data.

A representative clock-hour contains Q equal-length subperiods. Three sequential stages operate in order: a **forward** auction (Stage 1, $m = F$), a **spot** auction (Stage 2, $m = S$), and ex-post **balancing** settlement (Stage 3, $m = B$). Each stage has its own *granularity* $Q^m \in \{1, Q\}$ — it clears either once per hour or once per subperiod — defining a $2^3 = 8$-state policy space. The main analysis fixes the auction-side path $(Q^F, Q^S) = (1, 1) \to (1, Q) \to (Q, Q)$. The forward and spot markets map empirically to the electricity day-ahead and intraday auctions respectively. The agents, stages, and reform path are diagrammed in Figure 14 (Appendix D.1).

Bidders and primitives. The **strategic bidder** is a residual monopolist with constant marginal cost c; it delivers *exactly* (no production shock) and therefore does not internalise the imbalance penalty γ. Its offer per delivery product is a non-decreasing step function — a ladder of tranches $\{(p_k, \Delta q_k)\}$ — submitted before the product's clearing state is known and cleared at uniform price (§3.2). The **operational aggregator** is a price-taker with stochastic production $y_t = \bar{y}_t + \eta_t$ (forecast-error range Δ_η); it allocates total hourly MWh $\bar{Y}_h$ across forward and spot and *does* internalise the imbalance penalty, at the rate $\gamma(Q^B)$ set by the balancing granularity. The **arbitrageur** carries a net position s from forward to spot, in one of the four regimes of Ito and Reguant (§3.3); arbitrage transmits price impact but not imbalance risk.

The within-hour state. Write the strategic bidder's residual demand for delivery quarter t in market m as $q_{mt} = A_{mt} - b_{mt}\, p_{mt}$, net of arbitrage positions and aggregator supply (Appendix D.1

states the full Ito and Reguant specification). The intercept decomposes into three pieces,

$$A_{mt} \;=\; \bar{A} \;+\; \zeta_t \;+\; v_{mt}.$$

(i) ζ_t is the *deterministic within-hour profile* (the ramp): $\sum_t \zeta_t = 0$, dispersion $\sigma_\zeta^2 = Q^{-1}\sum_t \zeta_t^2$, large in critical hours and ≈ 0 in flat hours (§4.5), known to everyone at all times. (ii) v_{mt} is the *clearing state* the bidder screens at stage m: unknown at submission, embedded in the residual demand it faces at clearing. (iii) Between the forward and spot gates the within-hour *update* ω_t realises and becomes the spot's traded object: $\omega_t = \zeta_t + \mu_t$ while the forward is hourly (the profile is not yet contractible upstream), and $\omega_t = \mu_t$ once the forward is per-quarter, where μ_t is mean-zero redistributive news ($\sum_t \mu_t = 0$: ramp-timing shifts that move MW between adjacent quarters with no hour-level trace). We write $\sigma_\omega^2 = \sigma_\zeta^2 + \sigma_\mu^2$ and $\sigma_\omega^2 = \sigma_\mu^2$ for the two cases.

Granularity as exposure to states. Decompose the stage-m clearing state as $v_{mt} = \theta_m + \mu_{mt}$: an hour-level component and a redistributive component with $\sum_t \mu_{mt} = 0$. An hourly product clears against the within-hour average, $Q^{-1}\sum_t v_{mt} = \theta_m$ — redistributive components cancel mechanically — while a per-quarter product clears against its own v_{mt}. We assume the screened state is uniform at each granularity: $\theta_m \sim U[-\Delta_m, \Delta_m]$ for an hourly product, and $v_{mt} \sim U[-\rho_{\mathcal{X}}\Delta_m, +\rho_{\mathcal{X}}\Delta_m]$ for a quarter product, with an amplification $\rho_{\mathcal{X}} \geq 1$ specific to the hour class: $\rho \approx 1$ in flat hours and $\rho > 1$ in ramp hours, where ramp-timing uncertainty concentrates.[1] Granularity thus changes *which* states a product is exposed to.

Solution concept. We solve the game by **backward induction**: at Stage 2 the strategic bidder best-responds per spot product taking the forward outcome as given; the arbitrageur takes the implied price spread and chooses s within its regime; at Stage 1 the strategic bidder commits its forward ladder internalising the spot continuation value (the forward price anchors spot demand), which matters both for the forward price level and, through the slope of the stage-1 locus, for the forward ladder (§3.4; full backward induction in Appendices D.4–D.6).

Three deliberate abstractions — a single residual monopolist per product, no strategic supply-function interaction across large firms in the sense of Klemperer and Meyer (1989), and a passive balancing stage — and the model's relation to Ito and Reguant's implicit competitive fringe are discussed in Appendix D.1.

3.2 Bid ladders within a product: step offers along the monopoly locus

With linear residual demand and one-dimensional clearing-state uncertainty, a continuous supply schedule replicates the full-information monopoly outcome state by state, so a finite tranche count is the only friction. The optimal n-tranche ladder has closed forms for every measured bid-shape statistic, is level-neutral on the mean price, and interacts with granularity in one line (Corollary 2).

Fix one delivery product with residual demand $q = A - b\,p$, $A = \bar{a} + v$, $v \sim U[-\Delta, \Delta]$, and constant marginal cost c. The bidder submits a non-decreasing step offer before v realises; the auction clears at the crossing of the realised residual demand with the offer. When the crossing lands on a horizontal segment the marginal tranche sets the price; on a vertical riser the realised demand does. Both occur in equilibrium and both are visible in the OMIE clearing data.

The continuous benchmark is frictionless: $S^*(p) = b\,(p - c)$ clears at the full-information monopoly point $p^m(A) = (A + bc)/(2b)$, $q^m(A) = (A - bc)/2$ in *every* state and earns the unconditional monopoly value $\mathbb{E}[(A - bc)^2]/(4b)$ (Lemma 2, Appendix D.3) — the single-firm

[1] Exactly closed under the tiling construction of Lemma 1 (Appendix D.1): redistributive news on J symmetric pairs of atoms delivers $\rho_{\mathcal{X}} = 2J$ exactly. We carry a general $\rho_{\mathcal{X}}$.

benchmark of Klemperer and Meyer (1989). Real offers, however, are step functions with few tranches (OMIE caps tranche counts, and bid management is costly), so the relevant problem is the best n-step approximation of the monopoly locus. It has an exact solution.

Proposition 1 (Optimal n-tranche ladder)**.** *Let $w = 2\Delta/(2n+1)$ and let the monopoly quantity be interior in every state by a quarter-cell margin ($q^m(\bar{a} - \Delta) \geq \Delta/(2(2n+1))$, Assumption 1 in Appendix D.3). The optimal offer with n competing price levels consists of an inframarginal base of $q^m(\bar{a} - \Delta + w/2)$ MW priced at the floor, plus n competing tranches of exactly w MW each at prices $p_k = p^m\big(\bar{a} - \Delta + (2k - \frac{1}{2})\, w\big)$, evenly spaced with grid step $p_{k+1} - p_k = w/b$ and spanning $(n-1)\, w/b$: the ladder splits the state range into $2n+1$ equal intervals, each served at its midpoint monopoly point, and expected profit is*

$$\mathbb{E}[\pi_n] \;=\; \frac{\mathbb{E}[(A - bc)^2]}{4b} \;-\; \frac{\Delta^2}{12\, b\, (2n+1)^2}.$$

(Proof: Appendix D.3; Figure 15 there draws the ladder weaving across the locus.)

Corollary 1 (Bid-shape statistics)**.** *Computed over the competing tranches — the in-band construction of §4.3, which nets out the inframarginal base and parked tiers — the optimal ladder delivers in closed form*

$$HHI_p = \frac{1}{n}, \qquad \sigma_p = \frac{2\Delta}{(2n+1)\, b}\sqrt{\frac{n^2-1}{12}} \xrightarrow[n\to\infty]{} \frac{\sigma_v}{2b}, \qquad \hat{\beta} = \frac{1}{b}, \qquad \hat{\gamma} = 0,$$

with $\sigma_v = \Delta/\sqrt{3}$ the SD of the screened state; $\hat{\beta}$ identifies the inverse *per-product residual-demand slope exactly.*

Ladder geometry is also *level-neutral*: $\mathbb{E}[\text{clearing price}] = p^m(\bar{a})$ for every n, because within each interval of states the realised price averages to the monopoly price at the interval's midpoint (Remark 1, Appendix D.3). Bid-shape changes and price-level changes are therefore separate channels — a reform can regrade ladders strongly while leaving price levels untouched, exactly the DA15 pattern in §6.

With a linear cost $\kappa > 0$ per competing tranche, the optimal depth satisfies $2n^* + 1 = \big(\Delta^2/(3b\kappa)\big)^{1/3}$ (Proposition 2, Appendix D.3): ladders deepen with the screened range and flatten with market thickness and complexity cost, so a wider screened range raises σ_p and lowers HHI_p *together* — the joint signature the bid-shape DiD of §5.2 tests.

Corollary 2 (Granularity × within-hour variation, in closed form)**.** *Let a venue switch from hourly to per-quarter products, so that in hour class $\mathcal{X}$ the screened range moves from Δ to $\rho_{\mathcal{X}}\Delta$ and the per-product thinness from b to b'; write $\sigma_p(\Delta;\, b)$ for the fixed-n value in Corollary 1. Then, exactly: (i) the conditional monopoly-price range moves from Δ/b to $\rho_{\mathcal{X}}\Delta/b'$, and with it the dispersion (σ_p multiplies by $\rho_{\mathcal{X}}\, b/b'$ at fixed depth) and the tranche-price span; (ii) with the depth endogenous, $2n^* + 1$ multiplies by $\rho_{\mathcal{X}}^{2/3}\, (b/b')^{1/3}$, so $HHI_p = 1/n^*$ falls in critical hours exactly when σ_p rises; and (iii) the critical-minus-flat, post-minus-pre double difference nets out the thinness change, which is common to both hour classes: with $\rho_{flat} = 1$ and at fixed depth,*

$$\underbrace{\big[\sigma_p(\rho_{\mathcal{X}}\Delta;\, b') - \sigma_p(\Delta;\, b)\big]}_{\textit{critical hours}} \;-\; \underbrace{\big[\sigma_p(\Delta;\, b') - \sigma_p(\Delta;\, b)\big]}_{\textit{flat hours}} \;=\; (\rho_{\mathcal{X}} - 1)\, \sigma_p(\Delta;\, b'),$$

zero when the hour is internally flat and strictly increasing in the within-hour variation that $\rho_{\mathcal{X}}$ encodes (Lemma 1). (Proof: Appendix D.3.)

This is the mechanism in one display: the reform changes bid shape exactly where, and exactly as much as, within-hour variation amplifies the state a product screens; anything common to both hour classes — thinness changes, complexity costs, price-level seasonality — drops out of (iii) by construction.

Corollary 3 (Continuation slope and forward locus)**.** *Under the stage-1 second-order condition $B_2 \equiv \sum_t b_{2t} < 2b_1$ (Assumption 2, Appendix D.3), the slope of the stage-1 monopoly locus is $dp_1^*/dv = 1/(2b_1 - B_2/2)$, the measured forward ladder slope is $\hat{\beta}_1 = 1/(b_1 - B_2/2)$, and the stage-1 clearing-price range over the screened states is exactly $2\Delta_1/(2b_1 - B_2/2)$. As the spot venue becomes per-quarter, B_2 jumps from a single hourly slope to the sum of Q per-quarter slopes; the forward locus, the measured slope, and the price range all steepen and widen correspondingly — the cross-market anticipation channel of §3.4(M4).*

A price-taking unit, by contrast, posts its marginal-cost schedule — for a zero-marginal-cost renewable a single block at the floor, $\sigma_p = 0$, $\mathrm{HHI}_p = 1$, out of band at any granularity (Remark 2, Appendix D.3) — so the bid-shape margin is dispatchable-specific, and the renewable margin of the reform is the quantity re-scheduling of §3.4(M1).

3.3 Arbitrage regimes and the pre-reform baseline

Following Ito and Reguant (2016, §I.A and Table 1), arbitrage profit per matched product is $\pi^A(s) = s\,(p_1 - p_2)$, and four regimes pin the position: (1) *no arbitrage*, $s = 0$, with a positive forward premium $p_1 > p_2 > c$; (2) *full arbitrage*, where s closes the spread and the strategic bidder withholds more in response; (3) *limited arbitrage*, where a capacity ceiling $s \leq K_a$ binds and sustains a wedge that depends on K_a; and (4) *strategic arbitrage*, where the arbitrageur internalises its own price impact and stops short of parity at the closed-form benchmark

$$\boxed{p_1 - p_2 \;=\; \frac{p_1 - c}{4} \;>\; 0.} \tag{1}$$

Comparative-statics signs agree across (1), (3), and (4). Real arbitrageurs have bounded balance sheets and market power — the persistent forward premium observed across electricity markets rules out full arbitrage — so the limited regime is the empirical benchmark; once granularity makes the wedge per-product at (Q, Q), the per-product capacity constraint binds at K_a/Q and delivers the non-collapse of within-hour wedge dispersion documented in §6.4.

Pre-reform $(1,1)$: scarcity wedge, no within-hour objects. Both auctions hourly. The Ito and Reguant scarcity wedge $p_1 > p_2 > c$ obtains under any non-full-arbitrage regime (Appendix D.4); neither venue can carry the within-hour profile or the redistributive news, so the balancing discrepancy is $\Delta_t = \zeta_t + \mu_t + \eta_t + \varepsilon_t$, all within-hour dispersion objects (the $D^\bullet$ outcomes of §4.4, the within-hour wedge SD) are mechanically zero, and the competing tiers screen the hour-level ranges Δ_1 and Δ_2 — the pre-reform baseline of the bid-shape statistics. Whatever the contracting grid cannot carry loads onto the balancing residual.

3.4 Spot reform $(1, Q)$: contractibility arrives

The spot becomes per-period while the forward stays hourly; the first granular venue makes the within-hour state contractible, and four mechanisms activate.

(M1) The aggregator gains from granular spot scheduling. A granular spot lets the aggregator wait until its forecast update η_t resolves and re-schedule per period, driving its per-period imbalance to zero; an hourly spot forces one schedule before η_t is known, so the update loads onto the imbalance bill. Under uniform shocks the expected hourly hedging gain is the avoided cost

$$\boxed{G(\Delta_\eta) \;=\; \frac{\gamma Q\,\Delta_\eta}{2}, \qquad \frac{\partial G}{\partial \Delta_\eta} \;=\; \frac{\gamma Q}{2} \;>\; 0,} \tag{2}$$

strictly increasing in the forecast-error range — largest for wind and solar. The individual split is a corner solution, so the migration is an aggregate equilibrium statement: $G > 0$ is a participation premium producing the sell-share migration of Result 6.4 (Appendix D.2).

(M2) Price levels: venue re-clearing and cross-market transmission. The spot best response per quarter is $p^*_{2t} = (p_1 + c)/2 + (\omega_t + s_t - S^{op}_{2t})/(2b_{2t})$ (Appendix D.5): the spot level falls with migrated aggregator supply and with any per-product thickening of the slope through the markup identity (6) below. On the forward leg, arbitrage transmits the spot discount upstream; the same-magnitude DA-side drop (§6.1) says this channel dominates the withholding push from the strengthened continuation value.

(M3) Block parity reveals within-hour spot dispersion. The hourly forward can be arbitraged only against the *average* of the Q spot prices — block parity — so arbitrage disciplines the mean wedge but not the within-hour dispersion of the granular leg. The spot prices spread across quarters with the update:

$$\boxed{\mathrm{sd}_t(p_{2t}) \;=\; \frac{\sigma_\omega}{2\,b_2}, \qquad \sigma^2_\omega = \sigma^2_\zeta + \sigma^2_\mu,} \tag{3}$$

Since p_1 is one number per clock-hour, (3) is also the within-hour wedge SD — mechanically zero before the reform, largest in ramp hours where σ_ζ is large (thinness-asymmetry corrections in Appendix D.5).

(M4) Bid ladders: own-venue exposure and cross-market anticipation. Two exact ladder results. *(a) Spot tier, own venue.* Each quarter product screens the $\rho_{\mathcal{X}}$-amplified state, so by Corollary 2 the critical-hour spot tier widens and deepens while flat hours are unchanged and class-common thinness shifts net out — exactly the contrast the bid-shape DiD of §5.2 is built on. *(b) Forward tier, cross-market.* The stage-1 problem internalises the spot continuation value, and the slope of the stage-1 monopoly locus is

$$\boxed{\frac{dp^*_1}{dv} \;=\; \frac{1}{2b_1 - B_2/2}, \qquad B_2 \equiv \sum_t b_{2t},} \tag{4}$$

with B_2 the total slope of the spot products hanging off the forward product (we maintain $B_2 < 2b_1$, Assumption 2). At the spot reform B_2 jumps from the single hourly slope b_2 to the sum of Q per-quarter slopes — empirically of the same order as the hourly one (Table 16), so the jump is large — steepening the forward locus: the stage-1 clearing-price range is exactly $2\Delta_1/(2b_1 - B_2/2)$ and the measured ladder slope becomes $\hat\beta_1 = 1/(b_1 - B_2/2)$ (Corollary 3). This anticipation channel is the model's account of the day-ahead bid-shape response at ID15 (σ_p and $\hat\beta$ up in §6.5); the tranche-count response on the cross-market cell is not signed by this channel alone, and we do not claim it.

Balancing: the quarterly spot absorbs $\zeta_t + \mu_t$ and the aggregator hedges η_t, so the balancing discrepancy is $\Delta_t = \varepsilon_t$ already at this state (Appendix D.8). Contractibility arrives with the *first* granular venue.

3.5 Forward reform (Q, Q): relocation and limited per-product arbitrage

The forward reform *relocates* the already-contractible within-hour objects from spot to forward.

(i) The ramp becomes forward-contractible. Per-quarter forward products price ζ_t at the day-ahead margin: $\mathbb{E}[p_{1t}] = p_1^*(\bar{A} + \zeta_t)$ with pass-through $\lambda_1 \equiv (2b_1 - B_2/2)^{-1}$ per unit of ramp, so the across-quarter dispersion of forward prices turns on at $\mathrm{sd}_t(\mathbb{E}\, p_{1t}) = \lambda_1 \sigma_\zeta$ — the model counterpart of the jump in the within-hour dispersion outcomes $D^{\bar{p}}$ and D^{MCP} on the DA at DA15 (§4.5). The spot's update demand correspondingly shrinks from $\zeta_t + \mu_t$ to μ_t.

(ii) Per-product limited arbitrage — and arbitrageur market power. The arbitrageur earns $s_t\,(p_{1t} - p_{2t})$ per matched quarter. Full arbitrage gives $p_{1t} = p_{2t}$ (wedge SD zero); unlimited strategic arbitrage gives the per-product wedge (1) with wedge SD $\lambda_1 \sigma_\zeta / 4$. Both fail empirically: the per-product capacity K_a/Q binds tightly. The symmetric state is therefore in the **limited-arbitrage regime** (Ito and Reguant Result 3) applied per product: the per-product wedge becomes $(p_{1t} - c)/2$ rather than $(p_{1t} - c)/4$, doubling the wedge SD to

$$\boxed{\mathrm{sd}_h(p_{1t} - p_{2t}) \;=\; \frac{\lambda_1\, \sigma_\zeta}{2}} \tag{5}$$

(Appendix D.6.3). The wedge SD does *not* collapse at the symmetric state: arbitrageur market power generates the non-collapse documented empirically at DA15 (§6.4).

(iii) Bid ladders relocate, not duplicate. The forward screen widens from Δ_1 to $\rho_{\mathcal{X}} \Delta_1$ — the redistributive component becomes contractible at the forward — so by Corollary 2 the forward tier widens and deepens in critical hours exactly as the spot tier did at the spot reform (a critical-vs-flat contrast now on the DA). Symmetrically, the spot screen loses its ramp-timing component: writing $\rho^\mu_{\mathcal{X}} \le \rho_{\mathcal{X}}$ for the news-only amplification (Lemma 1 on the smaller support; $= 1$ in flat hours), the spot screen falls from $\rho_{\mathcal{X}} \Delta_2$ to $\rho^\mu_{\mathcal{X}} \Delta_2$ and the spot tier *tightens* by the same closed forms run in reverse.

(iv) Markup channel and the Cournot identity. The Cournot first-order condition for the strategic bidder against an inverse residual demand of slope b_{mt} is

$$p_t - c \;=\; \frac{q_m}{b_{mt}}, \qquad \mathrm{markup}_m \;=\; \frac{q_m}{b_{mt}}, \tag{6}$$

empirically tested via the strategic markup proxy q_f^{strat}/b_f of §6.6. The per-firm slope b_f thickens at every reform on every Big-4 dominant firm, so (6) delivers markup compression — the headline market-power result, operating through the markup identity and *not* through ladder geometry (Remark 1).

(v) Balancing unchanged. Nothing new loads onto the balancing residual: $\Delta_t = \varepsilon_t$ at both $(1, Q)$ and (Q, Q), so the model predicts no further imbalance gain at DA15 — the empirical null (§6.2).

3.6 Comparative statics and welfare across the reform path

Predictions to test empirically. The model delivers seven sign predictions across the reform path, pinned by the contractibility logic; the state-by-state closed forms are tabulated in Table 23 (Appendix D.7), and §6 verifies the predictions in turn: (P1) aggregator quantity shifts from forward to spot at the spot reform, larger in clock-hours with bigger forecast uncertainty; (P2) the granular-side equilibrium price drops at the spot reform via venue re-clearing and Cournot markup compression as per-firm b_f thickens (not through ladder geometry, Remark 1); (P3) within-hour wedge dispersion turns on at the spot reform at $\sigma_\omega/(2b_2)$, largest in ramp hours; (P4) the spot competing tier widens and deepens in critical hours at the spot reform (Corollary 2), and the forward competing tier steepens through the continuation-slope channel (Corollary 3); (P5) the forward competing tier widens and deepens at the forward reform (the ramp is now forward-contractible), and the spot competing tier correspondingly tightens — the relocation pattern; (P6) within-hour wedge SD does not collapse at (Q, Q) — the empirical asymmetric-persistence finding under per-product limited arbitrage; (P7) production-unit arbitrageurs exist and are capacity-constrained (production plants submitting both buy and sell offers).

New contractibility arrives only at the *spot* reform; the *forward* reform relocates it. This asymmetry of activation is the model basis for the empirical pattern of §6.2: −33% imbalance penalty per MWh at ID15, flat at DA15.

Welfare. Three channels raise welfare, one is distributional: the imbalance saving $\gamma Q(\sigma_\zeta^2+\sigma_\mu^2+\sigma_\eta^2)$, arriving entirely at the spot reform and net-positive whenever $\gamma > 1/(8b_2)$ after the small allocative cost of per-quarter monopoly pass-through; markup compression wherever per-firm b_f thickens, through (6) — by Remark 1 the only channel that moves mean prices; the discreteness loss $\Delta^2/(12b(2n^*+1)^2)$, shrinking where ladders deepen; and bid-ladder rent, mostly a CS$\rightarrow \pi^{SB}$ transfer. Per-channel derivations in Appendix D.9.

4 Data: OMIE bid-level offers, weather, redispatch

4.1 Sources

Three sources feed the analysis: bid-level OMIE data (every unit, every period, every offer tranche) covering 2024-06-14 to 2026-03-06; ENTSO-E load and renewable generation; and the ESIOS daily settlement of REE's adjustment-cost channels. The bid-level granularity is what lets the bid-shape DiD identify within-day shape changes; the system-level data feeds the BSTS counterfactual and the cost decomposition.

[To be written. Concepts to cover:]

- **OMIE** — bid-level offers in DA (PDBC, files *cab*/*det*) and IDA (PIBCI, files *icab*/*idet*); one row per (unit, date, period, tranche); structural type (simple, minimum-income condition, block); OMIE spec v1.37. Daily resolution 2022-01-01 onwards; bid-level detail 2024-06-14 onwards (post-IDA-reform 3-session regime).
- **ENTSO-E Transparency Platform** — 15-min actual total load (code A65), per-technology actual generation (A75: B16 solar PV, B18 wind onshore, B19 wind offshore), installed capacity, cross-border flows. Builds the residual-demand series (load − wind − solar) used for the critical/flat partition and the wind/solar covariates for the BSTS and OLS price specs.
- **ESIOS (REE)** — daily settlement of REE's adjustment-cost components (TR-up/dn, aFRR-up/dn, mFRR-up/dn, Fase I up/dn, Fase II), 139 curated indicators, per-BSP settlement detail (archives 17/203), aFRR offer curves, unit outages. Token-authenticated via `.env`.
- **Magnitude.** 420 GB raw, 200 GB processed parquet, ∼10 billion rows on a dedicated external SSD. Full automated reproducible pipeline.

4.2 Bid offers, reform path, and notation

Unit of observation and indices. The unit of observation is one offer curve: a unit u, on date d, for delivery period p — a clock-hour before the market's MTU15 cutover, a 15-minute quarter after it. Clock-hours are indexed $h \in \{1, \dots, 24\}$, with within-hour quarters $q \in \{1, \dots, 4\}$ defined post-MTU15 only; s indexes IDA sessions, t technologies, $m \in \{\mathrm{DA}, \mathrm{IDA}\}$ markets, and k the tranches of a step ladder. T_r is a reform cutover date and $\mathcal{X} \in \{\text{morning ramp, midday, evening ramp, flat}\}$ the hour class of §4.5.

Bid offer. Each (u, d, p) sell offer is a step ladder $\{(p_k, \Delta q_k)\}_{k=1}^{K_{u,d,p}}$ sorted by ascending price: Δq_k is the incremental MW the unit adds to its supply once the price reaches p_k, so cumulative supply at any price p is $S_{u,d,p}(p) = \sum_{k:\, p_k \le p} \Delta q_k$ — exactly the step offer of the model (§3.2). Offers carry one of three structural types: *simple* (a bare price–quantity ladder), *minimum-income condition* (MIC), or *block* (all-or-nothing across periods). Most technologies bid simple; CCGT is the outlier, with the bulk of its offers in MIC or block form.

Market clearing. $\mathrm{MCP}_{d,p}$ is the OMIE clearing price (EUR/MWh) of the auction in which the offer was submitted and $Q^\star_{d,p}$ the system cleared quantity; per-technology cleared MWh per clock-hour is written $Q^{\mathrm{cl}}_{t,m,d,h}$.

Reform path. The Spanish wholesale market underwent three granularity reforms in under a year: **ISP15** (2024-12-11), REE's settlement period moved from 60 to 15 min while the OMIE contract grid stayed at 60 min; **ID15** (2025-03-19), intraday auctions and continuous market moved from 60 to 15 min; **DA15** (2025-10-01), day-ahead moved from 60 to 15 min. The April 2025 blackout (2025-04-28) triggered a continuous *operación reforzada* regime (CCGT-led voltage support, post-DA redispatch) that overlaps DA15 calendar-wise but operates on a different mechanism. Figure 1 maps the sequence; analysis windows are built inside constant-regime spans, so the blackout regime is held fixed on both sides of each cutover.

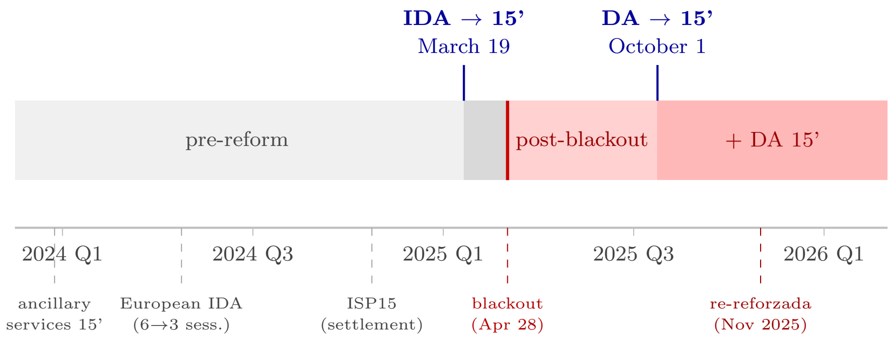


Figure 1: The Spanish reform sequence, 2024–2026. The two headline granularity reforms — intraday auctions to 15-minute products (ID15, 2025-03-19) and day-ahead to 15-minute products (DA15, 2025-10-01), blue markers — are preceded by three 2024 pre-reforms (ancillary-services products to 15 minutes, the European IDA consolidation from six to three sessions, and 15-minute imbalance settlement, ISP15) and interrupted by the April 2025 blackout, which triggered the reinforced-operation regime (red shading), tightened again in November 2025. Analysis windows are built inside constant-shading spans so the blackout regime is held fixed on both sides of each cutover.

4.3 In-band region and bid-shape outcomes

The bandwidth $\delta = p_{90} - p_{50}$ of the contemporary MCP distribution isolates the *competing tier* of each ladder — tranches close enough to clearing to be in real contention. Per in-band curve we compute five jointly informative bid-shape statistics: MW-weighted price dispersion σ_p, intercept $\hat{\alpha}$ (level), linear slope $\hat{\beta}$, curvature $\hat{\gamma}$ (convex > 0 / concave < 0), and bid-stack concentration HHI_p (MW-weighted Herfindahl of in-band tranches). σ_p is reported alongside the $(\hat{\alpha}, \hat{\beta}, \hat{\gamma})$ polynomial fit because it is well-defined on every curve — including single- and few-tranche curves where the OLS slope is noisy — and so serves as the robust price-side anchor of the bid-shape DiD.

Bandwidth. Let δ be a bandwidth that isolates the tranches in contention — close enough to the clearing price to compete in the *competing tier* of the offer (CCGT's two-tier structure makes this distinction explicit, Appendix C). We anchor δ to the contemporary contested margin: $\delta = p_{90} - p_{50}$ of the MCP distribution in the analysis window, market by market. The eight bandwidths (one per reform $\times$ {real, placebo} $\times$ {DA, IDA}) are reported below; wider-δ robustness in Appendix A.5.

Window	δ in DA (EUR/MWh)	δ in IDA (EUR/MWh)
ID15 real	50	62
ID15 placebo (2024)	45	46
DA15 real	50	58
DA15 placebo (2024)	45	49

Table 1: Bandwidths δ used to define the in-band region. δ is set to $p_{90} - p_{50}$ of the MCP distribution in each (window, market). Window-and-market-specific by construction. Bandwidth robustness in Appendix A.5.

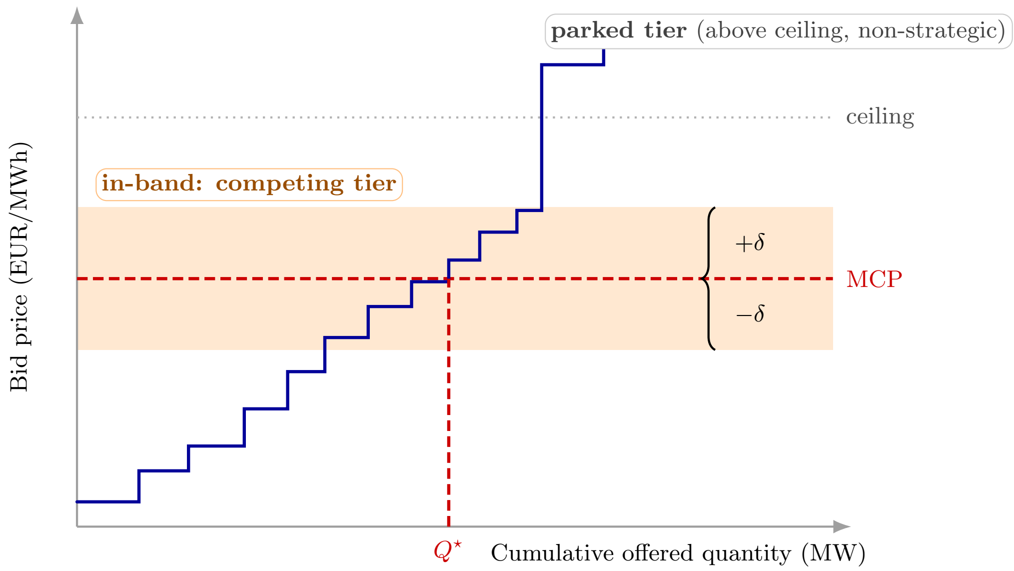


Figure 2: Bandwidth δ and the in-band region. The blue step function is the sorted sell-bid ladder for one quarter. Only tranches falling within $\pm\delta$ of the clearing price (orange band) enter the bid-shape outcomes; the parked tier above the clearing-price ceiling is excluded (Appendix C).

In-band set. For each (u, d, p) curve:

$$\mathcal{B}_{u,d,p} = \{k : |p_k - \text{MCP}_{d,p}| \le \delta\}, \qquad w_k = \frac{\Delta q_k}{\sum_{j \in \mathcal{B}_{u,d,p}} \Delta q_j}, \qquad \bar{p}_{u,d,p} = \sum_{k \in \mathcal{B}} w_k \, p_k.$$

Representative bid curves per technology in critical and flat hours are in Appendix C.4 (Figures 11–12); they illustrate the dispatchable-ladder vs renewable-block contrast.

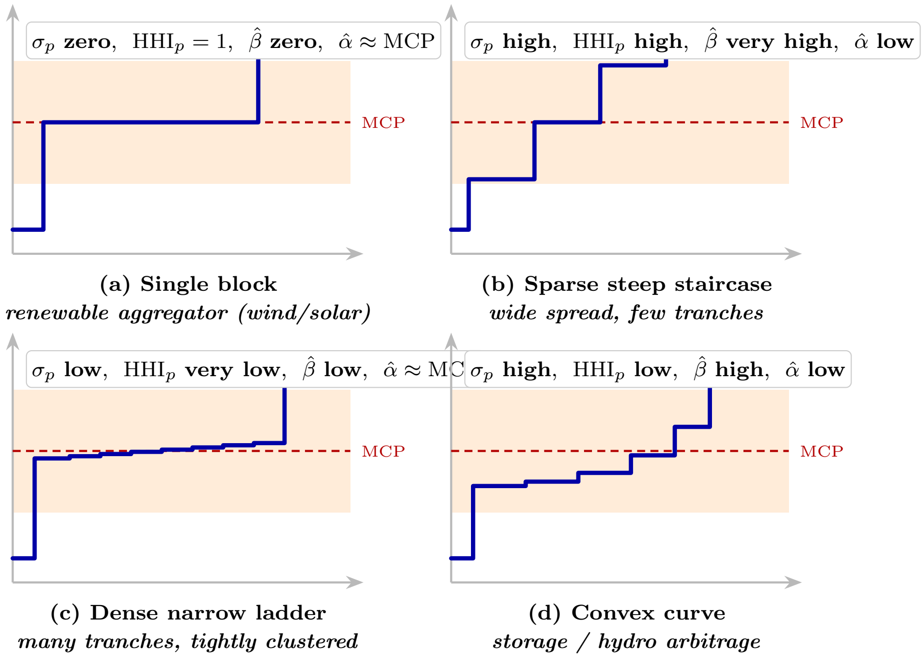


Figure 3: Four stylized in-band bid shapes spanning the $(\sigma_p, \mathbf{HHI}_p, \hat\beta, \hat\alpha)$ space. Each panel shows one bid ladder against a common MCP (red dotted) and bandwidth δ (orange band), with the implied outcome values noted in the corner.

Five bid-shape outcomes. Per in-band curve we summarise the ladder with a quintuple $(\sigma_p, \hat\alpha, \hat\beta, \hat\gamma, \mathrm{HHI}_p)$. Let $Q_k = \sum_{j \le k} \Delta q_j$ be the cumulative in-band MW after the kth tranche.

Price-side dispersion (the robust price-side anchor, well-defined on every curve including single- and few-tranche ones where the OLS slope is noisy):

$$\sigma_{p,\,u,d,p} \;=\; \Big[\sum_{k\in\mathcal{B}} w_k\,(p_k - \bar p_{u,d,p})^2\Big]^{1/2}.$$

Level and linear slope (unweighted OLS on in-band step endpoints):

$$p_k \;=\; \hat\alpha_{u,d,p} + \hat\beta_{u,d,p}\,Q_k + e_k.$$

$\hat\alpha$ is the geometric level of the ladder; $\hat\beta$ is the average slope (EUR/MWh per MW) and the direct empirical counterpart of the model's per-product residual-demand slope b_{mt} (§3.1).

Curvature (separate quadratic fit; we keep only the quadratic coefficient):

$$p_k \;=\; \tilde\alpha_{u,d,p} + \tilde\beta_{u,d,p}\,Q_k + \hat\gamma_{u,d,p}\,Q_k^2 + \tilde e_k.$$

$\hat\gamma > 0$ means convex (cheap MW packed at the bottom, marginal MW priced high); $\hat\gamma < 0$ means concave.

Bid-stack concentration (MW-weighted Herfindahl of in-band tranches):

$$\mathrm{HHI}_{p,\,u,d,p} \;=\; \sum_{k\in\mathcal{B}} w_k^2, \qquad w_k = \frac{\Delta q_k}{\sum_{j\in\mathcal{B}}\Delta q_j} \in \big[1/|\mathcal{B}|,\,1\big].$$

Three of the construction's choices deserve their reasons. First, σ_p is MW-weighted because profit uncertainty scales with tranche size: a 200-MW tranche priced five euros off the curve's mean moves the bidder's revenue risk, while a 0.1-MW decoy tranche at a distant price barely registers — unweighted dispersion would let micro-tranches dominate the statistic. Second, the level–slope pair $(\hat{\alpha}, \hat{\beta})$ and the curvature $\hat{\gamma}$ come from two separate fits so each coefficient keeps a clean reading: $\hat{\gamma}$ is the curvature left over after the linear trend, in the Frisch–Waugh sense, and is identified only on curves with at least three tranches (two-tranche curves identify $(\hat{\alpha}, \hat{\beta})$ alone). Third, every outcome is computed relative to the contemporaneous clearing price through the bandwidth δ, so none of the five inherits the price-level seasonality that dominates raw Spanish prices. Figure 3 illustrates all five on stylised ladders.

Of the five, σ_p and HHI_p carry the headline evidence: as first and second moments of the in-band price distribution they are robust to the mechanical changes in tranche counts and price-grid density that the European-IDA reform (2024-06-14) and ID15 introduced. $\hat{\beta}$ and $\hat{\gamma}$ are OLS objects on the tranche grid and inherit exactly those mechanical shifts, which is why their pre-trends drift (Appendix A) and why the analysis treats them as supporting rather than headline evidence.

4.4 Aggregate outcomes

The system-level outcomes are built off the bid stack and the cleared prices: in-band offered energy per (tech, market, day, hour-class); the day-ahead/intraday wedge $\Delta_{d,h}$ and its within-day SD Δ_d^{SD}; the intraday-auction in-band sell-share by tech; system ajuste cost (per day, EUR M); and five post-MTU15-only within-hour dispersion outcomes $D^{\bullet}_{d,h}$ that summarise across-quarter shifts in price level, volume, system MCP, ladder spread, and tranche count.

In-band offered energy (per (tech, market, day, hour-class)) measures how much capacity competes near the clearing price — the quantity-side counterpart of the bid-shape outcomes. Per-curve in-band MW $m_{u,d,p} = \sum_{k \in \mathcal{B}_{u,d,p}} \Delta q_k$ is aggregated with the period length τ_p (= 1 pre-MTU15, 1/4 post-MTU15), which converts MW to MWh so that pre- and post-reform quantities are comparable — a raw sum of per-period MW would mechanically quadruple after the cutover purely from the period count:

$$E^{\mathrm{band}}_{t,m,d,\mathcal{X}} \;=\; \frac{1}{|\mathcal{X}|} \sum_{h \in \mathcal{X}} \sum_{p \in h} \sum_{u \in t} m_{u,d,p} \cdot \tau_p \qquad \text{(MWh per class-hour)}.$$

Cross-market wedge (per clock-hour) is the arbitrage object of §3.3: its level measures the forward premium, and its within-day SD the dispersion that granularity unlocks and bounded arbitrage fails to close. With $\mathrm{MCP}^m_{d,h}$ the within-market mean of OMIE clearing prices over sub-hour structure on date d:

$$\Delta_{d,h} \;=\; \mathrm{MCP}^{\mathrm{DA}}_{d,h} - \mathrm{MCP}^{\mathrm{IDA}}_{d,h}, \qquad \Delta_d^{\mathrm{SD}} \;=\; \mathrm{sd}_h(\Delta_{d,h}).$$

IDA in-band sell-share (per (tech, day)) locates each technology between sell-heavy participation (rent capture on the granular leg) and buy-heavy participation (forecast-driven rebalancing) — the empirical counterpart of the aggregator-migration mechanism (M1):

$$\mathrm{SS}_{t,d} \;=\; \frac{\sum_{q,k \in \mathcal{B}} q^{\mathrm{sell}}_{t,d,q,k}}{\sum_{q,k \in \mathcal{B}} \big(q^{\mathrm{sell}}_{t,d,q,k} + q^{\mathrm{buy}}_{t,d,q,k}\big)}.$$

System ajuste cost (per day, EUR M) tracks what REE pays to keep the system feasible — the outcome the imbalance and reforzada questions of §6 are about. Components from ESIOS: up-direction (TR-up, aFRR-up, mFRR-up, Fase I) and down-direction (Fase II, RR-dn), stacked in Figure 4.

Within-hour dispersion outcomes (post-MTU15 only) trace within-hour discrimination directly: each is an across-quarter SD inside one clock-hour, so each is mechanically zero under hourly products. For each (unit, date, clock-hour) with four quarters $q = 1, \ldots, 4$:

$$D^{\bar{p}}_{d,h} = \mathrm{sd}_q(\bar{p}_q), \quad D^{m}_{d,h} = \mathrm{sd}_q(m_q)/\bar{m}, \quad D^{\mathrm{MCP}}_{d,h} = \mathrm{sd}_q(\mathrm{MCP}_q), \quad D^{\sigma_p}_{d,h} = \mathrm{sd}_q(\sigma_{p,q}), \quad D^{\mathrm{HHI}}_{d,h} = \mathrm{sd}_q(\mathrm{HHI}_{p,q}).$$

The five capture, in order: across-quarter shifts in the curve's price level, its offered volume (as a coefficient of variation), the system clearing price, the ladder spread, and the tranche concentration. Any positive value post-MTU15 is a unit actively treating the four quarters differently.

4.5 Critical / flat hour-class partition

Granular pricing only has economic content where within-hour residual demand[2] actually varies; if a clock-hour is internally flat, the four quarters look the same and there is nothing to discriminate on. We measure within-hour variation by $\sigma_{\mathrm{within}}(h,d) = \mathrm{sd}_q[\mathrm{load} - \mathrm{wind} - \mathrm{solar}]_{q,h,d}$ on 15-min ENTSO-E data (A65 load; A75 PSR codes B16, B18, B19 for wind+solar) and partition clock-hours by median σ_{within} post-2024-06-14:

- **Critical** $\mathcal{C} = \mathcal{C}_M \cup \mathcal{C}_E$ with $\mathcal{C}_M = \{5, \ldots, 8\}$ (morning ramp) and $\mathcal{C}_E = \{16, \ldots, 22\}$ (evening ramp); $\sigma_{\mathrm{within}} \in [522, 1303]$ MW.
- **Flat** $\mathcal{F} = \{1, 2, 3\}$: overnight; $\sigma_{\mathrm{within}} \in [336, 357]$ MW.
- **Midday** $\mathcal{M} = \{11, \ldots, 14\}$ (solar-marginal, $\sigma_{\mathrm{within}} \in [468, 502]$ MW): excluded from main specs; kept for falsification (Appendix A.2).

The bid-shape DiD of §5.2 uses the critical-vs-flat contrast as its within-day identification device.

Descriptive critical − flat differential. Let $D^{\bullet}_{d,h}$ stand for any one of the five within-hour dispersion outcomes of §4.4 on date d, clock-hour h. Fit per outcome on the post-MTU15 sample, $h \in \mathcal{C} \cup \mathcal{F}$:

$$D^{\bullet}_{d,h} \;=\; \eta_d \;+\; \theta\,\mathbf{1}\{h \in \mathcal{C}\} \;+\; \varepsilon_{d,h}$$

with date fixed effects η_d (distinct from the bandwidth δ). θ is the critical−flat dispersion gap; a descriptive contrast, not a causal estimate.

Outcome	Subsample	ID15	DA15	crit/flat ratio (IDA, DA)
$D^{\bar{p}}$ (EUR/MWh)	per unit (pooled)	**0.39***** (0.05)	**1.35***** (0.11)	4.1×, **7.4**×
D^{m} (CV)	per unit (pooled)	**0.024***** (0.003)	**0.033***** (0.002)	4.3×, 4.9×
D^{MCP} (EUR/MWh)	system	**4.55***** (0.40)	**3.97***** (0.37)	2.8×, 2.6×
D^{σ_p} (EUR/MWh)	CCGT	−0.19 (0.36)	**0.61***** (0.12)	0.8×, **6.5**×
	Hydro	**0.48***** (0.08)	**1.32***** (0.17)	2.2×, 3.3×
	Hydro pump	**0.14***** (0.03)	**0.70***** (0.17)	3.3×, 10.9×
	Wind	0.00 (0.01)	0.03** (0.01)	0.9×, 1.9×
$D^{1/\mathrm{HHI}}$	CCGT	0.17 (0.16)	**0.17***** (0.02)	1.1×, **14.3**×
	Hydro	**0.05***** (0.01)	**0.19***** (0.03)	2.0×, 2.8×
	Hydro pump	**0.03***** (0.01)	**0.13***** (0.03)	2.6×, 4.3×
	Wind	0.00 (0.00)	0.01 (0.00)	1.0×, 1.3×

[2] Spanish-market convention calls this the *hueco térmico* (thermal gap) — the load that thermal and other dispatchable generation must cover after subtracting wind and solar.

Table 2: Post-MTU15 critical−flat within-hour dispersion gap θ, by outcome. Post-only cross-sectional gap. Windows: ID15 post 2025-03-19 to 2025-04-27; DA15 post 2025-10-01 to 2025-12-31. SE clustered by date; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (applies to all subsequent tables). Bid-level rows ($D^{\bar{p}}$, D^{m}, D^{MCP}) are bandwidth-free; the per-tech D^{σ_p} and $D^{1/\mathrm{HHI}}$ rows use the window-specific p_{90} bandwidth (§4.3): ID15 column δ=62 (post-ID15 IDA); DA15 column δ=50 (post-DA15 DA). $D^{1/\mathrm{HHI}}$ uses the effective-tranche-count framing ($N_{\mathrm{eff}} = 1/\mathrm{HHI}_p$) because the within-hour SD of HHI is mechanically tiny on a $[0, 1]$ scale; the inverse-HHI quantity carries the same content on a more readable scale.

5 Empirical strategy

The empirical strategy pairs two specifications with two economic objects. For **level outcomes** (prices, wedge SD, imbalance penalty, cleared MWh) we run hourly OLS and a BSTS counterfactual on long pre-windows (2022 onwards) with year-by-year renewable interactions, and check robustness to convex (quadratic and cubic) renewable terms. For **bid-shape outcomes** (σ_p, $\hat{\alpha}$, $\hat{\beta}$, $\hat{\gamma}$, HHI_p) we run a bid-curve-level DiD with flat hours as the within-day control for critical hours.

5.1 Level variables: hourly OLS with convex-renewable controls and BSTS counterfactual

[To be written. Concepts to cover:]

- **Windows.** Long pre-window (2022-01-01 onwards) to capture full seasonality across years; short post (40 days for ID15, pre-blackout; 92 days for DA15). DA15 windows hold reforzada constant across the cutover by starting on 2025-04-28.
- **Headline OLS spec** (no linear trend, hourly): $y_t \sim \mathrm{post} + \mathrm{wind} + \mathrm{solar} + \mathrm{gas} + \mathrm{wind} \times \mathbf{1}\{\mathrm{year} \geq 2024\} + \mathrm{solar} \times \mathbf{1}\{\mathrm{year} \geq 2024\} + \mathrm{hour\,FE} + \mathrm{month\,FE} + \mathrm{DOW\,FE}$. Newey-West HAC SE lag $= 24 \times 7$.
- **Why year-by-renewable interactions are load-bearing.** Spanish solar PV grew +43% between Jan 2024 and Jan 2025; spring-2025 IDA prices clearing roughly flat would be a price drop net of solar's marginal merit-order push.
- **Convex-renewable robustness.** Replace linear renewables with quadratic (wind^2, solar^2, year-interacted on both linear and quadratic terms) and cubic specs. Quadratic spec absorbs the merit-order curvature; cubic spec adds essentially nothing on top.
- **BSTS counterfactual** (Brodersen et al., 2015), EPEX-style (Märkle-Hußet al., 2020): state-space level μ_t (random walk with stochastic slope) + 7-day seasonality + exogenous regressors with spike-and-slab prior. Same regressors as OLS. Treatment effect $\bar{\tau} = \frac{1}{|\mathrm{post}|} \sum_{t \geq T_r} (y_t - \hat{y}_t(0))$.
- **Same-calendar placebo** one year before each cutover guards against recurring seasonal artefacts. Real effect is retained only if the placebo is null or opposite-signed.
- **Per-firm residual-demand slope** $b_{f,t}$ following Ito and Reguant (2016) §III.A: rebuild rivals' aggregate supply $S_{-f}(p)$ from the OMIE offer book, form $Q_f^{\mathrm{res}}(p) = D - S_{-f}(p)$, read the slope at MCP via a quadratic fit on MCP ± 50 EUR/MWh.
- **Strategic markup proxy.** Cournot first-order condition: $(p - MC)/p = q_f^{\mathrm{strat}}/(b_f \cdot p)$, where q_f^{strat} subtracts the focal firm's own renewable cleared MW (infra-marginal at $p \approx 0$). Pre/post comparison of q_f^{strat}/b_f per firm captures both the residual-demand thickening and the loss of strategic capacity to renewables.

5.2 Bid-shape outcomes: per-curve DiD with critical-vs-flat hours

The bid-shape DiD identifies *within-day* changes in the bid curve: per (unit, date, period) we summarise the in-band ladder with the quintuple $(\sigma_p, \hat{\alpha}, \hat{\beta}, \hat{\gamma}, \mathrm{HHI}_p)$ and use flat hours as the within-day control for critical hours. The flat-hour partition is the identifying restriction; no time-series extrapolation is needed.

For per-tech bid-shape outcomes $Y \in \{\sigma_p, \hat{\alpha}, \hat{\beta}, \hat{\gamma}, \mathrm{HHI}_p\}$ we use the DiD:

$$Y_{u,d,p[,s]} = \alpha_u + \beta\, \mathbf{1}\{d \geq T_r\} + \xi\, \mathbf{1}\{h \in \mathcal{C}\} + \theta\, D_{d,h} + \varepsilon, \qquad \text{SEs clustered by date.}$$

[To be written. Concepts to cover:]

- **Why critical vs flat.** Differentiated bidding only pays off where residual demand is steep enough to reward it. In critical hours that condition holds and four quarter-prices unlock real discrimination; in flat hours the curve is shallow and the extra granularity is mostly wasted optionality. The critical−flat differential θ *is* the granularity-enabled discrimination, not a residual we are netting out.
- **Identification assumption.** Parallel trends in critical and flat hours absent the reform. σ_p and HHI_p satisfy this cleanly for CCGT and Hydro-pump (the headline cells); $\hat{\beta}$ and $\hat{\gamma}$ show pre-trend drift because they are mechanically sensitive to tranche-count and price-grid density (both shifted at 2024-06-14 and 2025-03-19). Treat $\hat{\beta}$ and $\hat{\gamma}$ as supporting, not headline.
- **Threats absorbed by construction.** Bid-shape outcomes are robust to seasonal level shifts (all computed relative to contemporaneous MCP via bandwidth δ). Main residual threat: post-reform shocks that affect critical or flat hours asymmetrically in *shape* (not level).
- **Each reform studied separately** — standard 2×2 DiD, no staggered-treatment complications (de Chaisemartin and D'Haultfœ uille, 2026, ch. 3).

6 Results

[To be written. Headline answer mirroring the slide deck: granularity made the market more competitive without moving aggregate quantities. Three first-order questions:]

- **Q1 Prices.** ID15 cut IDA prices substantially; DA15 was null. Cross-market response appears on the bid *shape*, not on the price level.
- **Q2 Imbalance and ancillary services.** Aggregate imbalance volume unchanged; penalty per MWh dropped by a third at ID15; ancillary-services cost rise is reforzada-driven, not granularity-driven.
- **Q3 Market power, bid shape, quantity migration.** Per-firm residual demand thickened at every reform; under a strategic-markup proxy that nets out infra-marginal own renewables, every Big-4 firm sees a markup squeeze at DA15 DA. Dispatchables grade their critical-hour ladders more finely; renewables compress on the reformed venue.

6.1 Q1 — Prices in each reform and cross-market effects

ID15 cut the granular-leg (IDA) clearing price by −30 to −46 EUR/MWh across linear specs and −23 to −24 EUR/MWh under a convex-renewable spec; the same magnitude appears on the DA cross-market response (anticipation through backward induction). DA15 own-venue and cross-market are null under every spec tried. The pricing-chain primitive — per-firm residual-demand slope b_f (Ito and Reguant, 2016) — thickens at every reform on every Big-4 dominant firm; under the Cournot first-order condition the implied markup squeezes.

[To be written. Concepts to cover:]

- **Headline robustness ladder** (ID15 IDA): OLS daily base -32^{***}; + year-by-renewable (2024+25 pooled) -39^{***}; OLS hourly + year-by-renewable -30^{***}; OLS hourly + quadratic renewables (D) -24^{***}; OLS hourly + cubic renewables (E) -23^{***}; BSTS daily -34^{*}.
- **Cross-market** (ID15 on DA): same magnitude −29 to −47 across specs. Once IDA is granular and elastic, the DA cleanly absorbs the discount through backward-induction arbitrage on the bid book.
- **DA15 null** on both own venue (−9 to +3) and cross-market IDA (−10 to +2). No spec rescues a level effect.
- **Convex-renewable robustness.** The quadratic spec adds wind^2, solar^2 with year-by-renewable interactions on both linear and quadratic terms; absorbs merit-order curvature in renewable price-suppression. Cubic adds essentially nothing (diminishing returns).
- **Renewable capacity vs production check, 2024 vs 2025.** Spring 2025 had +43% more solar capacity but only +13% more realised solar (spring 2024 already very sunny); wind essentially flat; gas +73%. Naively predicted spring-2025 price from gas alone would be much higher; the residual −23 to −39 EUR/MWh is the granularity-attributable channel.
- **Other robustness** in Appendix A: 90- and 180-day post-windows, hour-class heterogeneity, per-IDA-session breakdown, 2024 same-calendar placebo (null at ID15, residual signal at DA15).

Reform-side leg	**Per-firm residual-demand slope** *%-change in b_f* (4-firm range)	**Equilibrium clearing price** *Effect range across specs* (EUR/MWh)
ID15 IDA (granular spot)	+15 to +25%	−23 to −46
ID15 DA (cross-market)	**+93** to **+141**%	−23 to −47
DA15 DA (granular forward)	+19 to +42%	null (−9 to +3)
DA15 IDA (cross-market)	+13 to +20%	null (−10 to +2)

Table 3: Headline match between pricing-chain primitive and equilibrium price. Left: per-firm hourly OLS on $\log b_f$ (*no* linear trend), with hour FE, month FE, DOW FE, gas, wind, solar, year-by-renewable interactions. Newey-West HAC SE lag 24 × 7. Range across the four Big-4 dominant firms (Iberdrola, Endesa, Naturgy, EDP). Right: clearing-price effect range across the linear / quadratic / cubic OLS and BSTS specs (Appendix A.1). The per-firm slope thickens at every reform-side leg; ID15 DA cross-market is the largest single effect (every firm sees b_f roughly double). The price effect is robustly negative at ID15 (both legs) and null at DA15 (both legs).

6.2 Q2 — Imbalance penalty per MWh drops at ID15, aggregate volume unchanged

ID15 cut the imbalance penalty per MWh by a third; DA15 added nothing. Aggregate imbalance volume is essentially unchanged at both reforms: granularity reorganises within-hour timing of imbalance, not its size: the same MWh of residual imbalance is resolved with cheaper balancing reserves.

[To be written. Concepts to cover:]

- **Penalty per MWh of deviation** (the imbalance settlement price) is set ex-post by REE via formula, *not discretionary*: the MW-weighted price of the reserves used to balance the system.
- ID15 cuts it by −33%; DA15 is flat.

- Aggregate volume essentially unchanged (-0.1% ID15, -4% DA15). Forecast errors are physical (wind/solar/demand stochasticity); granularity reshuffles their within-hour timing, not their size.
- **Mechanism.** Granular spot reveals firms' positions earlier $\Rightarrow$ REE schedules cheaper reserves ahead of time and avoids dipping into expensive ones. Same MWh of residual imbalance, resolved with cheaper reserves.

Outcome	**Pre mean**	**ID15 (March)**	**DA15 (October)**
Penalty per MWh of |deviation| (EUR/MWh)	3.45	$\mathbf{-1.14}^{***}$ (-33%)	-0.05 (null, flat)
Aggregate |imbalance| (MWh/day)	1,676	-2 (null, -0.1%)	-67 (null, -4%)

Table 4: BSTS on imbalance penalty and aggregate volume. ID15 pre-window 2024-12-01 to 2025-03-18, post 2025-03-19 to 2025-04-27 (pre-blackout). DA15 pre 2025-04-28 to 2025-09-30, post 2025-10-01 to 2025-12-31. BSTS with seasonality, wind, solar, and gas covariates. Relative effect in parentheses. *** denotes posterior probability > 0.999 that the effect is non-zero. Penalty per MWh is the imbalance settlement price (the MW-weighted price of the reserves used to balance the system).

6.3 Q2 (cost evidence) — ancillary-services costs are dominated by reforzada, not granularity

REE's adjustment-cost lines balloon post-blackout, almost entirely on Fase I up-redispatch and almost entirely on CCGT. This is a reforzada effect, mechanistically distinct from MTU15. Pre-blackout (blue zone in Figure 4), real-time channels (TR-up, aFRR-up) shrink slightly across ISP15-win and MTU15-IDA, but our identification is too tight to pin a granularity-attributable component on the EUR cost aggregates.

[To be written. Concepts to cover:]

- The system operator is paying CCGT to produce (expensive) through pre-IDA Fase I redispatch and curtailing solar (cheap). Fase I up-redispatch costs balloon by EUR ~5 M/day post-blackout, almost entirely CCGT voltage support.
- Granularity vs reforzada: the real-time channels shrink at the same time the pre-real-time Fase I rises. The displacement reading (procurement timing shifts upstream) and the efficiency reading (granular auction absorbs intra-hour forecast errors) are observationally equivalent on the EUR aggregates; we do not separate them.
- Independent corroboration of the reforzada cost rise: Brown and Reguant (2026); Daví-Arderius and Graf (2026).

6.4 Q3 (wedge evidence) — within-hour wedge SD jumps at ID15 and per-IDA-session levels diverge

The within-day SD of the DA−IDA wedge jumps at ID15 because four IDA quarter-prices spread across each hourly DA price (mechanical). At DA15, with both auctions clearing at 15 min, the SD does *not* collapse — the empirical fingerprint of bounded per-product arbitrage capacity. On per-session levels, ID15 widens the session-1 wedge by roughly 3 EUR/MWh and narrows session 3 by roughly 7 (both robust, session 2 null); DA15 is null across sessions.

[To be written. Concepts to cover:]

- ID15 within-day wedge SD: overall $+1.1^{**}$, critical hours $+2.1^{***}$, morning ramp $+2.3^{***}$, evening ramp $+1.2^{*}$, midday -1.6^{**}, flat null. Sharpest in the morning ramp.

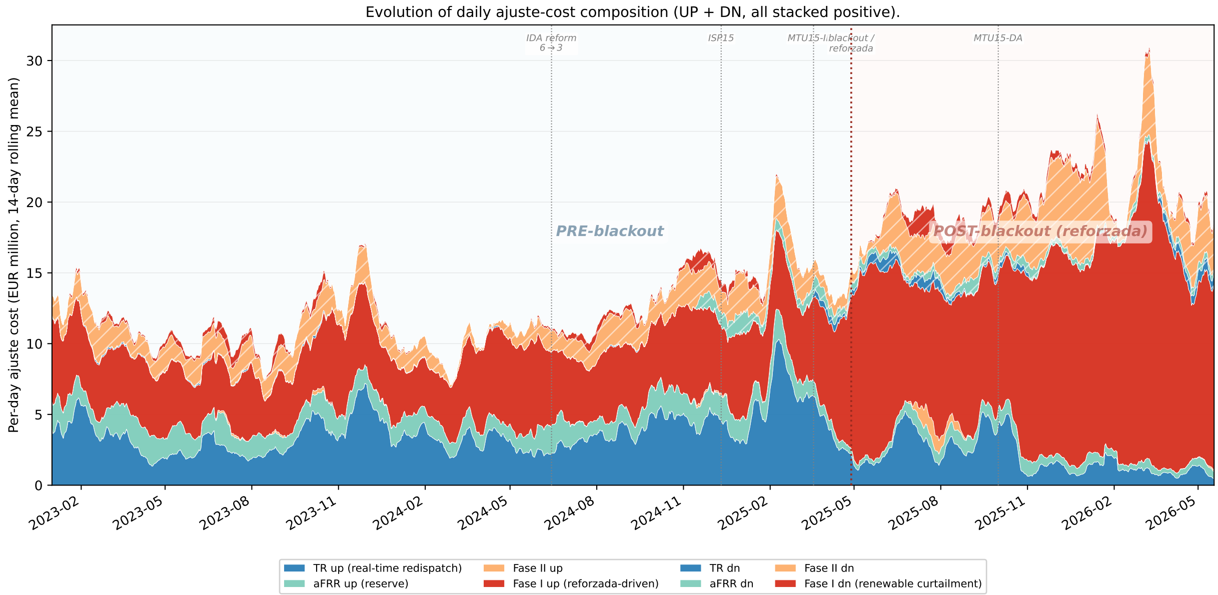


Figure 4: Daily adjustment-cost composition, 2023–2026 (EUR M / day, 14-day rolling mean). Each channel is the daily sum of the pay-as-bid system-cost series REE publishes via ESIOS: *TR up/dn* is real-time technical restrictions under PO 3.7 (cost of energy REE buys/sells to enforce grid security between gate closure and delivery); *aFRR up/dn* is the automatic Frequency Restoration Reserve energy cost; *Fase I up/dn* is the pre-IDA technical-restrictions process applied to PDBF (pay-as-bid of a separate restriction offer per PO 3.2 and PO 14.4 ap. 20.1, distinct from the day-ahead market offer); *Fase II up/dn* is the post-Fase-I rebalance that re-absorbs the up-redispatch into the schedule. All series are EUR sums collected at native 15-min granularity (raw daily totals smoothed only by the 14-day moving average), with down-direction channels taken as absolute values so the chart reads as gross costs REE pays providers in both directions. Up-direction stacked solid (TR-up, aFRR-up, Fase II up, Fase I up); down-direction stacked hatched. Blue background: pre-blackout (granularity reforms shrink the up-direction TR and aFRR channels through the ISP15-win and MTU15-IDA cutovers, visible against the stable 2023 baseline). Red background: post-blackout reforzada (Fase I up-redispatch balloons by EUR ∼5 M/day, almost entirely CCGT voltage support; this is the level shift between the 2023 baseline and 2025-Q4 totals).

- DA15 wedge SD: no significant effect across hour classes.
- Per-IDA-session wedge level at ID15: S1 +3.0** (robust across OLS daily, hourly, BSTS), S3 −7.0* (robust); ID15 placebo null. DA15 per-session null with residual signal at the placebo cutover.

6.5 Q3 (quantity-shift evidence) — bid-shape widening in critical hours

Dispatchables grade their critical-hour ladders more finely; renewables compress on the reformed venue. In the headline cells of the bid-curve-level DiD (critical vs flat, day-clustered SE), CCGT widens DA σ_p at ID15 by +4.45*** EUR/MWh, with HHI deconcentrating −0.18***; same direction at DA15 DA (+1.05***, −0.17***). Hydro-pump's DA ladder coarsens at ID15 (σ_p −1.48***): once IDA is granular, storage withholds upstream. Renewables compress: wind, biomass, cogen.

[To be written. Concepts to cover:]

- Headline metrics are σ_p and HHI: robust to mechanical tranche-count and price-grid-density changes. Slope $\hat{\beta}$ and curvature $\hat{\gamma}$ in Appendix — pre-trends drift on those, treat as supporting.
- CCGT DA widens at both reforms: ID15 σ_p +4.45***, HHI −0.18***; DA15 σ_p +1.05***, HHI −0.17***. Marginal-cost units grade their critical-hour ladder into many small price steps.
- Strategic renewables compress on the reformed venue: ID15 DA wind −0.17***, biomass −0.23***;

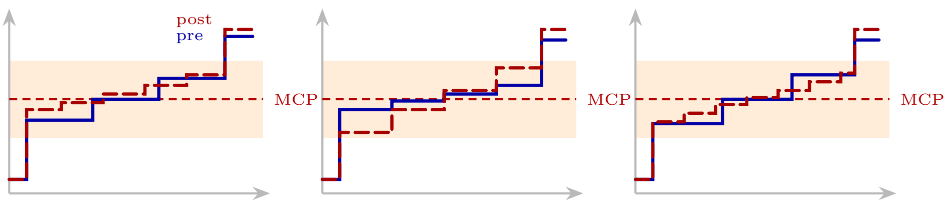
DA15 IDA CCGT cross-market -1.01^{***}.

- Hydro-pump anomaly: ID15 DA σ_p -1.48^{***}. Once IDA is granular, pump storage withholds upstream and the DA ladder coarsens.
- HHI: CCGT DA deconcentrates at both reforms; cogen, wind, biomass IDA HHI rises at ID15 (fewer fatter tranches).
- Robustness: morning-ramp-only DiD reproduces the headline signs; pre-trend figures clean for σ_p and HHI on the headline cells (Appendix).

		ID15		**DA15**	
Outcome (DiD θ)	**Tech**	DA	IDA	DA	IDA
σ_p (EUR/MWh)	CCGT	$+\mathbf{4.45}^{***}$	$+0.24$	$+\mathbf{1.05}^{***}$	$-\mathbf{1.01}^{***}$
	Hydro	-0.35^{**}	$+0.24$	$+\mathbf{0.85}^{***}$	$+\mathbf{0.44}^{***}$
	Pump-storage	$-\mathbf{1.48}^{***}$	$+0.03$	$+0.13$	-0.02
HHI_p	CCGT	$-\mathbf{0.18}^{***}$	-0.04	$-\mathbf{0.17}^{***}$	-0.00
	Hydro	$+0.010^{*}$	$-\mathbf{0.034}^{***}$	$-\mathbf{0.037}^{***}$	-0.01
	Pump-storage	0.00	-0.02	0.00	$+0.01$

Slope and curvature in Appendix — $\hat\beta$, $\hat\gamma$ are mechanically sensitive to tranche-count and price-grid density; treat as supporting.

Table 5: Bid-curve-level DiD on headline bid-shape outcomes. Critical-flat DiD on price dispersion σ_p and bid-stack concentration HHI_p. Tight pre/post windows (reforzada held constant): ID15 pre 2024-12-19 to 2025-03-18, post 2025-03-19 to 2025-04-27 (40 days, pre-blackout); DA15 pre 2025-07-01 to 2025-09-30, post 2025-10-01 to 2025-12-31. Bandwidth $\delta = p_{90} - p_{50}$ window-specific. SEs clustered by date. A negative HHI DiD means the bid stack *deconcentrates* in critical hours relative to flat. Full $(\hat\alpha, \hat\beta, \hat\gamma)$ structural decomposition, additional technologies (Cogen, Hybrid, Wind, Biomass), morning-ramp-only robustness, and parallel-trends figures in Appendix A.



DA15 DA CCGT $\hat\alpha\uparrow$, $\hat\beta\downarrow$, $\hat\gamma\uparrow$ **ID15 DA CCGT (anticipation)** $\hat\alpha\downarrow$, $\hat\beta\uparrow$, $\hat\gamma\downarrow$ **ID15 IDA hydro / pump-storage** $\mathbf{HHI}_p\downarrow$ **(denser ladder)**

Figure 5: Stylised retained bidding shifts. Three in-band ladder reshapes observed in the data: DA15 day-ahead CCGT (level up, slope flatter, convex bend); ID15 day-ahead CCGT anticipation (level down, steeper, concave bend); ID15 intraday-auction hydro and pump-storage (denser ladder, level and slope unchanged). Blue solid: pre-reform; red dashed: post-reform.

6.6 Synthesis — structural markup squeeze, aggregator vs strategic-bidder split

DA15 delivers a clean Cournot markup squeeze: per-firm residual demand thickens by +24 to +28% on the day-ahead while the firm's strategic capacity (total cleared MW minus its own renewable cleared MW) shrinks, so the markup proxy q_f^{strat}/b_f falls −12 to −27% on every Big-4 firm. ID15 also thickens b_f but the markup-proxy signs are compositional (small post window with nuclear out and CCGT heavily called). The two reforms engage two distinct agents: the aggregator (price-taker hedging forecast error) at the spot reform, the strategic bidder (residual monopolist) at the forward reform.

[To be written. Concepts to cover:]

- Strategic markup proxy (Cournot FOC, q_f^{strat} = total cleared minus own renewable cleared, infra-marginal at $p \approx 0$): DA15 DA -12 to -27% across IB/GE/GN/HC; ID15 markup signs mixed due to compositional shifts in the 40-day window.
- Aggregator footprint at the spot reform (ID15): aggregate imbalance volume unchanged, penalty per MWh drops, granular spot reveals firms' positions earlier.
- Strategic-bidder footprint at the forward reform (DA15): CCGT day-ahead ladder widening, residual demand thickening for all four firms.
- Pricing chain: b_f thicker $\Rightarrow$ Cournot markup compresses $\Rightarrow$ clearing prices fall. Delivered at ID15 (IDA and cross-market DA); null at DA15.

7 Conclusion

[To be written. Concepts to cover:]

- **Aggregate effects of DA15 are small; ID15 delivered the headline gains.** Significant changes in bidding behaviour at both reforms, but bid-shape signals are mixed across techs. Evidence of rising solar cannibalisation — already a serious profitability issue.
- **Open question:** did market microstructure affect grid stability? Three facts in the open: (i) REE's post-blackout response is technology-specific (CCGT doubles in restriction dispatch); (ii) realised renewable supply was similar in spring 2024 vs 2025, yet only 2025 blacked out; (iii) nuclear's auction-plus-bilateral output dropped sharply in the 40 days between ID15 and the blackout (Appendix A.8). The ENTSO-E Expert Panel does not list ID15 as a root cause, but CNMC flags quarter-hourly trading as contributing to abrupt voltage variations.
- **The biggest welfare losses live in the ancillary markets**, not in the wholesale auctions (Brown and Reguant, 2026; Daví-Arderius and Graf, 2026).
- **Urgent need to strengthen network security** to bring down retail prices — not finer time resolution.

CNMC report on the blackout flags quarter-hourly trading among contributors to within-hour voltage variability (CNMC, 2026, p. 44):

> *The implementation of the 15-minute trading unit in the Spanish intraday market was completed on 18 March 2025, one month before the incident. . . . The evolution to quarter-hourly trading and settlement has made the appearance of transitions between negative and positive prices increasingly frequent, leading to strong changes in production volumes in certain zones.*

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A Robustness

A.1 Year-by-year solar coefficient: the renewable response is non-stationary

The BSTS uses long pre-windows (2022 onwards) with $\text{wind}_t \times \mathbf{1}\{\text{year} = y\}$ and $\text{solar}_t \times \mathbf{1}\{\text{year} = y\}$ interactions. The economic reason is that Spanish wind+solar installed capacity grew by a factor of ~ 6 over 2018-2025, and the marginal effect of an additional GWh of renewable output on price is not constant across that period. A flat-coefficient spec mis-fits the cross-year merit-order shift and reports artificially confident treatment effects (§5.1).

Table 6 reports the cumulative solar coefficient (base + year-dummy) by calendar year, extracted from the BSTS posterior of the price spec under both windows (ID15 pre ends 2025-03-18, DA15 pre ends 2025-09-30). **Conditional posterior means** (the coefficient when the spike-and-slab includes the regressor). The wind base coefficient is stable around -0.50 EUR/MWh per GWh; the wind$\times$year interactions all have inclusion probability below 1.5%. The solar coefficient is the one that varies.

Year	**ID15 IDA spec**	**DA15 IDA spec**	**DA15 DA spec**
2022 (base)	-0.22	-0.23	-0.22
2023	-0.02	-0.23	-0.17
2024	$\mathbf{-0.46}$	$\mathbf{-0.47}$	$\mathbf{-0.40}$
2025 †	$\mathbf{-0.29}$ *(Jan-Mar only on ID15)*	$\mathbf{-0.50}$	$\mathbf{-0.49}$
Implied weighted average for the post-window prediction			
2024+2025 weighted avg ‡	-0.43	-0.45	-0.44

Table 6: Cumulative solar coefficient (EUR/MWh per GWh of solar) by year, extracted from the BSTS posterior of the corresponding price spec. Cumulative coefficient = base + year-dummy interaction, conditional posterior mean among MCMC draws that include the regressor. †The ID15 spec pre-window ends 2025-03-18, so its 2025-by-solar coefficient is fitted on Jan-Mar 2025 only, when Spanish solar production is at its lowest seasonal level; the DA15 spec pre-window extends through 2025-09-30 and includes summer, giving the more representative estimate. ‡Months-weighted average of the 2024 and 2025 cumulative coefficients (12 months of 2024 + available months of 2025): this is the coefficient one would extrapolate forward if 2024 and 2025 were pooled into a single dummy for the post-window prediction; the per-year BSTS instead extrapolates using the 2025-row coefficient alone.

Reading the table: the DA15 spec (which sees Jan-Sep 2025) tells the cleaner story. Solar's marginal price-suppressing effect is roughly constant from 2022 to 2023, then nearly doubles from 2023 to 2024 (Spain added $\sim$5 GW of solar capacity in 2024 alone), and continues to deepen in 2025 as solar saturates more clock-hours and negative-price events become routine. The ID15 spec's 2025 coefficient looks deceptively small (Jan-Mar winter sun is weak) but the underlying renewable response in 2025 is the most negative on record. The non-stationarity is exactly what the year interaction is designed to absorb: a flat-coefficient spec would average -0.22 (2022) and -0.50 (2025) into a single value around -0.35, mis-fitting both ends of the window.

Calibrated-solar robustness: how much of the price drop is solar vs the reform? A natural concern about the headline BSTS counterfactual is that Spanish solar capacity grew rapidly through 2024 and 2025, so the marginal price-suppressing effect of solar is itself non-stationary. The BSTS per-year spike-and-slab attempts to capture this via year-by-solar interactions; the conditional posterior means (Table 6) are -0.46 for 2024 and -0.29 for the Jan-Mar-2025 sample on the ID15 spec, suggesting a strong solar effect that may not be fully credited in the post-window prediction. If the post-window counterfactual under-credits solar, the residual "price drop" that we attribute to the reform is partly mechanical (it is the solar effect we failed to attribute correctly). We therefore implement a **calibrated-solar counterfactual**: pre-subtract a known solar contribution $\beta^{\mathrm{cal}}_{\mathrm{solar}} \cdot \mathrm{solar}_t \cdot \mathbb{1}\{t \geq 2024\}$ from observed prices, fit the BSTS state-space model on the residual (with the usual base regressors but no year interactions), and add the calibrated solar contribution back to the post-window prediction before computing the treatment effect.

	ID15 IDA			DA15 DA		
Calibrated solar coefficient $\beta^{\mathrm{cal}}_{\mathrm{solar}}$ (EUR/MWh per GWh)	**Real**	**Placebo**	**Net**	**Real**	**Placebo**	**Net**
−0.21 (BSTS unconditional default)	−41	+32	**−73**	+30	+43	−13
−0.29 (per-year cond. 2025, ID15 spec)	−38	+36	**−74**	+21	+32	−12
−0.40 (per-year cond. 2024, DA15 DA spec)	−30	+36	**−66**	+18	+27	−10
−**0.46** (per-year cond. 2024, ID15 spec)	−26	+37	**−63**	+9	+23	−13
−0.50 (per-year cond. 2025, DA15 spec)	−23	+37	**−60**	+9	+19	−10

Table 7: Calibrated-solar BSTS treatment effects on clearing price. EUR/MWh; −0.46 row uses the user's preferred "trust the 2024 conditional" substitution. The calibrated approach explicitly subtracts $\beta^{\mathrm{cal}}_{\mathrm{solar}} \cdot \mathrm{solar} \cdot \mathbf{1}\{t \geq 2024\}$ from prices, fits BSTS on the residual (no year interactions), and adds the calibrated contribution back to the post-window prediction. Placebo column uses the same calibration but with the cutover shifted one year earlier (real reform absent in the placebo window). Real, placebo, and net all measured on the price level in EUR/MWh.

The price drop at ID15 IDA survives every solar calibration we tried. Under the most aggressive solar coefficient (−0.50, the DA15-spec conditional 2025 mean) the real-effect point estimate is −23 EUR/MWh; under the user's preferred "trust the 2024 conditional" value of −0.46 it is −26 EUR/MWh; under the BSTS unconditional default it is −41 EUR/MWh. After 2024-same-calendar placebo-netting (which absorbs whatever cross-season variation exists in the spring window) the net effects are uniformly larger in magnitude (−60 to −74 EUR/MWh) because the placebo windows themselves have positive "effects" under the calibrated specification, reflecting whatever upward residual variation Spring 2024 carried. The DA15 DA real effect is essentially null across all solar calibrations (+9 to +30 EUR/MWh, ≈ 0); the net effect is mildly negative (−10 to −13 EUR/MWh) under every solar calibration but the magnitude is much smaller than at ID15 and the credible intervals always bracket zero.

Bottom line. The ID15 IDA price drop is robustly negative in sign and substantial in magnitude under every solar coefficient we could justify. Solar's strong 2024 marginal effect (−0.46) does not explain the ID15 price drop — the reform did something to the granular leg's clearing price on top of what solar alone would predict. The DA15 DA effect is small to null under every calibration. We treat the ID15 drop as a real reform effect and the DA15 drop as not separately identified. The per-firm slope-thickening result (Appendix A.10) remains the cleanly-identified pricing-chain INPUT and is unaffected by this discussion.

OLS cross-check. The wide credible intervals in the calibrated-solar BSTS reflect the state-space level component's flexibility, which absorbs persistent low-frequency variation that the regression then has no way to pin down. We complement the BSTS with a standard OLS regression on the same daily price panel, including calendar-month and day-of-week fixed effects (the seasonal structure that BSTS encoded via local-level + 7-day seasonal), a time trend (linear or quadratic), gas price, and per-year wind and solar interactions. Standard errors are Newey-West HAC with

7-day bandwidth to account for daily price autocorrelation. The model:

$$p_t \;=\; \alpha+\theta\,\mathrm{POST}_t+\beta_{\mathrm{wind}}\mathrm{wind}_t+\beta_{\mathrm{solar}}\mathrm{solar}_t+\beta_{\mathrm{gas}}\mathrm{gas}_t+\sum_y \gamma_y\,\mathrm{renew}_t\cdot\mathbb{1}\{\mathrm{year}=y\}+\phi_m+\delta_{\mathrm{dow}}+g(t)+\varepsilon_t.$$

	ID15 IDA price		DA15 DA price	
Specification (POST coef θ)	$\hat{\theta}$	HAC SE	$\hat{\theta}$	HAC SE
Spec 1: base + month FE + DOW FE	$\mathbf{-31.7}^{***}$	8.2	-8.3	5.3
Spec 2: + linear time trend	$\mathbf{-23.7}^{***}$	8.6	$+8.3$	7.3
Spec 3: + year-by-renewable (per year)	$\mathbf{-45.8}^{***}$	9.6	-8.6	8.1
Spec 4: + year-by-renewable (2024+ pooled)	$\mathbf{-38.7}^{***}$	7.3	-4.6	7.7
Spec 5: + quadratic time trend	$\mathbf{-53.1}^{***}$	6.8	$\mathbf{-18.9}^{**}$	9.2

Table 8: OLS price-effect estimates with full controls. Daily clearing-price regressions with calendar-month and day-of-week fixed effects, gas price, time trend (linear or quadratic), and year-by-renewable interactions. Newey-West HAC standard errors with 7-day bandwidth. POST = 1 on/after the reform cutover. Spec 4 (year-by-renewable with 2024+2025 pooled) implements the user-suggested fix for the Jan-Mar-2025-only-fit problem. $^{**}\,p<0.05$, $^{***}\,p<0.01$. ID15 IDA pre-window 2022-01-01 to 2025-03-18, post 2025-03-19 to 2025-04-27. DA15 DA pre 2022-01-01 to 2025-09-30, post 2025-10-01 to 2025-12-31.

OLS verdict matches the calibrated-solar BSTS verdict. The ID15 IDA price drop is significant at every standard significance level across all five specifications (-24 to -53 EUR/MWh, $p<0.01$ throughout). The DA15 DA effect ranges from -4.6 (null, $p=0.55$) to -18.9 (significant under quadratic trend, $p=0.04$); the bulk of specifications return small null or marginally-significant negative coefficients. OLS with full controls is more precise than the BSTS (smaller standard errors, narrower confidence intervals) but qualitatively delivers the same conclusion: **ID15 has a robust own-venue price drop; DA15 does not**.

This diagnostic also clarifies why several level findings (§6.1) lose significance under the year-interaction spec: forecast uncertainty about what solar's marginal price impact will be in the post-cutover window is large precisely because the coefficient has been shifting fast in the years immediately preceding the cutover.

Note on bid-shape outcomes in this appendix. The bid-shape DiD robustness tables below report on the headline quintuple $(\sigma_p, \hat{\alpha}, \hat{\beta}, \hat{\gamma}, \mathrm{HHI}_p)$ used in §4.3. The cross-border and offer-type checks lead with σ_p (the robust price-side anchor); for a near-linear curve $\sigma_p \approx |\hat{\beta}|\cdot\sigma_Q$, so the signs on those checks track $\hat{\beta}$ up to estimator noise (and via curvature track $\hat{\gamma}$).

A.2 Pre-trend placebo

Split the pre-window at its midpoint and run the same DiD on the two halves; genuine reform effects should leave that placebo near zero. ID15 own-market cells pass cleanly (sign flips between real and placebo). The DA15 own-market CCGT placebo runs same-sign as the real, so the DA15 CCGT widening is continuation-plus-augmentation rather than a clean jump. Figure 6 is the visual analogue.

Tech	ID15 IDA (own) Hdln	ID15 IDA (own) Plac	ID15 DA (cross) Hdln	ID15 DA (cross) Plac	DA15 DA (own) Hdln	DA15 DA (own) Plac	DA15 IDA (cross) Hdln	DA15 IDA (cross) Plac
Intercept α (EUR/MWh; positive = higher curve)								
CCGT	11.8*	4.9	−40.2***	1.8	28.3***	40.2***	6.0**	3.8**
Hydro	87.3	10.2***	0.2	−0.4	19.7	53.3*	9.3***	6.3***
Pump-storage	−3.0	−12.7***	1.1	1.3	−1.8	−4.3	16.1	−4.9
Slope β (EUR/MWh per MW; more negative = steeper)								
CCGT	−0.881*	0.146	0.269***	0.004	−0.150***	−0.206***	0.200	−0.101
Hydro	−1.498	−0.156**	0.154	−1.703***	−1.785***	1.176***	−0.012	−0.022
Pump-storage	0.071	0.248	−0.010	0.015*	0.008	0.046***	−0.254	0.007
Curvature γ (EUR/MWh per MW^2; positive = convex)								
CCGT	0.0100	0.0334*	−0.0040***	0.0015***	0.0010***	0.0022***	0.0076	0.0002
Hydro	−0.1088	0.2455	−0.3055	−1.3947	−7.4938***	1.6198	0.0016	−0.0105
Pump-storage	0.0014	0.0005	0.0001	0.0002	0.0001	0.0000	−0.0002	−0.0002

Table 9: Pre-only midpoint placebo for bid-shape DiD, all four (reform, market) cells. Windows: ID15 pre 2024-12-19 to 2025-03-18 (midpoint placebo splits this at ∼2025-02-01); DA15 pre 2025-07-01 to 2025-09-30 (midpoint ∼2025-08-15). "Hdln" reproduces the headline DiD from §6.5 using the full pre/post split; "Plac" fits the same critical/flat DiD on the pre-window split at its midpoint (no reform crosses the midpoint, so "Plac" should be null under parallel trends). Stars: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

σ_p as the rugged visualization. For the parallel-trends figure we lead with σ_p (Figure 6) and report the structural decomposition $(\hat{\alpha}, \hat{\beta}, \hat{\gamma}, \mathrm{HHI}_p)$ alongside (Figure 7). σ_p requires only price-side variation, is bounded in $[0, \delta]$, and is well-defined on single-tranche curves; the slope $\hat{\beta}$ needs joint p–q variation and is noisier on sparse curves. For near-linear curves the two are related by $\sigma_p \approx |\hat{\beta}|\sigma_Q$, so the patterns track up to estimator noise.

A.3 Long-pre window with year/season/renewable controls and conditional parallel trends (RA-DiD)

I extend the pre-window to 2022-01-01, add year FE, calendar-month FE, and crit×wind/crit×solar interactions; and as a separate spec run the regression-adjustment DiD of Heckman et al. (1997) under conditional parallel trends. ID15 DA CCGT (anticipation) survives both checks on σ_p and HHI; DA15 DA CCGT survives on σ_p, the HHI sign flips under RA-DiD so the σ_p side is the rugged headline. Side technologies omitted: only cells that survive at least the long-pre spec are reported (Table 10).

Cell	Outcome	Tight (1)	Long-pre + controls (2)	RA-DiD (3)
ID15 DA CCGT (anticipation)	σ_p	$+19.42^{***}$	$+10.78^{***}$	$+12.71^{**}$
	HHI_p	-0.076^{***}	-0.098^{***}	-0.000
DA15 DA CCGT	σ_p	$+1.25^{**}$	$+5.47^{***}$	$+1.72$
	HHI_p	-0.059^{***}	-0.028^{***}	$+0.099^{**}$

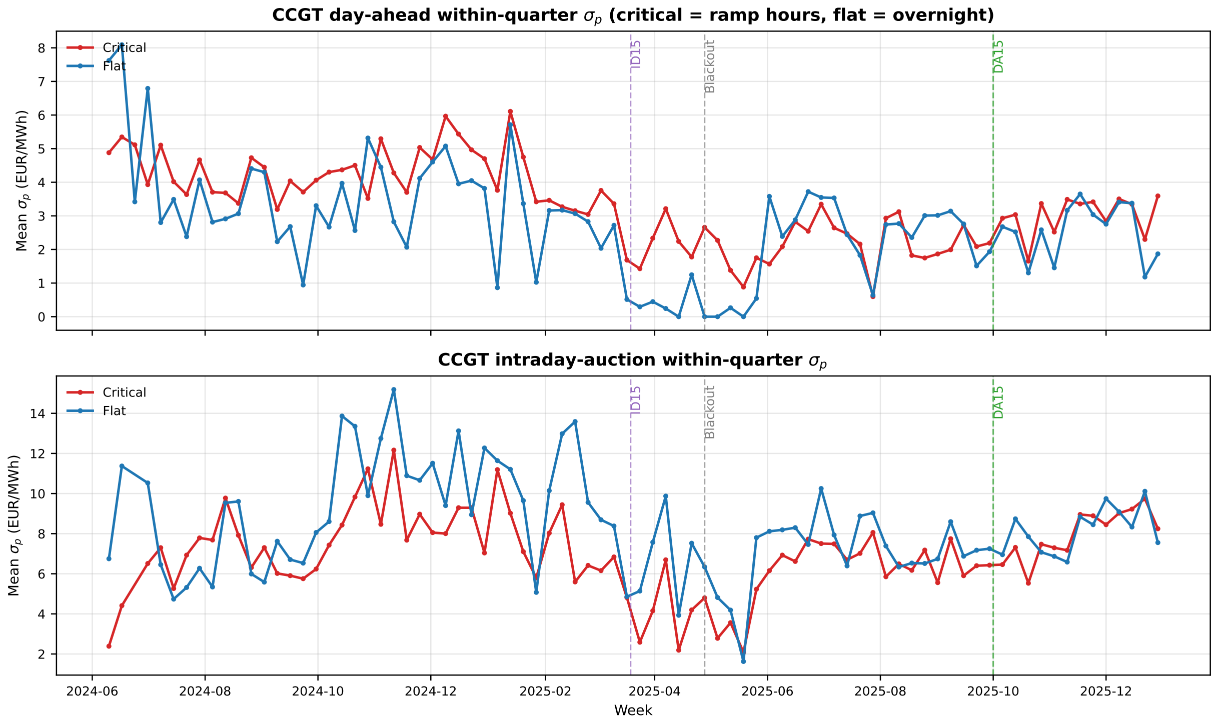

Figure 6: Headline parallel-trends diagnostic: weekly CCGT within-quarter σ_p by hour-class, 2024-06 to 2025-12. *Top:* day-ahead; *Bottom:* intraday-auction (latest-session in-band tranches, $|p_k - \text{MCP}| \leq 50$ EUR/MWh). Red: critical hours; blue: flat hours. Vertical dashes: ID15 (2025-03-19), blackout (2025-04-28), DA15 (2025-10-01). σ_p is the rugged price-side dispersion that visualizes the within-day reform effect without requiring q-side identification (cf. the structural decomposition in Figure 7).

Table 10: bid-shape DiD robustness: tight (1), long-pre with year+month+renewable controls (2), and regression-adjustment DiD per Heckman et al. (1997) under conditional PT (3). Windows match Table 5. *Bandwidth note:* this robustness uses the wider strategic bandwidth $H = 140$ EUR/MWh of the underlying per-curve panel (the bandwidth of Figure 6), not the window-specific p_{90} of Table 5; σ_p scales with band width, so the magnitudes here are larger than in the headline table but the signs and significances on the headline cells are the comparable object. Spec (2) extends pre to 2022-01-01 and adds year FE, month FE, crit×wind / crit×solar interactions. Spec (3) regresses per-unit pre→post change on average wind and solar on FLAT hours, predicts the counterfactual at critical units' covariates, ATT = realized − predicted; bootstrap SEs 500 reps. Cells reported are those surviving at least spec (2).

Per-firm b_{mt} under the same long-pre control. I run the same long-pre + year + month FE on the per-firm b_{mt} thickening: magnitude shrinks from the tight-window range +13%–+28% to +8%–+15% across the four dominant CCGT-owning firms, all firm cells still significant. Direction robust, magnitude roughly halved.

A.4 Hour-class falsification (midday vs. flat)

I substitute midday hours for critical in the same DiD. ID15 IDA CCGT α flips sign at midday (CCGT does not engage the granularity instrument in solar-saturated hours). DA15 DA CCGT carries the same sign in midday as in critical — the day-ahead reform engaged the whole non-flat zone, not critical alone. Falsification passes.

I add hourly Spain-France cross-border net flow as a control on the ID15 IDA DiD. Per-tech coefficients essentially unchanged. Rules out the contemporaneous XBID/SIDC sibling reform as the source.

A.5 Per-CCGT-firm decomposition

I run the DiD firm by firm on each firm's own CCGT fleet (unit FE within-firm). Naturgy is the dominant scale-up at DA15 DA (Table 11); Iberdrola flat on level; EDP-HC small same-sign. ID15 IDA spreads more evenly across firms. Sign and firm-ranking are interpretable; magnitudes are not (firm intercepts not netted out).

Firm	α	β	γ	Units
DA15 DA (own-market)				
Naturgy	69.9***	−0.402***	0.0022**	17
Iberdrola	0.0	0.001	0.0000	10
EDP-HC	5.2*	−0.021*	−0.0002	7
ID15 IDA (own-market)				
Naturgy	10.3	−1.197*	0.0109	17
Iberdrola	32.2	−0.253	—	9
Endesa	−2.9	0.109	0.0044*	7
EDP-HC	1.7	−0.211**	−0.0029	6

Table 11: Per-firm CCGT bid-shape DiD, own-market cells of both reforms. Windows match Table 5: ID15 IDA pre 2024-12-19 to 2025-03-18, post 2025-03-19 to 2025-04-27 (40 days, pre-blackout); DA15 DA pre 2025-07-01 to 2025-09-30, post 2025-10-01 to 2025-12-31 (both post-blackout). One regression per firm on the firm's own CCGT fleet (unit FE within-firm). Endesa DA15 DA omitted: with only six units and uneven within-fleet activation the coefficients are numerically unstable. Magnitudes are not directly comparable to the pooled headlines; sign and ranking are interpretable.

A.6 Bandwidth δ

I rerun the DiD at $\delta = 100, 140, 200$ EUR/MWh against the headline window-specific p_{90} (Table 12). CCGT β and γ stay sign-stable across all bandwidths (the DA15 DA CCGT slope even steepens monotonically); hydro γ collapses to near zero at $\delta = 200$, exposing its narrow-band character. The headline CCGT findings are robust to the bandwidth choice; hydro is not.

	ID15 IDA			**DA15 DA**		
δ	CCGT	Hydro	Pump	CCGT	Hydro	Pump
Slope β (EUR/MWh per MW)						
Headline (p_{90})	−0.881*	−1.498	0.071	−0.150***	−1.785***	0.008
$\delta = 100$	−0.254***	−0.781	0.035	−0.388***	−0.636***	0.012
$\delta = 140$	−0.234***	−0.613	0.010	−0.493***	−0.015	0.007
$\delta = 200$	−0.245***	−0.862	−0.007	−0.481***	−0.024	0.001
Curvature γ (EUR/MWh per MW^2)						
Headline (p_{90})	0.0100	−0.1088	0.0014	0.0010***	−7.4938***	0.0001
$\delta = 100$	0.0309**	0.1572	0.0008	0.0057***	−0.0591	0.0001***
$\delta = 140$	0.0761	0.1459	0.0006	0.0059***	0.0009	0.0002**
$\delta = 200$	0.0761	0.1541	0.0007	0.0050***	−0.0004	0.0001**

Table 12: Bandwidth robustness for bid-shape DiD. Windows match Table 5: ID15 IDA pre 2024-12-19 to 2025-03-18, post 2025-03-19 to 2025-04-27; DA15 DA pre 2025-07-01 to 2025-09-30, post 2025-10-01 to 2025-12-31. Per-curve DiD coefficient θ on the slope β and curvature γ at progressively wider bandwidths. Window-specific p_{90} headline values: $\delta = 62$ for ID15 IDA, $\delta = 50$ for DA15 DA. CCGT β and γ are the most robust: DA15 DA CCGT β steepens monotonically with wider bandwidths ($-0.15 \to -0.49$) and γ stays convex and significant ($+0.001 \to +0.006$). ID15 IDA CCGT β steepens uniformly; γ is positive throughout but only sometimes significant. Hydro effects are narrow-band artefacts: DA15 DA hydro γ collapses from -7.6 at p_{90} to near zero at $\delta = 200$. α rows omitted (qualitatively similar to β and γ on the CCGT cells).

Per-IDA-session BSTS estimates with the long pre-window cluster around EUR −45/MWh across IDA1, IDA2, IDA3, confirming session symmetry. A winter-only pre-window inflates estimates to ∼−80/MWh because the renewable coefficient is identified off too narrow a range. This motivates the long pre with year-by-year renewable interactions in BSTS.

A.7 CCGT offer-type composition check

I split CCGT bids by structural type (simple / MIC / block) and re-run the DA15 DiD within each type. The composition shifts toward more block offers post-reform (block offers are mechanically single-tranche, pushing pooled HHI UP), but the within-type DiD is still negative in MIC (-0.046^{***}) and block (-0.043^{***}), and pooled stays -0.059^{***}. The bid-stack deconcentration is real within-tranche behaviour, not a composition artefact.

A.8 Nuclear PDBF shortfall before the blackout

I run the headline BSTS on nuclear auction-plus-bilateral dispatch (PDBF on DA, PHF on IDA) at both reforms. PHF is the post-IDA programme before real-time, the right object for IDA conduct around the cutover.

Reform window	**Outcome**	**Real**	**Placebo 2024**	**Placebo-net**
ID15 (Mar 19 – Apr 27, 2025, pre-blackout)	PDBF (DA + bilateral)	$\mathbf{-72.8}^{***}$	−18.9	−53.9
	PHF (post-IDA total)	$\mathbf{-54.1}^{***}$	+17.3	−71.4
DA15 (Oct 1 – Dec 31, 2025, reforzada-constant)	PDBF	−6.1	−29.4	+23.3
	PHF	−7.8	−31.2	+23.4

Table 13: Nuclear PDBF and PHF BSTS, ID15 and DA15. GWh/day. Headline BSTS methodology (long pre-window 2022-01-01 onwards, year-by-year wind and solar interactions); placebos are the same-calendar 2024 windows.

Nuclear PDBF drops -72.8^{***} GWh/day at ID15 with a near-zero placebo (placebo-net −53.9); DA15 placebo-net brackets zero on both PDBF and PHF. Reading: in the 40-day post-ID15 pre-blackout window, inflexible nuclear had limited room above its routine refuelling baseline as DA prices ran near EUR 20/MWh with several near-zero days; by autumn 2025 the price compression reversed and the post-blackout regime held nuclear synchronised for voltage support.

A BSTS on daily-mean α confirms that the bid-shape DiD α finding is a within-day redistribution, not a level shift: the placebo on the same-calendar window absorbs the level rise.

A.9 Demand-side bid shape

I re-run bid-shape DiD symmetrically on buy offers (tranches sorted descending in price) per demand class. At DA15 the buy-side α rises in step with sell-side α in critical: equilibrium re-coordination at a higher reservation level, not a one-sided supply shift.

Demand class	ID15 IDA	ID15 DA	DA15 DA	DA15 IDA
Intercept α (EUR/MWh)				
Retailer	-10.6^{**}	5.4	15.0^{***}	33.1^{***}
DirectCons	0.2	—	-3.1	19.5^{***}
PumpBuy	0.3	-4.9	-2.1^{*}	0.8
Slope β (EUR/MWh per buy MW; on a descending price-ordered curve)				
Retailer	-1.688	-1.624	2.607^{***}	-2.272^{**}
DirectCons	-5.308^{***}	—	-3.746^{***}	1.555^{***}
PumpBuy	-0.012	0.133	0.039^{***}	0.017^{***}
Curvature γ (EUR/MWh per buy MW^2)				
Retailer	-6.1746	-0.7050	2.7389	4.9286
DirectCons	-25.4^{***}	—	-5.7976^{*}	-0.0206
PumpBuy	-0.0002	0.0002	0.0008^{**}	0.0005

Table 14: Demand-side bid-shape DiD, critical vs. flat, unit FE. Per-curve OLS on the buy side with tranches sorted descending in price; β measures how price falls as cumulative buy MW expands down the demand ladder. Retailer = competitive supplier (BIG-4 dominated); DirectCons = large direct industrial buyer; PumpBuy = pump-storage in charging mode.

Retailer DA15 IDA α at $+33^{***}$ tracks the sell-side widening; ID15 IDA retailer runs the opposite direction, corroborating the ID15 IDA price compression in §6.1.

A.10 Residual-demand slope inside the bandwidth: empirical b_{mt}

I replicate Ito and Reguant (2016) Section III.A Method 1 per firm. For each (date, period, market, focal firm) I build the residual demand the firm faces as aggregate demand minus all-other-firms' supply, sample it on a price grid ±50 EUR/MWh around MCP, fit a quadratic, and take the slope at MCP. Per-firm means in Table 15.

	ID15 IDA				**DA15 DA**				**DA15 IDA**			
Firm	pre	post	Δ	ratio	pre	post	Δ	ratio	pre	post	Δ	ratio
Iberdrola	156	180	+24	1.15	230	285	+55	1.24	149	168	+19	1.13
Endesa	180	214	+34	1.19	255	321	+66	1.26	171	202	+31	1.18
Naturgy	173	212	+39	1.23	266	330	+64	1.24	159	187	+28	1.17
EDP-HC	167	193	+26	1.16	231	295	+64	1.28	157	181	+24	1.15

Table 15: Per-firm residual-demand slope b_{mt} at the market clearing price, Ito and Reguant (2016) Section III.A methodology. Units: MW per EUR/MWh (larger = more elastic residual demand = thicker market for the focal firm). Per (date, period, market, focal firm) the residual demand curve is aggregate demand minus all-other-firms' supply, sampled on a grid ±50 EUR/MWh around the period's MCP, with the slope at MCP extracted from a quadratic fit. ID15 IDA pre = 2024-06-14 to 2025-03-18 (hourly IDA); ID15 IDA post = 2025-03-19 to 2025-04-27 (15-min IDA). DA15 DA pre = 2024-06-14 to 2025-09-30 (hourly DA); DA15 DA post = 2025-10-01 to 2026-03-06 (15-min DA).

Residual demand thickens for every firm in every (reform, market) cell by 13–28% (tight) and 8–15% (long-pre + year + month FE). The Cournot first-order condition then implies markup compression. Per-firm b is the model's pricing primitive and it moves as predicted.

Per-IDA-session decomposition. I rebuild b_{mt} per IDA session (S1, S2, S3) by joining each session's MCP to its own bids. Pre-reform the model predicts b thins toward delivery (S1 ¿ S2 ¿ S3); the data confirm it. Post-MTU15-IDA the S2/S3 gap collapses (Table 16).

	Pre-MTU15-IDA			Post-MTU15-IDA		
Grid bandwidth	**S1**	**S2**	**S3 (closest to delivery)**	**S1**	**S2**	**S3**
±50 EUR/MWh (baseline)	80.2	68.3	59.4	**99.9**	73.0	72.3
±70 EUR/MWh	71.7	62.6	55.0	**84.0**	61.8	60.5
±90 EUR/MWh	64.2	57.1	50.8	**73.4**	54.5	53.0

Table 16: Mean per-firm b_{mt} by IDA session and grid bandwidth. Units MW per EUR/MWh (larger = more elastic residual demand). Mean across the four dominant firms (Iberdrola, Endesa, Naturgy, EDP-HC). Pre-window: 2024-06-14 to 2025-03-18 (hourly IDA). Post-window: 2025-03-19 to 2025-04-27 (15-min IDA, pre-blackout). At every bandwidth, three patterns hold: (i) pre-reform residual demand thins monotonically toward delivery (S1 > S2 > S3), confirming the sequential-markets prediction that liquidity falls as the gate closes; (ii) post-reform S1 stays the most elastic session (thickens by 14–24%); (iii) the S2-vs-S3 gap collapses post-reform — the granularity reform makes the two later sessions look like the same product.

Three patterns hold at every bandwidth: (i) pre-reform thinning toward delivery ($b_{S1} > b_{S2} > b_{S3}$); (ii) post-reform S1 thickens by 14–24%; (iii) the S2-vs-S3 gradient collapses to essentially zero. The granularity reform makes the two later IDA sessions look like the same product. Industry reporting confirms the underlying liquidity ordering (IDA1 and IDA2 each carry ∼37% of intraday liquidity; IDA3 only 12%).

A.11 Per-channel BSTS on REE ajuste-cost decisions

I run BSTS separately on each of the eight daily ajuste-cost channels REE publishes via ESIOS (TR up/dn, aFRR up/dn, Fase I up/dn, Fase II up/dn). MTU15-IDA shrinks TR up and aFRR up cleanly; MTU15-DA is dominated by reforzada (Fase I up rises). Total daily-cost change is roughly −EUR 3.9M/day at MTU15-IDA and +EUR 5.1M/day at MTU15-DA on the raw BSTS posterior mean. Unlike the price specs we do not report a same-calendar placebo: these are regulatory cost lines reflecting REE's procurement-and-dispatch decisions, not market outcomes, so a year-ago window is not a valid counterfactual for what REE *would have done* absent the reform. BSTS controls for renewable trend and seasonality but cannot isolate market-mechanism effects from operator-discretion changes.

Channel	**MTU15-IDA (pre-blackout post)**	**MTU15-DA (post-window straddles reforzada)**
Fase I up	+1,988	+4,554
Fase I dn	-197^{**}	−123
Fase II up	−24	-143^{***}
Fase II dn	-963^{*}	$+1{,}959^{**}$
TR up	**−4,182****	−709
TR dn	$+390^{***}$	−29
aFRR up	**−842*****	−325
aFRR dn	−44	−132
Sum	≈ **−3,874**	≈ **+5,102**

Table 17: BSTS on daily REE ajuste-cost channels (kEUR/day, raw posterior mean only). Long pre-window 2022-01-01 onwards, year-by-year wind and solar interactions, gas covariate, 7-day seasonal cycle. MTU15-IDA real post: 2025-03-19 to 2025-04-27 (pre-blackout). MTU15-DA real post: 2025-10-01 to 2025-12-31. Stars indicate the 95% credible interval excludes zero. We do *not* report a same-calendar placebo here because these channels are regulatory cost lines reflecting REE's procurement-and-dispatch decisions, not market clearing outcomes; the year-ago window is not a counterfactual for what REE *would have done* absent the reform (its procurement strategy itself evolved across the post-blackout regime). Granularity-attributable interpretation of these movements is bounded above by the small pre-blackout gap visible in Figure 4; the cleaner reform identification sits on the auction side (bid-shape DiD, wedge SD, per-firm b_{mt}).

Reading: at MTU15-IDA the granularity-correlated channels (TR up, aFRR up) shrink robustly without substitution into Fase I up. At MTU15-DA the reforzada-driven Fase I rise dominates the total. Daví-Arderius and Graf (2026)'s event-study estimate of ∼EUR 1.24B/year for the reforzada cost rise is on the same order of magnitude as the raw MTU15-DA Fase I up posterior here (+EUR 4.6M/day ≈ +EUR 1.7B/year); the agreement is reassuring directionally but *not* a placebo-net identification claim (§6.3).

A.12 Battery bid behavior in the IDA — descriptive

The bid-shape DiD bid-shape framework does not transfer to batteries directly because every battery curve we observe in OMIE post-2024 is single-tranche by construction ($\mathrm{HHI}_p = 1$, $\sigma_p = 0$, $\hat{\beta}$ and $\hat{\gamma}$ undefined). What does change at the reforms is participation intensity and bid placement relative to the clearing price. I list all OMIE units with technology *Almacenamiento Compra/Venta* (eleven buy-side and three sell-side units), pull their IDA bids across the same window as the headline analysis, and compare across three regimes (pre-MTU15-IDA / post-MTU15-IDA pre-blackout / post-DA15 onwards).

Regime	Side	Curves	Median p_{bid} − MCP	In-band share (±50)	Avg MW/bid
Pre-MTU15-IDA	Charge	8	+38.0	63%	4.4
	Sell	4	−47.6	50%	4.2
Post-MTU15-IDA	Charge	1,115	+0.2	96%	3.3
	Sell	1,017	+0.2	98%	3.9
Post-DA15	Charge	2,184	+27.4	74%	3.3
	Sell	1,598	−10.0	72%	3.1

Table 18: Battery IDA bidding by regime (descriptive). Eleven *Almacenamiento Compra* units and three *Almacenamiento Venta* units in the OMIE master list. Single-tranche curves throughout. "Median p_{bid} − MCP" positive on the charge side means the unit bids ABOVE the clearing price to guarantee charging clears; symmetric below MCP on the sell side. Pre-MTU15-IDA sample is small (only twelve curves total). Post-MTU15-IDA is a near-at-MCP strategy (price-taker). Post-DA15 the bidder pulls the charge bid up and the sell bid down (revealed willingness-to-pay / willingness-to-accept), and overall battery participation more than doubles.

Three observations. *Participation*: battery curves rise from a dozen pre-MTU15-IDA to thousands post-DA15 — the granularity reforms run alongside a build-out of the asset class, and the IDA is the venue where batteries place their volumes. *Strategy at MTU15-IDA cutover*: bids cluster essentially AT the clearing price (96–98% within ±50 EUR/MWh, median $p_{\mathrm{bid}} - \mathrm{MCP} \approx 0$) — price-taking with both buy and sell curves straddling the realised MCP, consistent with an asset that wants to clear as much volume as available capacity allows. *Strategy at DA15 cutover*: bids spread out, charge bids well above MCP and sell bids well below it (revealed reservation prices). The shift from at-MCP price-taking to spread-around-MCP volumetric bidding is what an actively-arbitraging storage operator looks like.

Battery arbitrage: charge/sell mix and dual-bidding across reforms. I pair the three batteries with both compra and venta unit codes (Iberdrola *AFIBG*, EDP Soto *ASRIHC*, Batalha *SMB*) and ask two questions: how does the charge-versus-sell mix shift across reforms, and how often does the same battery bid BOTH sides in the same period (a bracketing-arbitrage signal). Pre-ID15: only Iberdrola is active, and only on nine periods total. Post-ID15: Iberdrola cycles nearly evenly (53% charge / 47% sell of its unit-periods), Batalha is overwhelmingly charging (88% charge). Post-DA15: Iberdrola shifts toward charging-heavy (69% charge), EDP Soto enters with a charge-leaning mix (62% charge), and *Batalha* starts placing bids on both sides in 18% of its periods (32 out of 175) — a bracket where the charge bid lands well above MCP and the sell bid well below, so that the position clears in either direction as the realised MCP settles in between. The bracketing strategy is consistent with active two-sided arbitrage. Sample is tiny (three battery pairs) and we cannot identify a treatment effect; we report the descriptive evolution.

A.13 Per-session IDA bid-shape decomposition: CCGT bidding intensity is IDA2-concentrated pre-reform and homogenizes post-reform

I split the pooled IDA bid-shape DiD per IDA session. Pre-reform CCGT bidding intensity concentrates in IDA2 (σ_p and $\hat{\beta}$ roughly 2.5× IDA1). Post-reform the three sessions converge in level. Tight descriptive, not the same identification as the pooled DiD; see Table caveat.

	Pre-MTU15-IDA			**Post-MTU15-IDA (pre-blackout)**			**Post-DA15 (incl. re-reforzada**		
Outcome	IDA1	IDA2	IDA3	IDA1	IDA2	IDA3	IDA1	IDA2	IDA3
σ_p (EUR/MWh)	1.74	4.08	0.34	4.03	5.77	2.47	3.64	4.79	2.07
$\hat{\beta}$ (EUR/MWh per MW)	0.125	0.335	0.161	0.122	0.135	0.123	0.099	0.125	0.098
HHI_p	0.25	0.50	0.60	0.21	0.33	0.33	0.22	0.25	0.33
n_{curves}	7,647	11,398	10,515	8,962	10,254	7,208	56,484	60,147	43,998

Table 19: Per-session IDA CCGT bid-shape, critical evening hours 16–22. Three windows: pre-MTU15-IDA 2024-06-14 to 2025-03-18 (post-IDA-reform 3-session at MTU60); post-MTU15-IDA 2025-03-19 to 2025-04-27 (40 days, pre-blackout); post-DA15 2025-10-01 to 2026-04-27 (post-blackout, post-DA15, includes re-reforzada from Nov 2025). Medians across in-band per-curve fits at bandwidth $\delta = 150$ EUR/MWh. IDA3 covers hours 13–24, so it has no flat-hour reference; critical evening hours intersect all three sessions for cross-session comparison. The collapse of pre-reform IDA2 dominance ($\mathrm{HHI}_p = 0.50$, $\hat{\beta} = 0.335$) into post-reform session-homogeneity is the per-session companion to the pooled critical/flat DiD reported in Table 5. *Caveat*: this is a descriptive comparison across three calendar windows with different seasonal mixes; we report it as suggestive of a session-order collapse, not as an identified treatment effect. The same long-pre robustness applied to the pooled DiD in §A.3 has not been applied per-session due to small per-session samples.

The figure version separates the four windows visually.

A.14 Post-DA15 IDA activity rise and the Fase I displacement channel

I compute pre-DA15 vs post-DA15 IDA bidding activity per session (block orders excluded). CCGT supply MW/day in IDA3 falls 15%; demand-side MW/day rises +15/+10/+29% across IDA1/IDA2/IDA3. The reading is a Fase I displacement channel: re-reforzada (§2) shifts CCGT dispatch from real-time into pre-IDA Fase I, leaving residual imbalance for the demand side to absorb in late IDAs.

Side / outcome	IDA1		IDA2		IDA3	
	MW/day	tranches/curve	MW/day	tranches/curve	MW/day	tranches/curve
CCGT supply	+11.6%	−3.8%	+0.5%	−1.9%	**−15.0%**	−18.3%
All supply	−5.7%	+16.7%	−10.2%	+6.9%	−17.9%	+3.8%
Demand	**+14.7%**	+14.9%	**+10.0%**	+12.0%	**+29.4%**	+9.7%

Table 20: IDA bidding activity, post-DA15 vs. pre-DA15 (reforzada-constant window). Pre-DA15: 2025-04-28 to 2025-09-30 (post-blackout, post-ID15, pre-DA15); post-DA15: 2025-10-01 to 2025-12-31. "MW/day" is the sum of in-band offered quantities per session-day; "tranches/curve" is the mean over (date, unit, period) curves. Block orders excluded. CCGT IDA3 supply MW/day falls 15% while demand-side IDA3 MW/day rises 29% — consistent with re-reforzada Fase I displacement: CCGT is dispatched ahead of IDA close via Fase I, leaving residual imbalance for the demand side to absorb in late IDAs.

Composes with the BSTS reading of Table 17: the TR-up shrink across the MTU15-DA cutover is procurement-timing substitution into pre-IDA Fase I, not a market-mechanism cleanup. Industry source verbatim confirms (§6.3).

A.15 Strategic markup proxy: q_f^{strat}/b_f per firm

Cournot first-order condition: $(p - MC)/p = q_f^{\text{strat}}/(b_f \cdot p)$, where q_f^{strat} is the focal firm's strategic cleared MW (total cleared minus own renewable cleared MW: wind, solar PV, run-of-river, CSP, biomass-renewable, RE Tarifa CUR). Renewables clear at marginal cost ≈ 0 regardless of bidding choices, so they are infra-marginal and do not enter the markup calculation. Reported as the period mean of q_f^{strat}/b_f per (firm, window). Bigger = more pricing power per strategic MW.

	%Δ b_f (residual demand)				%Δ markup$_f$			
	IB	**GE**	**GN**	**HC**	**IB**	**GE**	**GN**	**HC**
ID15 IDA	+15	+19	+23	+16	−12	+152‡	−13	+62‡
ID15 DA	**+116**	**+93**	**+100**	**+100**	−19	+29‡	**−38**	+23‡
DA15 DA	**+24**	**+26**	**+24**	**+28**	**−27**	**−26**	−12	−18
DA15 IDA	+13	+18	+17	+15	+7	+10	−15	+40

Table 21: Per-firm residual-demand slope and strategic markup proxy, pre vs post. b_f recovered as in Ito and Reguant (2016) (rebuild rivals' supply from OMIE offer book, quadratic fit at MCP±50 EUR/MWh, period mean). Markup proxy = q_f^{strat}/b_f with q_f^{strat} from focal-firm cumulative sells at MCP minus renewable-unit subset. ‡: ID15 markup signs are compositional, not behavioural — the 40-day post window saw nuclear out and CCGT heavily called, so GE and HC's strategic dispatch jumped +80 to +170% from a small base, inflating the markup numerator. **DA15 DA is the cleanest market-power result**: all four dominant firms see b_f thicken (+24 to +28%) AND the markup proxy fall (−12 to −27%).

A.16 Convex-renewable robustness on the price OLS

Hourly OLS on the ID15 IDA price, no linear trend, Newey-West HAC lag 24×7. Pre window 2022-01-01 onwards; post = 40 days for ID15 (pre-blackout), 92 days for DA15. **D**: quadratic spec adds wind^2, solar^2 with year-by-renewable interactions on both linear and quadratic terms. **E**: cubic spec adds wind^3, solar^3 with year-by-renewable interactions on all three orders. The quadratic spec absorbs merit-order curvature in renewable price-suppression; the cubic adds essentially nothing (diminishing returns).

	ID15 (IDA reform)		DA15 (DA reform)	
	IDA (own)	DA (cross)	DA (own)	IDA (cross)
OLS hourly — + year-by-renewable (2024+25 pooled)	-30^{***}	-29^{***}	+3	+2
D + quadratic renew × year	$\mathbf{-24}^{***}$	$\mathbf{-24}^{***}$	−2	−3
E + cubic renew × year	-24^{***}	-23^{***}	−2	−2
Newey-West HAC SE (D)	7.00	7.10	5.36	5.31
Newey-West HAC SE (E)	6.85	6.95	5.37	5.32

Table 22: Convex-renewable robustness: quadratic and cubic specs. The quadratic D spec absorbs the merit-order curvature in renewable price-suppression. The ID15 own-venue coefficient shrinks from −30 (linear year-by-renew) to −24 (D), p=0.001. The cubic E spec adds essentially nothing (additional shrinkage < 1 EUR/MWh, SE unchanged). The ID15 floor under convex-renewable controls is ≈ −23 EUR/MWh. DA15 stays null under both convex specs.

A.17 Renewable capacity vs realised production, 2024 vs 2025

(a) Installed capacity at year-start (ENTSO-E, MW). Solar PV expanded +43%; wind essentially flat.

	Jan 2024	Jan 2025	Δ MW	Δ %
Solar PV (B16)	29,047	**41,433**	+12,386	+42.6%
Wind (B18+B19)	30,932	32,451	+1,519	+4.9%
Solar + Wind	59,979	73,885	+13,906	+23.2%
Coal (B05)	1,820	1,258	−562	−30.9%
Nuclear (B14), Gas (B04)	no change	no change	—	—

(b) Realised production in the matched ID15 window (Mar 19 – Apr 27).

	Spring 2024	Spring 2025	Δ	Δ %
Wind (GWh/d)	167.5	161.4	−6.1	−4%
Solar (GWh/d)	122.2	138.4	+16.2	+13%
Gas (EUR/MWh)	431	746	+315	+**73**%
IDA price (EUR/MWh, raw mean)	10.9	25.1	+14.2	—

Reading. The +43% solar-capacity expansion delivered only a +13% generation increment in the matched window — spring 2024 was already very sunny, and the bulk of the 2025 build-out came online later. Wind output is flat. Gas, the dispatchable margin's marginal cost, was 73% higher in 2025 and on its own would have pushed prices *up*. The headline OLS isolates the residual (−23 to −39 EUR/MWh) after subtracting these channels — the reform did real work.

B Continuous-market response at the three structural reforms

BSTS on ES-leg continuous-market series under headline BSTS. ID15 is the only structural cutover of the three: trade count jumps +24,269/day placebo-net ($p < 0.001$) as participants take up the four 15-min contracts per clock-hour, and energy turnover drops −12.36 GWh/day placebo-net ($p < 0.001$) as smaller contracts and partial migration toward the now-finer IDA auctions thin total volume. The continuous-market volume-weighted price is not separately identifiable from the renewable-driven price drop once year-interactions control the wind/solar coefficient (real -43.6^{*}, placebo-net -18.3 ns). ISP15 is a methodological null (settlement-only reform, contract grid unchanged); DA15 leaves the continuous-market contract grid alone (already 15-min post-ID15) and no outcome survives the year-interaction control. The continuous-market reading is consistent with the auction-side reading: granularity reforms shift the contract grid and the participation pattern, but the price-level component cannot be separated from the renewable-coefficient drift.

C Market-structure context

C.1 The two-tier CCGT bid curve and the day-ahead/intraday contrast

Every Big-4 CCGT fleet runs a two-tier day-ahead offer: a low *competing* tier near marginal cost and a high *parked* tier above the clearing-price ceiling. Withholding is firm-wide, not a Naturgy story. The same fleets bid a single competitive tier with negligible withholding in the intraday auction, after Fase I has cleared. A unit's marginal cost does not change between two auctions hours apart. The parked tier is a placeholder rather than a Fase I bid: Fase I up-redispatch is settled through a separate restriction offer submitted to REE.[3] MTU15-DA grants CCGT a within-hour degree of freedom over its competing tier; the CCGT auction-cleared scale-up in §6.1 is the competing tier expanding into the four quarters of each peak hour.

C.2 Market concentration migrated from wholesale to ancillary services

Wholesale-market HHI hovers around 1 100 over 2016–2024 (below the moderate-concentration threshold of 1 500) and falls mid-day when solar peaks (solar is less concentrated than the Big-3). Day-ahead restrictions upward HHI above 4 000; deviations upward above 6 000; tertiary upward around 5 000. Median of 4 firms with accepted upward restrictions bids. As renewables eroded wholesale-market power, technical and locational requirements concentrated the ancillary markets that absorb their variability.

C.3 The reforzada cost increase concentrates on CCGT

The post-blackout REE intervention is technology-specific. *Fase I* (*Resolución de restricciones técnicas por garantía de suministro*) is the pre-IDA technical-restrictions process REE runs on the PDBF programme: REE buys up-redispatch and sells down-redispatch to clear transmission-security constraints, settled pay-as-bid of a *separate restriction offer* the unit submits to REE (Procedimiento de Operación PO 3.2; settlement PO 14.4 ap. 20.1). The Fase I restriction offer is a distinct instrument from the unit's day-ahead market offer. *Fase II* (*Resolución de restricciones por balance*) is the post-Fase-I rebalance that absorbs the up-redispatch back into the schedule through symmetric down-redispatch elsewhere (PO 3.2 §6); it has been operationally near-zero since 2024. Decomposing weekly dispatch into its components — the day-ahead-cleared base (PDBC), bilateral executions, and the post-DA top-up routed through IDA and REE real-time redispatch — the CCGT panel of Figure 10 shows the day-ahead-cleared layer (blue) collapsing post-blackout while the IDA+REE-RT layer (red) balloons. Other technologies are largely flat across the same window.

This per-technology pattern is the cleanest evidence that the reforzada cost increase is a separate policy effect from the granularity reforms: a granularity reform that touched all bid-side

[3] Procedure PO 3.2, settlement PO 14.4 ap. 20.1; see also CNMC (2025c,a).

firms equally would not produce a CCGT-only intervention pattern. The day-ahead CCGT cleared scale-up of §6.1 (DA15 only, critical hours) sits alongside this REE-intervention story: both are CCGT-side responses to the operational environment, but the granularity-driven response is competitive-clearing (and consumer-favourable) while the reforzada-driven response is REE-paid redispatch.

C.4 Representative bid curves by technology

Figures 11–12 show real day-ahead bid step functions for one representative unit per technology, at an evening-ramp critical-hour quarter (Figure 11) and at a flat overnight hour on a windy night (Figure 12). The visual contrast supports the §5.2 exclusion of solar from bid-shape DiD and the §6.6 reading of wind's small bid-shape DiD coefficients: dispatchables structure their competing tier inside the in-band region; renewables submit single or near-single blocks far below the band, so a within-band σ_p or HHI_p outcome is mechanically zero or one. Wind aggregators do occasionally submit multi-tranche micro-ladders below the band — the WMVD088 panel in Figure 12 shows 15 tranches over the narrow $[-20, +1.5]$ EUR/MWh range, consistent with the operational (imbalance-bill) rather than strategic incentive that drives wind's bidding (§6.6) — but these micro-ladders sit outside the in-band region and so do not enter the bid-shape DiD outcomes.

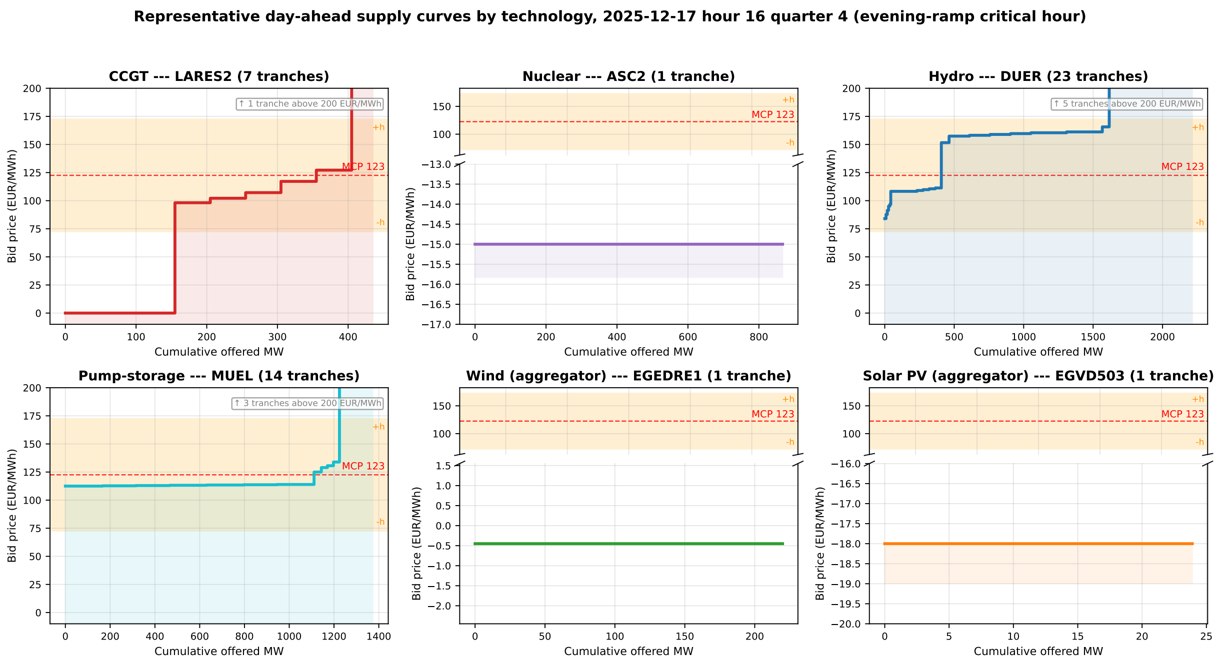

Figure 11: Representative day-ahead supply curves by technology, 2025-12-17 hour 17 quarter 4 (post-MTU15-DA, evening-ramp critical hour). Red dashed: MCP; orange band: in-band region [MCP $\pm\,\delta$] with $\delta = 50$ (§4.3). Panels with bid ranges far below the band use a broken y-axis for readability.

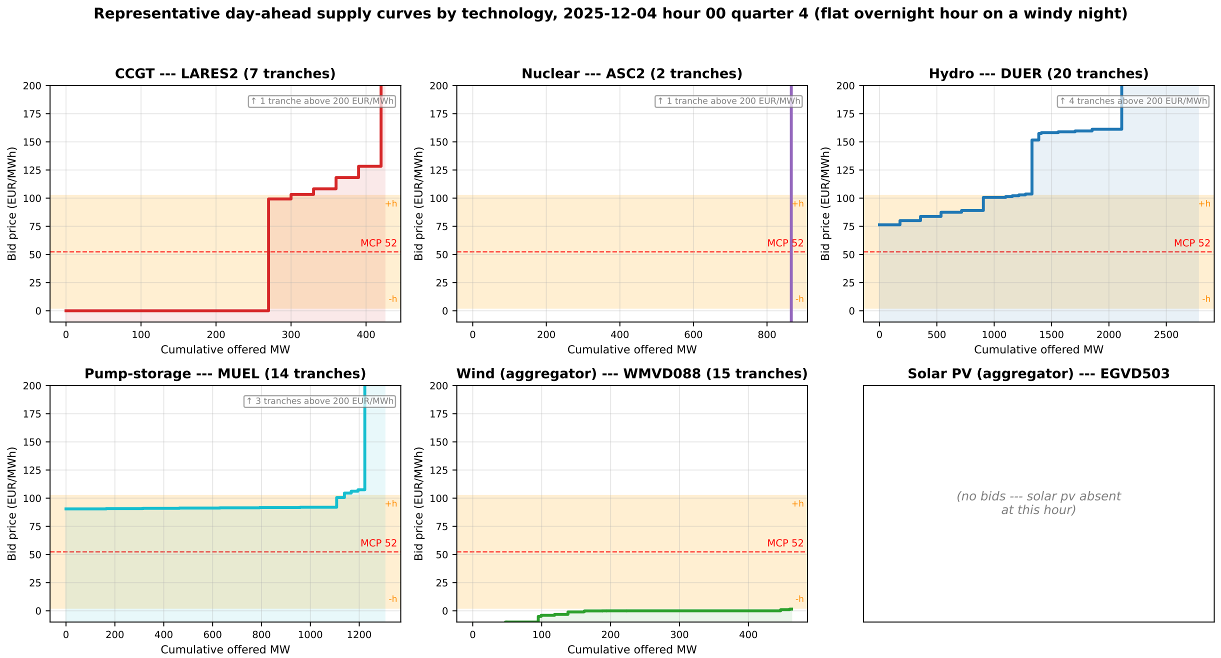

Figure 12: Representative day-ahead supply curves by technology, 2025-12-04 hour 01 quarter 4 (flat overnight hour on a windy night). Same colour scheme and broken y-axis convention as Figure 11. The wind panel (WMVD088) shows a 15-tranche aggregator micro-ladder concentrated in $[-20, +1.5]$ EUR/MWh, well below the band.

Buy vs sell IDA bid shapes (pre-ID15). Figure 13 shows representative IDA bid curves with the sell side (blue, ascending) and the buy side (red, descending) overlaid for one unit per tech in the pre-ID15 window, when buy-side activity was largest (§6.4). The shapes differ sharply across techs: CCGT and pump-storage submit dense sell ladders with a thin one-tranche buy book (sellers with a small rebuy option); hydro and renewable aggregators write multi-tranche curves on *both* sides (forecast-driven rebalancing that can clear in either direction); nuclear barely participates on the buy side (the entire 9-month pre-ID15 window contains a single nuclear (date, session, period) cell with both sides). The renewable-aggregator panels are the clearest illustration of the buy-side rebalancing role of IDA: the buy and sell curves intersect near the realised MCP, and the aggregator clears on whichever side its quarter-level forecast falls.

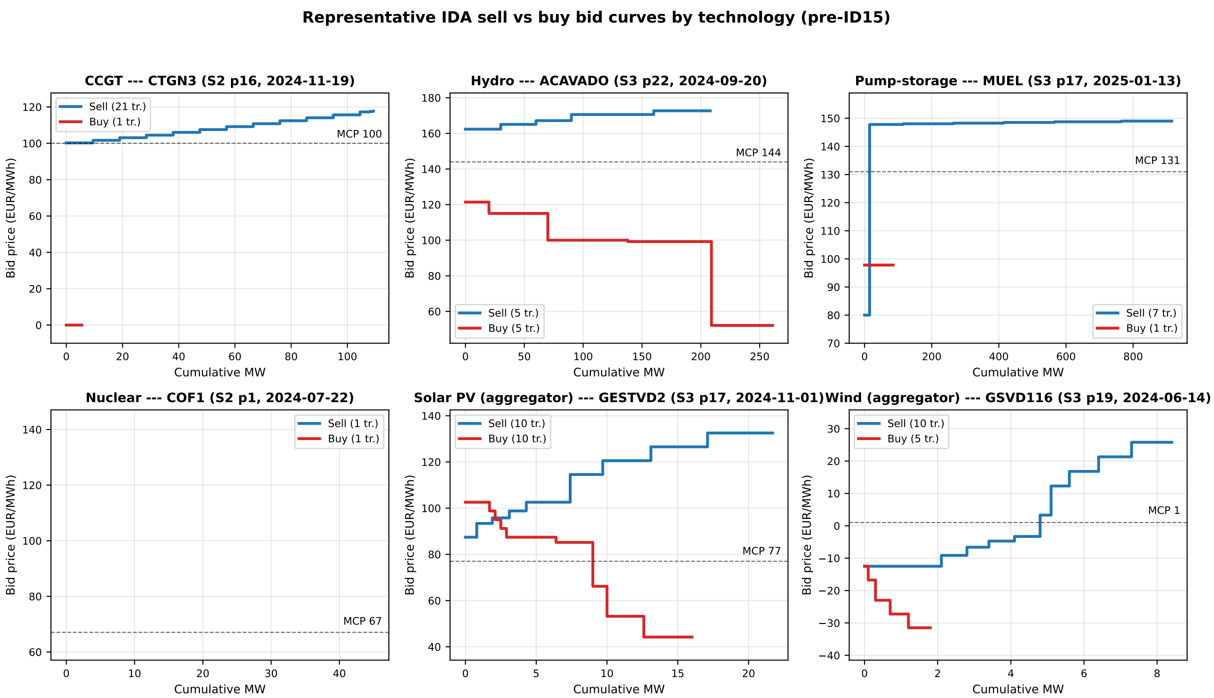

Figure 13: Representative IDA sell vs buy bid curves by technology, pre-ID15. One representative (unit, date, session, period) per tech. Blue: sell curve (supply, ascending in price). Red: buy curve (demand, descending in price). Dashed: MCP. Pre-ID15 window 2024-06-14 to 2025-03-18.

D Theoretical derivations

D.1 Setup details and information arrival

This subsection records the timing of information arrival, the Ito and Reguant residual-demand specification used throughout, and how that specification specialises when one or both markets are hourly.

Timing. Stage 1 (forward gate): $\bar{A}$, the profile $\{\zeta_t\}$, and all distributions are common knowledge; each forward product clears against its realised clearing state v_{1t}, screened by the stage-1 ladder. Between the gates the within-hour update ω_t realises: $\omega_t = \zeta_t + \mu_t$ when the forward is hourly, $\omega_t = \mu_t$ when it is per-quarter. Stage 2 (spot gate): each spot product clears against ω_t and its own clearing state v_{2t}, screened by the stage-2 ladder. Stage 3 (balancing): the unpriceable noise ε_t realises and the settlement prices residual deviations at the penalty γ.

Residual demand (Ito and Reguant (2016) form). The strategic residual monopolist with constant marginal cost c serves what the operational aggregator does not cover, net of any arbitrage position s_t carried from forward to spot. Per delivery quarter t, with all quantities in MWh:

$$q_{1t} \;=\; \bar{A} + \zeta_t\,\mathbb{1}\{Q^F = Q\} + v_{1t} - b_{1t}\,p_{1t} - s_t - S^{op}_{1t}, \qquad q_{2t} \;=\; b_{2t}\left(p_{1(t)} - p_{2t}\right) + \omega_t + s_t - S^{op}_{2t}, \tag{7}$$

where $p_{1(t)}$ is the forward price covering quarter t (one number per clock-hour when $Q^F = 1$), S^{op}_{mt} is the aggregator's offered MWh in market m at quarter t (uniform S^{op}_m/Q when market m is hourly), and b_{mt} is market thinness with per-quarter asymmetry allowed (when market m is hourly, $b_{mt} \equiv b_m$ and $p_{mt} \equiv p_m$).

Hourly residual demand by specialisation. When market m is hourly, its product clears against the within-hour *average* of the per-quarter objects in (7): the profile and every redistributive component drop out by $\sum_t \zeta_t = \sum_t \mu_{mt} = 0$, the screened state is the hour-level $\theta_m = Q^{-1} \sum_t v_{mt}$, and the spot's traded update is the hour-level $\bar{\omega} = Q^{-1} \sum_t \omega_t$. The within-hour variation that this averaging suppresses does not vanish from the model: it loads onto the balancing residual (§D.8) and shows up in the system imbalance cost C^I_{sys} in the welfare derivations (§D.9).

Tiling lemma: uniform amplification under aggregation. The body of §3.1 assumes the screened state is uniform at each granularity ($\theta_m \sim U[-\Delta_m, \Delta_m]$ for hourly products, $v_{mt} \sim U[-\rho_{\mathcal{X}}\Delta_m, +\rho_{\mathcal{X}}\Delta_m]$ for per-quarter products). The following lemma shows the uniform family is exactly closed under the aggregation $v_{mt} = \theta_m + \mu_{mt}$, with ρ pinned by the support of the redistributive component.

Lemma 1 (Tiling). *Let $\theta_m \sim U[-\Delta_m, \Delta_m]$ and let μ_{mt} be supported on $\{-\Delta_m, +\Delta_m\}$, taking each value with probability $1/2$ in each quarter (with $\sum_t \mu_{mt} = 0$ enforced across quarters). Then $v_{mt} = \theta_m + \mu_{mt} \sim U[-2\Delta_m, +2\Delta_m]$ exactly: $\rho_{\mathcal{X}} = 2$. More generally, supporting μ_{mt} on the integer multiples $\{-(2j-1)\Delta_m, +(2j-1)\Delta_m\}$ for $j = 1, \ldots, J$ delivers $v_{mt} \sim U[-(2J)\Delta_m, +(2J)\Delta_m]$ exactly, i.e. $\rho_{\mathcal{X}} = 2J$, an arbitrary even integer.*

Proof. For the $J = 1$ case: $\theta_m + \Delta_m \sim U[0, 2\Delta_m]$ when $\mu_{mt} = +\Delta_m$, and $\theta_m - \Delta_m \sim U[-2\Delta_m, 0]$ when $\mu_{mt} = -\Delta_m$. Each sub-distribution has density $1/(2\Delta_m)$ on its support; the equiprobable mixture has density $1/(4\Delta_m)$ on $[-2\Delta_m, 2\Delta_m]$, which is $U[-2\Delta_m, 2\Delta_m]$. The two sub-supports tile $[-2\Delta_m, 2\Delta_m]$ without overlap or gap. The $J > 1$ case is the same tiling argument with J pairs of sub-supports. □ □

The body carries a general $\rho_{\mathcal{X}}$ rather than the even-integer values delivered by the tiling; we treat $\rho_{\mathcal{X}}$ as a primitive of the hour class, pinned by the support of the redistributive component the venue's per-quarter object carries. The same construction with a smaller support delivers the news-only value $\rho^{\mu}_{\mathcal{X}} \leq \rho_{\mathcal{X}}$ used in §3.5(iii).

Model abstractions: residual monopolist, supply functions, and the passive balancing stage. Three simplifications deserve flagging. First, the strategic bidder is a single residual monopolist per product, even though Spanish wholesale concentration is moderate (HHI ∼1 100) and several firms compete on the bid stack. The residual-monopolist is a stylised device that delivers qualitative comparative statics on within-band conduct; its empirical counterpart is the per-firm residual demand of Ito and Reguant (2016) recovered in Appendix A.10. Second, within a product the bid *is* a supply function, and with a single strategic bidder facing one-dimensional residual-demand uncertainty the optimal schedule is exact (§3.2); what we do not model is strategic interaction in supply functions across several large firms — the Klemperer and Meyer (1989) equilibrium concept for uniform-price multi-unit auctions — which gets intractable in a sequential-market setting. Third, the balancing stage in the model is a passive settlement of the residual imbalance, *not* an active auction with bidding. The aggregator internalises its own deviation through the penalty γ; the strategic bidder produces with certainty and so carries no individual imbalance; the unpriced within-hour residual collects in C^I_{sys} for the welfare account (Appendix D.8, D.9). In reality REE runs aFRR/mFRR auctions where firms bid actively; we abstract from that to keep the focus on how forward/spot granularity reshapes the auction stages. The model's prediction that finer auctions absorb the within-hour state upstream and shrink the balancing residual is what §6.3 reads against the empirical ajuste-cost decomposition.

Renewable fringe and our relation to Ito and Reguant. Ito and Reguant's residual-demand interpretation already absorbs a renewable (or thermal) competitive fringe into the slope b_m: the strategic bidder serves $A - b_m p_m$, the implicit fringe serves the complement. Their setup delivers the strategic-bidder, arbitrage, and price-wedge mechanisms but is silent on two objects our data record directly: *why* the fringe's willingness to supply changes with granularity, and *what shape* the strategic bidder's offer takes. We make the fringe explicit (the operational aggregator with forecast errors and an imbalance penalty γ), so the hedging gain from per-period scheduling has the closed form $G(\Delta_\eta) = \gamma Q \Delta_\eta / 2$ (Appendix D.2); and we let the strategic bidder choose the step offer the OMIE files record, so the empirical bid-shape quintuple of §4.3 acquires exact model counterparts (§3.2) rather than serving as proxies.

D.2 Operational aggregator: closed-form gain from per-period scheduling

We derive the aggregator's expected hedging gain $G(\Delta_\eta)$ from per-period (rather than hourly) spot scheduling in closed form under uniform shocks. We then explain why the aggregator's *individual* forward-vs-spot allocation is not pinned down by an interior first-order condition under standard penalty specifications, and why the empirical migration onto the granular leg (Result 3) is therefore an equilibrium statement about G being positive, not an aggregator-level Δ^* identification.

D.2.1 Problem statement and primitives

The aggregator allocates total hourly expected production $\bar{Y}_h$ between forward and spot. Forecast-error shock $\eta_t \sim \text{Uniform}[-\Delta_\eta, \Delta_\eta]$, i.i.d. across t, with variance $\sigma_\eta^2 = \Delta_\eta^2/3$. We set the structural per-subperiod expected production constant ($\bar{y}_t = \bar{Y}_h/Q$) for the closed-form derivation; the structural variation $\sigma_a^2 = \text{Var}(\bar{y}_t - \bar{Y}_h/Q)$ adds linearly to G and is handled in the closing remark of §D.2.2. Imbalance penalty is $\gamma|\mathcal{I}_t|$ per period (linear in the magnitude of the gap).

What the reform changes in the aggregator's choice set. The reform changes what the aggregator can choose at Stage 2:

	Hourly spot ($Q^S = 1$)	**Granular spot** ($Q^S = Q$)
At Stage 2 the aggregator chooses	ONE number S_2 for the whole hour	Q numbers $\{S_{2,t}\}$ per period
Per-period schedule	uniform $(S_1 + S_2)/Q$	$S_1/Q + S^*_{2,t}$
Can respond to η_t per period?	no	yes, exactly
Per-period imbalance	η_t	0

D.2.2 Expected penalty under each regime

Granular spot ($Q^S = Q$)**.** At Stage 2 the aggregator observes η_t and sets the per-subperiod allocation $S_{2t}^{op*} = \bar{y}_t + \eta_t - S_1^{op}/Q$. The realised per-period imbalance is 0 (modulo post-Stage-2 noise ε_t, which is regime-invariant and we drop for clarity):

$$\mathbb{E}[\mathcal{C}_{\text{gran}}] \;=\; 0.$$

Hourly spot ($Q^S = 1$)**.** The aggregator commits $S_1^{op} + S_2^{op} = \bar{Y}_h$ before η_t resolves; the per-subperiod schedule is therefore uniform at $\bar{Y}_h/Q$. With $\bar{y}_t = \bar{Y}_h/Q$ the per-period imbalance is exactly η_t. For $\eta_t \sim \text{Uniform}[-\Delta_\eta, \Delta_\eta]$:

$$\mathbb{E}|\eta_t| \;=\; \frac{\Delta_\eta}{2}, \qquad \mathbb{E}[\mathcal{C}_{\text{hour}}] \;=\; \gamma Q \cdot \frac{\Delta_\eta}{2}.$$

Closed-form hedging gain. The aggregator's expected hourly saving from per-period (granular) spot scheduling relative to hourly spot scheduling is

$$\boxed{G(\Delta_\eta) \;=\; \mathbb{E}[\mathcal{C}_{\text{hour}}] - \mathbb{E}[\mathcal{C}_{\text{gran}}] \;=\; \frac{\gamma Q\,\Delta_\eta}{2}.} \tag{8}$$

Comparative statics are immediate:

$$\frac{\partial G}{\partial \Delta_\eta} \;=\; \frac{\gamma Q}{2} \;>\; 0, \qquad \frac{\partial G}{\partial \sigma_\eta^2} \;=\; \frac{\gamma Q}{2\sqrt{3}\,\sigma_\eta} \;>\; 0. \tag{9}$$

The gain is strictly positive and strictly increasing in the forecast-error variance. *Closing remark (structural variation).* When $\bar{y}_t \neq \bar{Y}_h/Q$, the hourly regime's uniform schedule additionally misses the deterministic profile, adding $\gamma \sum_t |\bar{y}_t - \bar{Y}_h/Q|$ to its bill; granular scheduling removes this term too, so it enters G additively and we omit it from the display.

D.2.3 *Why Δ^* is not pinned down by an individual FOC*

A natural follow-up is to ask whether G pins down the aggregator's allocation S_1^{op} via a first-order condition. It does not. The aggregator's expected profit under either regime is

$$\Pi(S_1^{op}) \;=\; (p_1 - p_2)\,S_1^{op} \;+\; p_2\,\bar{Y}_h \;-\; \mathbb{E}[\mathcal{C}],$$

where the expected imbalance cost $\mathbb{E}[\mathcal{C}]$ is independent of S_1^{op} once we impose the no-average-bias condition $S_1^{op} + S_2^{op} = \bar{Y}_h$. The FOC reduces to $p_1 - p_2 = 0$, a corner solution rather than an interior Δ^*. This is a structural feature of the price-taker setup: the imbalance cost depends on the resolution of information (whether the aggregator can react to η_t), not on how forward and spot allocations split.

The migration onto the granular leg is therefore an *equilibrium* statement: across aggregators, $G(\Delta_\eta) > 0$ generates a positive premium for participation on the granular spot; equilibrium prices adjust so that the wedge $p_1 - p_2$ partially incorporates this premium; and the empirical aggregate shift in supply onto the granular leg (Result 3, sell-share rise of nuclear and wind at the March reform) is the equilibrium counterpart. We claim only the direction of the shift via $G > 0$; we do not derive a level prediction for Δ^* at the individual aggregator's optimum.

D.3 Strategic bidder: the optimal n-tranche ladder along the monopoly locus

We prove the bid-ladder propositions of §3.2 in turn: the continuous benchmark (Lemma 2), the optimal n-tranche ladder (Proposition 1), the closed-form bid-shape statistics (Corollary 1), level neutrality (Remark 1), the optimal tranche depth under a complexity cost (Proposition 2), the granularity–variation interaction (Corollary 2), and the stage-1 forward locus slope (Corollary 3).

Fix one delivery product. Residual demand is $q = A - b\,p$ with $A = \bar{a} + v$ and $v \sim U[-\Delta, \Delta]$; the bidder has constant marginal cost c. The auction is uniform-price; the bidder submits a non-decreasing step function $S(p) = \sum_k \Delta q_k \cdot \mathbf{1}\{p \geq p_k\}$ and the clearing pair $(p^\star, q^\star)$ solves $S(p^\star) = A - b\,p^\star$ on the relevant segment.

Assumption 1. $q^m(\bar{a} - \Delta) \geq \Delta/(2(2n+1))$: the monopoly quantity is interior in every state, by a margin of a quarter cell.

Lemma 2 (Continuous benchmark)**.** *The supply schedule $S^*(p) = b\,(p - c)$ clears at the full-information monopoly point $p^m(A) = (A + bc)/(2b)$, $q^m(A) = (A - bc)/2$ in every state A, and is the unique non-decreasing schedule that does so on the contested range. Expected profit equals $\mathbb{E}[(A - bc)^2]/(4b)$, the unconditional monopoly value.*

D.3.1 Continuous benchmark (proof of Lemma 2)

In state A, the full-information monopoly point maximises $(p-c)(A-b\,p)$ over p; the FOC gives $p^m(A)=(A+b\,c)/(2b)$ and $q^m(A)=(A-b\,c)/2$. Eliminating A, the monopoly locus is $\{(q,p): q=b(p-c)\}$, a line of slope b through $(p,q)=(c,0)$. Define $S^\star(p)=b(p-c)$. For any state A,

$$S^\star(p^m(A)) \;=\; b\Big(\frac{A+b\,c}{2b}-c\Big) \;=\; \frac{A-b\,c}{2} \;=\; q^m(A),$$

so $(p^m(A),q^m(A))$ is on $S^\star$ in every state. *Uniqueness on the contested range*: a non-decreasing schedule that clears at the monopoly point in every state must contain the locus, which is strictly increasing; a monotone schedule containing a strictly increasing curve coincides with it on its range. Expected profit at $S^\star$ is $\mathbb{E}[(p^m(A)-c)q^m(A)]=\mathbb{E}[(A-bc)^2]/(4b)$, the unconditional monopoly value. □

D.3.2 Optimal n-tranche ladder (proof of Proposition 1)

Clearing geometry under a step offer. Index the n competing tranches $1,\dots,n$ by ascending price $p_1<\cdots<p_n$, let q_0 be the inframarginal base priced at the floor, and let $Q_k=q_0+\sum_{j\le k}\Delta q_j$ be cumulative quantity ($Q_0=q_0$). Three clearing regimes are possible per state A, alternating riser–horizontal–riser:

(V_0) *Base riser.* For $A\in[\bar a-\Delta,\ Q_0+b\,p_1]$: quantity is Q_0 and the realised demand sets the price, $p=(A-Q_0)/b$ between the floor and p_1.

(H_k) *Horizontal segment at p_k.* For $A\in[Q_{k-1}+b\,p_k,\ Q_k+b\,p_k]$: clearing is $(p_k,\ A-b\,p_k)$ — the marginal tranche is partially filled.

(V_k) *Riser at Q_k.* For $A\in[Q_k+b\,p_k,\ Q_k+b\,p_{k+1}]$ (the top riser V_n closing at $\bar a+\Delta$): clearing is $\big((A-Q_k)/b,\ Q_k\big)$ — the realised demand sets the price.

This partitions the state space into $2n+1$ full cells: one base riser, n horizontals of length Δq_k (in A-units), and n risers of length $b(p_{k+1}-p_k)$. Profit is continuous at every switch: both adjacent regimes deliver the same price–quantity pair at the boundary state.

Each instrument serves an interval of states and loses a squared distance. Profit in state A is a quadratic hill around the monopoly point. On a horizontal segment at price p, completing the square gives $\pi(p;\,A)=\pi^m(A)-b\,(p-p^m(A))^2$; on a riser at quantity q (the base included), $\pi(q;\,A)=\pi^m(A)-(q-q^m(A))^2/b$. Each instrument is therefore exactly right in one state — the step at p in the state whose monopoly price is p, the riser at q in the state whose monopoly quantity is q; call that state the instrument's *target* $\hat A$. Because p^m and q^m are linear in A with slopes $1/(2b)$ and $1/2$, both losses reduce to the same expression,

$$\pi^m(A)-\pi \;=\; \frac{(A-\hat A)^2}{4\,b},$$

so the bidder's problem is to place $2n+1$ targets so that whichever state realises, it lands close to a target.

Equal intervals are optimal. An instrument serving an interval of length L does best with its target at the interval's midpoint, since the loss is symmetric in the distance; the mean squared distance of a uniform draw from the midpoint is $L^2/12$ (the variance of a uniform), and the interval is reached with probability $L/(2\Delta)$. Its contribution to the expected loss is therefore at least

$$\frac{L^2}{12}\cdot\frac{1}{4b}\cdot\frac{L}{2\Delta} \;=\; \frac{L^3}{96\,b\,\Delta},$$

and the total expected loss is at least $\sum_j L_j^3/(96\,b\,\Delta)$ over the $2n+1$ intervals with $\sum_j L_j = 2\Delta$. Since x^3 is convex, moving length from a longer interval to a shorter one always lowers the sum, so the minimum splits the state range into *equal* intervals of width $w \equiv 2\Delta/(2n+1)$.

The clearing rule places the boundaries halfway by itself. The bidder chooses only the targets; the interval boundaries then follow from the clearing geometry, and they land exactly halfway between adjacent targets. The switch from the step at p_k to the riser at Q_k occurs at the state $A^\star = Q_k + b\,p_k$; substituting $p_k = p^m(\hat{A}_{H_k})$ and $Q_k = q^m(\hat{A}_{V_k})$,

$$A^\star \;=\; \frac{\hat{A}_{H_k} + bc}{2} + \frac{\hat{A}_{V_k} - bc}{2} \;=\; \frac{\hat{A}_{H_k} + \hat{A}_{V_k}}{2},$$

and likewise at every other switch. Placing the targets at the equal-grid midpoints $m_j = \bar{a} - \Delta + (j - \frac{1}{2})\,w$, $j = 1, \ldots, 2n+1$, therefore reproduces exactly the equal-interval partition with every target at its interval's midpoint: the lower bound is attained.

Closed forms and feasibility. The alternation assigns V_0 to interval 1, H_k to interval $2k$, V_k to interval $2k{+}1$, so

$$q_0 = q^m\big(\bar{a} - \Delta + \tfrac{w}{2}\big), \qquad p_k = p^m(m_{2k}) = p^m\big(\bar{a} - \Delta + (2k - \tfrac{1}{2})\,w\big), \qquad Q_k = q^m(m_{2k+1}):$$

every competing tranche carries $\Delta q_k = Q_k - Q_{k-1} = (m_{2k+1} - m_{2k-1})/2 = w$ MW (the first measured from the base) and the price grid is even with step $p_{k+1} - p_k = (m_{2k+2} - m_{2k})/(2b) = w/b$. Feasibility under Assumption 1: the base $q_0 = q^m(\bar{a} - \Delta) + w/4 \geq w/4 > 0$, and the lowest realised price (in V_0 at $A = \bar{a} - \Delta$) is $p^m(\bar{a} - \Delta - w/2) \geq c$ iff $q^m(\bar{a} - \Delta) \geq w/4$ — the stated condition.

Expected profit. With equal intervals the mean squared distance to the target is $w^2/12$ everywhere, so the total expected loss is $w^2/(48\,b) = \Delta^2/(12\,b\,(2n+1)^2)$ and

$$\mathbb{E}[\pi_n] \;=\; \frac{\mathbb{E}[(A - b\,c)^2]}{4\,b} \;-\; \frac{\Delta^2}{12\,b\,(2n+1)^2}. \qquad \square$$

D.3.3 Bid-shape statistics (proof of Corollary 1)

The competing tranches carry equal MW w, so $w_k \equiv \Delta q_k / \sum_j \Delta q_j = 1/n$. Hence

$$\mathrm{HHI}_p \;=\; \sum_{k=1}^{n} w_k^2 \;=\; \sum_{k=1}^{n} \frac{1}{n^2} \;=\; \frac{1}{n}.$$

The tranche prices form an arithmetic progression $p_k = p^m(\bar{a}) + (k - (n+1)/2)\,(w/b)$ centred on $p^m(\bar{a})$. The MW-weighted mean price is $\bar{p} = (1/n)\sum_k p_k = p^m(\bar{a})$ by symmetry, and the MW-weighted variance is

$$\sigma_p^2 \;=\; \frac{1}{n}\sum_{k=1}^{n}(p_k - \bar{p})^2 \;=\; \frac{1}{n}\left(\frac{w}{b}\right)^2 \sum_{k=1}^{n}\left(k - \tfrac{n+1}{2}\right)^2 \;=\; \frac{1}{n}\left(\frac{w}{b}\right)^2 \cdot \frac{n(n^2-1)}{12} \;=\; \left(\frac{w}{b}\right)^2 \frac{n^2-1}{12}.$$

Substituting $w = 2\Delta/(2n+1)$,

$$\sigma_p \;=\; \frac{2\Delta}{(2n+1)\,b}\sqrt{\frac{n^2-1}{12}}.$$

As $n \to \infty$, $\sigma_p \to \Delta/(2b\sqrt{3}) = \sigma_v/(2b)$ since $\sigma_v = \mathrm{sd}_v(v) = \Delta/\sqrt{3}$ for $v \sim U[-\Delta, \Delta]$.

For $\hat{\beta}$, the cumulative quantity at tranche k is $Q_k = q_0 + k\,w$ and the price is $p_k = p^m(\bar{a}) + (k - (n+1)/2)\,(w/b)$. Both are linear in k, so the OLS slope of p_k on Q_k is exactly the ratio of slopes:

$$\hat{\beta} \;=\; \frac{p_{k+1} - p_k}{Q_{k+1} - Q_k} \;=\; \frac{w/b}{w} \;=\; \frac{1}{b}.$$

For $\hat{\gamma}$: p_k is an exact affine function of Q_k, so the least-squares quadratic fit reproduces the affine relation with zero residual and zero curvature coefficient. Hence $\hat{\gamma} = 0$. □

Remark 1 (Level neutrality). $\mathbb{E}[\text{clearing price}] = p^m(\bar{a})$ for every n: ladder geometry reshapes dispersion and surplus, but never the mean price.

D.3.4 Level neutrality (proof of Remark 1)

Take expectation of the clearing price over states. Each cell contributes its midpoint-monopoly price weighted by cell length: horizontal cells contribute $p_k = p^m(m_{2k})$ over length w; riser cells contribute the average realised price over the riser, $\mathbb{E}[(A - Q_k)/b \mid V_k] = (m_{2k+1} - q^m(m_{2k+1}))/b = p^m(m_{2k+1})$, using $A - b\,p^m(A) - q^m(A) \equiv 0$; the base riser contributes $p^m(m_1)$ by the same demand-sets-price computation. The resulting weighted average over all $2n+1$ cells (each of length w) is the simple average of the $2n+1$ midpoint-monopoly prices, which — because p^m is affine and the midpoints m_j are symmetric around $\bar{a}$ — equals $p^m(\bar{a})$. So $\mathbb{E}[\text{clearing price}] = p^m(\bar{a})$, independent of n. □

Distributional consequence. The step structure reshapes the variance and the per-state surplus split, but never the mean clearing price. Bid-shape changes (regrading the ladder) are therefore *level-neutral* on price; mean price changes come only from changes in residual demand (the slope b or the venue supply S^{op}), which is the channel underpinning the markup identity (6) in the body.

Proposition 2 (Tranche budget)**.** *With a linear cost $\kappa > 0$ per competing tranche (treating n as continuous; the integer optimum is one of the two adjacent integers), the optimal depth satisfies*

$$2n^* + 1 \;=\; \Bigl(\frac{\Delta^2}{3\,b\,\kappa}\Bigr)^{1/3}:$$

ladders deepen with the screened range, $n^* + \frac{1}{2} \;=\; \frac{1}{2}\,\Delta^{2/3}\,(3\,b\,\kappa)^{-1/3}$, and flatten with market thickness b and complexity cost κ.

D.3.5 Tranche budget (proof of Proposition 2)

With a complexity cost $\kappa > 0$ per competing tranche, the objective is

$$\mathbb{E}[\pi_n] - \kappa\,n \;=\; \frac{\mathbb{E}[(A - bc)^2]}{4b} \;-\; \frac{\Delta^2}{12\,b\,(2n+1)^2} \;-\; \kappa\,n.$$

Treating n as continuous, the FOC in n is

$$\frac{d}{dn}\Bigl[-\,\frac{\Delta^2}{12b\,(2n+1)^2}\Bigr] \;=\; \kappa \quad\Longleftrightarrow\quad \frac{\Delta^2}{3b\,(2n+1)^3} \;=\; \kappa \quad\Longleftrightarrow\quad 2n^* + 1 \;=\; \Bigl(\frac{\Delta^2}{3b\,\kappa}\Bigr)^{1/3}.$$

The integer optimum is one of the two adjacent integers; either way $n^* + \frac{1}{2} = \frac{1}{2}\,\Delta^{2/3}\,(3\,b\,\kappa)^{-1/3}$ at the continuous optimum. Joint signature with Corollary 1: a wider screened range Δ raises σ_p and lowers $\mathrm{HHI}_p = 1/n^*$ — they move together. □

D.3.6 Granularity × within-hour variation (proof of Corollary 2)

(i) Under the continuous benchmark the clearing price is $p^m(A) = (A + bc)/(2b)$, affine in A with slope $1/(2b)$; over $A \in [\bar{a} - \Delta, \bar{a} + \Delta]$ its range is $2\Delta/(2b) = \Delta/b$, and substituting $(\rho_{\mathcal{X}}\Delta, b')$ gives $\rho_{\mathcal{X}}\Delta/b'$. The span $\frac{2(n-1)\Delta}{(2n+1)b}$ and the dispersion $\sigma_p = \frac{2\Delta}{(2n+1)b}\sqrt{(n^2-1)/12}$ of §D.3.3 are linear in Δ at fixed n, so the same substitution multiplies both by $\rho_{\mathcal{X}}\, b/b'$; $\mathrm{HHI}_p = 1/n$ involves neither Δ nor b.

(ii) Substituting $(\rho_{\mathcal{X}}\Delta, b')$ into Proposition 2: $2n^*{+}1 = \big((\rho_{\mathcal{X}}\Delta)^2/(3b'\kappa)\big)^{1/3} = \rho_{\mathcal{X}}^{2/3}\left(\Delta^2/(3b'\kappa)\right)^{1/3}$.

(iii) The flat-hour change is $\sigma_p(\Delta; b') - \sigma_p(\Delta; b)$ — the pure thinness shift, since $\rho_{\text{flat}} = 1$ leaves the screen at Δ. Subtracting it from the critical-hour change $\rho_{\mathcal{X}}\,\sigma_p(\Delta; b') - \sigma_p(\Delta; b)$ leaves

$$(\rho_{\mathcal{X}} - 1)\,\sigma_p(\Delta; b') \;=\; (\rho_{\mathcal{X}} - 1)\,\frac{2\Delta}{(2n+1)\,b'}\sqrt{\frac{n^2-1}{12}},$$

zero iff $\rho_{\mathcal{X}} = 1$ and strictly increasing in $\rho_{\mathcal{X}}$. □

Assumption 2 (Stage-1 second-order condition)**.** $B_2 \equiv \sum_t b_{2t} < 2b_1$: the total continuation slope of the spot products hanging off the forward product is strictly less than twice the forward slope.

D.3.7 Continuation slope and forward locus (proof of Corollary 3)

Apply backward induction at the spot stage. With Q spot products $t = 1, \ldots, Q$ each of slope b_{2t}, the per-quarter strategic best response is $p_{2t}^*(p_1) = (p_1 + c)/2$ at the symmetric solution (omitting arbitrage and aggregator terms which are p_1-independent for the slope calculation). The per-quarter cleared quantity is $q_{2t}^*(p_1) = b_{2t}(p_1 - p_{2t}^*) = b_{2t}(p_1 - c)/2$. Spot profit at the best response is

$$\pi_{2t}^*(p_1) \;=\; (p_{2t}^* - c)\,q_{2t}^* \;=\; \frac{p_1 - c}{2}\cdot\frac{b_{2t}\,(p_1 - c)}{2} \;=\; \frac{b_{2t}\,(p_1 - c)^2}{4}.$$

Summing across spot products, the stage-1 total profit in state v at forward intercept $A_1 \equiv \bar{A} + v$ is

$$\Pi_1(p_1, v) \;=\; (p_1 - c)(A_1 - b_1\,p_1) \;+\; \frac{(p_1 - c)^2}{4}\sum_t b_{2t} \;=\; (p_1 - c)(A_1 - b_1\,p_1) \;+\; \frac{B_2\,(p_1 - c)^2}{4},$$

with $B_2 \equiv \sum_t b_{2t}$. The stage-1 FOC:

$$\frac{\partial \Pi_1}{\partial p_1} \;=\; (A_1 - b_1\,p_1) + (p_1 - c)(-b_1) + \frac{B_2(p_1 - c)}{2} \;=\; 0 \quad\Longleftrightarrow\quad A_1 - 2b_1 p_1 + b_1 c + \frac{B_2}{2}(p_1 - c) \;=\; 0.$$

Solving for p_1^*:

$$p_1^*(v) \;=\; \frac{A_1 + c(b_1 - B_2/2)}{2b_1 - B_2/2} \;=\; \frac{\bar{A} + v + c(b_1 - B_2/2)}{2b_1 - B_2/2}.$$

The stage-1 monopoly-locus slope in v is therefore

$$\boxed{\frac{dp_1^*}{dv} \;=\; \frac{1}{2b_1 - B_2/2}}$$

exactly as claimed in the body (4). Assumption 2 ($B_2 < 2b_1$) ensures the denominator is positive, so the locus is increasing in v and the stage-1 problem is concave at the interior optimum.

Price range. Over the screened states $v \in [-\Delta_1, \Delta_1]$ the stage-1 clearing-price range under the continuous benchmark is exactly

$$2\Delta_1 \cdot \frac{dp_1^*}{dv} \;=\; \frac{2\Delta_1}{2b_1 - B_2/2},$$

so the forward tier's price span widens one-for-one with the locus slope.

Measured ladder slope $\hat{\beta}_1$. The cleared quantity at the monopoly point in state v is $q_1^*(v) = A_1 - b_1\, p_1^*(v)$. Substituting,

$$q_1^*(v) \;=\; \frac{(b_1 - B_2/2)(\bar{A} + v - b_1 c)}{2b_1 - B_2/2}, \quad \frac{dq_1^*}{dv} \;=\; \frac{b_1 - B_2/2}{2b_1 - B_2/2}.$$

Assumption 2 also makes $dq_1^*/dv > 0$: the locus is increasing in *both* coordinates, hence implementable by a non-decreasing ladder. Building the step offer along this locus as in §D.3.2, tranche prices and cumulative quantities both advance linearly with the state — at the rates dp_1^*/dv and dq_1^*/dv — so both are evenly spaced sequences and the OLS slope of p_k on Q_k is the ratio of the two rates, independent of the grid width:

$$\hat{\beta}_1 \;=\; \frac{dp_1^*/dv}{dq_1^*/dv} \;=\; \frac{1/(2b_1 - B_2/2)}{(b_1 - B_2/2)/(2b_1 - B_2/2)} \;=\; \frac{1}{b_1 - B_2/2}.$$

Spot-reform jump. At the asymmetric state $(1, Q)$, the spot venue becomes per-quarter, so B_2 jumps from a single hourly slope b_2 to the sum $\sum_t b_{2t}$ of Q per-quarter slopes. Empirically the per-quarter spot slopes are of the same order as the hourly one (Table 16), so the continuation coefficient B_2 jumps from b_2 to roughly $Q \cdot b_2$ — a large factor. The denominators $2b_1 - B_2/2$ and $b_1 - B_2/2$ shrink correspondingly (within the SOC): the stage-1 price range $2\Delta_1/(2b_1 - B_2/2)$ widens and the measured forward ladder slope $\hat{\beta}_1 = 1/(b_1 - B_2/2)$ steepens, both in exact closed form. This is the model's cross-market anticipation channel of §3.4(M4); the proof is complete. □

Remark 2 (Price-takers post blocks). A price-taking unit optimally offers its marginal-cost schedule; for a zero-marginal-cost renewable that is a single block at the floor — $\sigma_p = 0$, $\mathrm{HHI}_p = 1$, out of band, for any granularity (panel (a) of Figure 3). The bid-shape margin is therefore dispatchable-specific; the renewable margin of the reform is the quantity re-scheduling of §3.4(M1).

Numerical example. With $\Delta = 200$ MW, $b = 50$ MW per EUR/MWh, and $n = 5$: tranches of 36 MW, a grid step of 0.73 EUR/MWh, a competing-tier span of 2.9 EUR/MWh, $\sigma_p = 1.03$ EUR/MWh, and $\mathrm{HHI}_p = 0.2$ — the order of magnitude of the CCGT critical-hour cells in Table 5. In a ramp hour with $\rho_{\mathcal{X}} = 2$ the same product as a quarter contract spans 5.8 EUR/MWh with $\sigma_p = 2.06$ EUR/MWh at the same depth, and the endogenous depth rises to $2n^* + 1 = 2^{2/3} \cdot 11 \approx 17.5$ (so $n^* \approx 8$, $\mathrm{HHI}_p \approx 0.12$); in a flat hour ($\rho = 1$) nothing moves.

D.4 $(1, 1)$ equilibrium

We solve the pre-reform game by backward induction: the Stage 2 best response (§D.4.1), then the four arbitrage regimes at Stage 1 (§D.4.2). The scarcity wedge $p_1 > p_2 > c$ survives in regimes (1), (3), and (4); only competitive full arbitrage closes it. *Convention:* throughout this subsection we report the stage-1 problem in the Ito and Reguant form, suppressing the stage-1 continuation term; Corollary 3 restores it and shows it tilts the stage-1 locus (slope $1/(2b_1 - b_2/2)$ here, since $B_2 = b_2$ at this state) without altering the regime classification or the wedge formulas, which depend on s and the spot best response only.

D.4.1 Stage 2 best response

The strategic bidder's Stage 2 problem, taking p_1, q_1 and s as given:

$$\max_{p_2} (p_2 - c) q_2 \quad \text{s.t.} \quad q_2 = b_2(p_1 - p_2) + s - S_2^{op}.$$

FOC: $b_2(p_1 - p_2) + s - S_2^{op} + (p_2 - c)(-b_2) = 0$, giving

$$p_2^*(p_1, s) \;=\; \frac{p_1 + c}{2} + \frac{s - S_2^{op}}{2\, b_2}. \tag{10}$$

D.4.2 Stage 1 best response and equilibrium in (p_1, s)

Substituting (10) into the strategic bidder's total profit and using $q_1 = \bar{A} - b_1 p_1 - s - S_1^{op}$,

$$\Pi(p_1, s) \;=\; (p_1 - c)(\bar{A} - b_1 p_1 - s - S_1^{op}) \;+\; (p_2^*(p_1, s) - c)q_2^*(p_1, s).$$

The four arbitrage cases pin s differently.

(R1) No arbitrage ($s = 0$). The strategic bidder solves a Cournot problem in each market separately. The forward FOC gives $p_1^* = (\bar{A} - S_1^{op} + b_1 c)/(2b_1)$. Combining with (10) yields $p_1^* > p_2^* > c$ and $q_1, q_2 > 0$ (Ito and Reguant Result 1).

(R2) Full arbitrage ($p_1 = p_2$). Setting $p_1 = p_2$ in (10) gives $s = b_2(p_1 - c) + S_2^{op}$. Substituting back, total cleared MWh $q_1 + q_2 = \bar{A} - b_1 p_1 - S_1^{op} - S_2^{op} = \bar{A} - b_1 p_1 - \bar{Y}_h$ (using $S_1^{op} + S_2^{op} = \bar{Y}_h$). The strategic bidder maximises $(p_1 - c)(\bar{A} - b_1 p_1 - \bar{Y}_h)$, giving

$$p_1^{*,F} \;=\; \frac{\bar{A} - \bar{Y}_h + b_1 c}{2b_1}, \qquad p_2^{*,F} \;=\; p_1^{*,F}. \tag{11}$$

Note that $s > 0$ *reduces* the strategic bidder's total output relative to no arbitrage (Ito and Reguant Result 2): even competitive arbitrage does not restore competitive prices, because the bidder withholds more in response.

(R3) Limited arbitrage ($s \leq K_a$ binding). If K_a is below the full-arbitrage level, $s = K_a$ binds and $p_1 - p_2 > 0$. From (10), $p_1 - p_2 = (p_1 - c)/2 - (K_a - S_2^{op})/(2b_2)$; the wedge depends on K_a and on S_2^{op} (Ito and Reguant Result 3). The smaller K_a, the larger the residual wedge.

(R4) Strategic arbitrage. The arbitrageur maximises $\pi^A(s) = s(p_1 - p_2)$ subject to (10):

$$\pi^A(s) \;=\; s\left[\frac{p_1 - c}{2} - \frac{s - S_2^{op}}{2\,b_2}\right].$$

FOC in s:

$$\frac{p_1 - c}{2} - \frac{2s - S_2^{op}}{2\,b_2} \;=\; 0 \quad\Longleftrightarrow\quad s^{*,S}(p_1) \;=\; \frac{b_2(p_1 - c)}{2} + \frac{S_2^{op}}{2}.$$

Substituting into (10),

$$p_2^{*,S} \;=\; \frac{p_1 + c}{2} + \frac{p_1 - c}{4} - \frac{S_2^{op}}{4\,b_2} \;=\; \frac{3p_1 + c}{4} - \frac{S_2^{op}}{4\,b_2}, \qquad p_1 - p_2^{*,S} \;=\; \frac{p_1 - c}{4} + \frac{S_2^{op}}{4\,b_2}.$$

With the aggregator's spot leg netted into the spot intercept ($S_2^{op} = 0$ in the display — the $S_2^{op}/(4b_2)$ term is a level shift common across regimes), this is the boxed body benchmark (1): $p_1 - p_2 = (p_1 - c)/4$ (Ito and Reguant Result 4).

D.5 $(1, Q)$ equilibrium

Forward hourly, spot per-period. Per-period spot best response, then within-block dispersion under full and strategic arbitrage, then the within-hour wedge SD.

The strategic bidder's Stage 2 problem now has Q spot products. Directly from the spot residual demand in (7) (suppressing the update term, restored in §D.5.3) and the FOC $q_{2t} = b_{2t}(p_{2t} - c)$, the strategic per-period spot best response is

$$p_{2t}^*(p_1, s_t) \;=\; \frac{p_1 + c}{2} \;+\; \frac{s_t - S_{2t}^{op}}{2\,b_{2t}}. \tag{12}$$

D.5.1 Full arbitrage: block parity and within-block dispersion

Block parity requires $p_1 = (1/Q)\sum_t p_{2t}$. Substituting (12),

$$p_1 \;=\; \frac{p_1+c}{2} + \frac{1}{2Q}\sum_t \frac{s_t - S^{op}_{2t}}{b_{2t}} \quad\Longleftrightarrow\quad p_1 - c \;=\; \frac{1}{Q}\sum_t \frac{s_t - S^{op}_{2t}}{b_{2t}}.$$

Setting $S^{op}_{2t} \equiv 0$ and uniform arbitrage $s_t \equiv \bar{s}$ to isolate the thinness mechanism, and $Q = 2$ for concreteness,

$$p_1 - c \;=\; \frac{\bar{s}}{2}\left(\frac{1}{b_{21}} + \frac{1}{b_{22}}\right) \;=\; \frac{\bar{s}}{2}\cdot\frac{b_{21}+b_{22}}{b_{21}b_{22}} \quad\Longleftrightarrow\quad \bar{s} \;=\; \frac{2(p_1-c)\,b_{21}b_{22}}{b_{21}+b_{22}}.$$

The within-block dispersion follows from differencing p^*_{2t} across t:

$$p_{21} - p_{22} \;=\; \left(\frac{\bar{s}}{2b_{21}} - \frac{\bar{s}}{2b_{22}}\right) \;=\; \frac{\bar{s}}{2}\cdot\frac{b_{22}-b_{21}}{b_{21}\,b_{22}}.$$

Substituting $\bar{s}$ from above,

$$\boxed{p_{21} - p_{22} \;=\; (p_1 - c)\cdot\frac{b_{22}-b_{21}}{b_{21}+b_{22}}} \qquad \text{(full arbitrage).} \tag{13}$$

D.5.2 Strategic arbitrage: block wedge and attenuated dispersion

The strategic arbitrageur internalises price impact and stops at

$$p_1 - (1/Q)\sum_t p_{2t} \;=\; \frac{p_1 - c}{4}.$$

The block-wedge condition is the average of (1) across the Q matched spot products. Repeating the algebra of §D.5.1 with this modified parity, the arbitrage position $\bar{s}$ falls to exactly half of the full-arbitrage value:

$$\bar{s}^S \;=\; \frac{(p_1-c)\,b_{21}b_{22}}{b_{21}+b_{22}} \;=\; \frac{\bar{s}^F}{2}.$$

The within-block dispersion correspondingly becomes

$$\boxed{p_{21} - p_{22} \;=\; \frac{p_1 - c}{2}\cdot\frac{b_{22}-b_{21}}{b_{21}+b_{22}}} \qquad \text{(strategic arbitrage).} \tag{14}$$

The dispersion is attenuated by a factor of two relative to the full-arbitrage case, reflecting the smaller arbitrage volume traded under strategic incentives.

D.5.3 Wedge SD in the asymmetric state

Update channel: $\sigma_\omega/(2b_2)$ (derivation of body eq. 3). Decompose the spot residual demand at the per-quarter level using the within-hour state of §3.1: $A_{2t} = \bar{A} + \zeta_t + v_{2t}$. While the forward is hourly, the update $\omega_t = \zeta_t + \mu_t$ realises between the gates and becomes the spot-side traded object (plus an hour-level component common across t, which is absorbed into the mean and does not affect dispersions). The strategic spot best response (12) acquires an ω_t term:

$$p^*_{2t}(p_1, s_t, \omega_t) \;=\; \frac{p_1 + c}{2} \;+\; \frac{\omega_t + s_t - S^{op}_{2t}}{2\,b_{2t}}.$$

With p_1 constant across t (hourly forward), the per-subperiod wedge $w_t = p_1 - p^*_{2t}$ inherits the cross-period variation in p^*_{2t}. Taking variance across t at uniform $b_{2t} \equiv b_2$, s_t, S^{op}_{2t} (so the only t-varying object is ω_t):

$$\mathrm{sd}_t(p^*_{2t}) \;=\; \frac{\mathrm{sd}_t(\omega_t)}{2\,b_2} \;=\; \frac{\sigma_\omega}{2\,b_2}, \qquad \sigma^2_\omega \;=\; \sigma^2_\zeta + \sigma^2_\mu,$$

and the within-hour wedge SD equals $\mathrm{sd}_t(p^*_{2t})$ exactly because p_1 contributes no cross-period variation. This is the main-body (3). Largest in ramp hours where σ_ζ is large; mechanically zero before the reform because the hourly spot averages ω_t to its hour-level component $\bar{\omega}$. (Arbitrage on the hourly forward is a block position, $s_t \equiv \bar{s}$: it disciplines the mean wedge but cannot move the dispersion of the granular leg.)

Thinness-asymmetry channel (block-parity arbitrage). The $\sigma_\omega/(2b_2)$ channel above holds at uniform b_{2t}. With per-subperiod thinness b_{2t} varying across quarters, an additional dispersion term emerges through the block-parity arbitrage of §D.5.1–§D.5.2. For two matched products ($Q = 2$) the population SD contribution from thinness asymmetry alone (setting $\omega_t = 0$) is

$$\mathrm{sd}_t^{F,\mathrm{thin}}(w_t) \;=\; \frac{(p_1 - c)\,|b_{22} - b_{21}|}{2(b_{21} + b_{22})}, \qquad \mathrm{sd}_t^{S,\mathrm{thin}}(w_t) \;=\; \frac{(p_1 - c)\,|b_{22} - b_{21}|}{4(b_{21} + b_{22})}, \tag{15}$$

the strategic value exactly half of the full-arbitrage value. Under limited arbitrage with per-product capacity $s_t = K_a/Q$ binding, the thinness-asymmetry SD becomes

$$\mathrm{sd}_t^{L,\mathrm{thin}}(w_t) \;=\; \frac{K_a\,|b_{22} - b_{21}|}{8\,b_{21}\,b_{22}} \qquad (Q = 2,\ s_t = K_a/2), \tag{16}$$

which carries K_a rather than $(p_1 - c)$. The total wedge SD in the asymmetric state is the sum (in quadrature) of the update channel $\sigma_\omega/(2b_2)$ and the thinness-asymmetry channel above; the update channel dominates empirically because σ_ζ is large in ramp hours.

D.6 (Q, Q) equilibrium

Both markets per-period. Per-product forward best response, three arbitrage regimes, Cournot cleared-MWh scale-up, and the stage-1 value of the granular continuation.

The strategic per-period forward best response is the continuation-adjusted locus of Corollary 3 with the matched single spot product ($B_2 = b_{2t}$); arbitrage and aggregator legs shift the intercept only and are omitted from the display:

$$p^*_{1t}(v) \;=\; \frac{\bar{A} + \zeta_t + v + c\,(b_{1t} - b_{2t}/2)}{2\,b_{1t} - b_{2t}/2}, \qquad \lambda_1 \;\equiv\; \frac{1}{2\,b_{1t} - b_{2t}/2}. \tag{17}$$

The ramp ζ_t and the screened state v both enter at the locus slope λ_1, so the across-quarter dispersion of expected forward prices is $\mathrm{sd}_t(\mathbb{E}\,p^*_{1t}) = \lambda_1\,\sigma_\zeta$, and the per-product forward screen is the $\rho_{\mathcal{X}}$-amplified $v \sim U[-\rho_{\mathcal{X}}\Delta_1, \rho_{\mathcal{X}}\Delta_1]$.

D.6.1 Full arbitrage: product parity

Per-product parity $p_{1t} = p_{2t}$ pins s_t at $s_t = b_{2t}(p_{1t} - c) + S^{op}_{2t}$ (same structure as the $(1, 1)$ full-arbitrage condition, but per product). The wedge SD is zero by construction.

D.6.2 Strategic arbitrage: product wedge and SD

Per-product strategic arbitrage — the (R4) derivation of §D.4 applied product by product — gives $p_{2t} = (3p_{1t} + c)/4$ (aggregator leg netted). The per-product wedge is $(p_{1t} - c)/4$. Cross-period SD:

$$\mathrm{sd}_t(p_{1t} - p_{2t}) \;=\; \tfrac{1}{4}\,\mathrm{sd}_t(p^*_{1t}) \;=\; \frac{\lambda_1\,\sigma_\zeta}{4},$$

using $\mathrm{sd}_t(\mathbb{E}\,p^*_{1t}) = \lambda_1\sigma_\zeta$ from (17) — the unlimited-strategic benchmark of the body.

D.6.3 Limited arbitrage at the symmetric state (empirical regime)

With Q matched products per clock-hour the arbitrageur's capacity K_a is spread over Q positions, so the effective per-product capacity is K_a/Q. When this capacity binds — the empirically relevant case in electricity (see §3.6) — the per-product equilibrium is Ito and Reguant Result 3 (limited arbitrage) applied product by product.

Strategic spot best response under binding per-product capacity. Setting $s_t = K_a/Q$ in the strategic Stage 2 FOC (12), with the matched per-product forward price p_{1t} in place of the hourly p_1:

$$p_{2t}^* \;=\; \frac{p_{1t}^* + c}{2} + \frac{K_a/Q - S_{2t}^{op}}{2\, b_{2t}}.$$

The per-product wedge is

$$p_{1t}^* - p_{2t}^* \;=\; \frac{p_{1t}^* - c}{2} \;-\; \frac{K_a/Q - S_{2t}^{op}}{2\, b_{2t}}.$$

Wedge SD: $\lambda_1\sigma_\zeta/2$. At (Q, Q) the per-quarter forward absorbs the ramp upstream: by (17) the forward price across quarters has SD $\mathrm{sd}_t(p_{1t}^*) = \lambda_1\, \sigma_\zeta$. The capacity term $(K_a/Q - S_{2t}^{op})/(2b_{2t})$ is common across quarters when b_{2t} and S_{2t}^{op} are, so only the first term disperses, feeding the wedge at half the rate:

$$\boxed{\mathrm{sd}_t(p_{1t}^* - p_{2t}^*) \;=\; \tfrac{1}{2}\,\mathrm{sd}_t(p_{1t}^*) \;=\; \frac{\lambda_1\, \sigma_\zeta}{2}.}$$

This is the main-body (5). Under unlimited strategic arbitrage the wedge SD halves further to $\lambda_1\sigma_\zeta/4$ (the doubling under limited per-product arbitrage is the empirical-asymmetric-persistence channel). If per-period thinness b_{2t} also varies across t the wedge inherits an additional thinness-asymmetry component on top of $\lambda_1\sigma_\zeta/2$; the empirically observed sustaining of the within-hour wedge SD at granular-state magnitudes (§6.4) is consistent with this combined effect.

Per-product capacity binding. For the limited regime to bind, K_a/Q must be smaller than the strategic-arbitrage equilibrium position $s_t^{\text{strat}} = b_{2t}(p_{1t} - c)/2$ (aggregator spot leg netted into the intercept). Binding therefore requires $K_a \le Q \cdot \bar{s}^{\text{strat}}$ where $\bar{s}^{\text{strat}}$ is the average per-product strategic position. Empirically, hourly arbitrage capacity is set by participation rules and balance-sheet limits that do not scale with Q when granularity reforms multiply products; the constraint binds at the granular state by construction.

D.6.4 Cournot cleared-MWh under per-period thinness

To isolate the thinness channel, shut the continuation and arbitrage channels in both states ($B_2 = 0$, $s_t = 0$) — a channel isolation, not an approximation: both states are evaluated under the same convention. The per-period cleared monopoly quantity is then $q_{1t}^* = (\bar{A} + \zeta_t - b_{1t}c - S_{1t}^{op})/2$. Summing across t and using $\sum_t \zeta_t = 0$:

$$\mathbb{E}\Big[\sum_t q_{1t}^*\Big] \;=\; \frac{Q\bar{A} - c\sum_t b_{1t} - \sum_t S_{1t}^{op}}{2}.$$

The hourly counterpart is $Q \cdot q_1^*|_{(1,1)} = Q \cdot (\bar{A} - b_1 c - S_1^{op})/2$. Comparing at equal aggregator totals ($\sum_t S_{1t}^{op} = S_1^{op}$):

$$\mathbb{E}\Big[\sum_{t=1}^{Q} q_{1t}^*\Big]\Big|_{(Q,Q)} \;-\; Q \cdot q_1^*\big|_{(1,1)} \;=\; \frac{c}{2}\left(Q\, b_1 - \sum_{t=1}^{Q} b_{1t}\right). \tag{18}$$

Positive whenever $\sum_{t=1}^{Q} b_{1t} < Q\, b_1$, as expected (each 15-min product is thinner than the hourly aggregate it replaces). This is the Cournot scale-up of main-body §3.5(iv).

D.6.5 *Stage-1 value of the granular continuation*

Maximised monopoly profit is convex in the demand intercept ($\pi^m(A) = (A - bc)^2/(4b)$), so a bidder that can re-optimise per quarter values the update as a mean-preserving spread: expected spot profit exceeds spot profit at the mean update by exactly $\mathbb{E}[\mu_t^2]/(4\,b_{2t}) = \Delta_\mu^2/(12\,b_{2t})$ per product under the uniform update — an exact equality, because the profit is quadratic. The term raises the stage-1 *value* of holding a granular continuation (part of why the granular state is privately valuable to the strategic bidder), but it does not involve p_1, so it leaves the stage-1 first-order condition unchanged: the forward-*ladder* response to the spot reform operates entirely through the continuation slope of Corollary 3, not through an option-value tilt.

D.7 Comparative statics across the three states

Table 23 collects the state-by-state closed forms behind the predictions of §3.6.

Table 23: Comparative statics across the reform path under limited per-product arbitrage (empirical regime). Notation: $\lambda_1(B) \equiv (2b_1 - B/2)^{-1}$ is the stage-1 locus slope at continuation coefficient B (Corollary 3; with the venue's own forward slope, b_{1t} per product at (Q,Q)); $\rho^\mu_{\mathcal{X}} \le \rho_{\mathcal{X}}$ is the news-only amplification of §3.5(iii), and $\rho^\mu_{\mathcal{X}} = \rho_{\mathcal{X}} = 1$ in flat hours. In the two ladder rows, twice the product of the entries is the product's conditional monopoly-price range (Corollary 2(i)); σ_p, the tranche-price span, and HHI_p follow in closed form from Corollary 1 and Proposition 2 evaluated at the screen and slope shown. Imbalance row: quadratic penalty $\gamma\,\mathbb{E}[\Delta_t^2]$, omitting the unpriceable σ_ε^2 common to all states; for uniform shocks $\sigma_\mu^2 = \Delta_\mu^2/3$ and $\sigma_\eta^2 = \Delta_\eta^2/3$ (§3.1, §3.6).

Outcome	$(1,1)$	$(1,Q)$	(Q,Q)
Aggregator hedging gain $G(\Delta_\eta)$	0	$\dfrac{\gamma Q\Delta_\eta}{2}$	$\dfrac{\gamma Q\Delta_\eta}{2}$
Spot tier: screen × locus slope	$\Delta_2 \times \dfrac{1}{2b_2}$	$\rho_{\mathcal{X}}\Delta_2 \times \dfrac{1}{2b_{2t}}$	$\rho^\mu_{\mathcal{X}}\Delta_2 \times \dfrac{1}{2b_{2t}}$
Forward tier: screen × locus slope	$\Delta_1 \times \lambda_1(b_2)$	$\Delta_1 \times \lambda_1\big(\sum_t b_{2t}\big)$	$\rho_{\mathcal{X}}\Delta_1 \times \lambda_1(b_{2t})$
Within-hour wedge SD	0	$\dfrac{\sigma_\omega}{2b_2}$, $\sigma_\omega^2 = \sigma_\zeta^2 + \sigma_\mu^2$	$\dfrac{\lambda_1\,\sigma_\zeta}{2}$ + thin. asym.
Within-hour DA dispersion $D^{\bar{p}}$	0	0	$\lambda_1\,\sigma_\zeta$
Balancing Δ_t	$\zeta_t + \mu_t + \eta_t + \varepsilon_t$	ε_t	ε_t
Imbalance cost $C^I/(\gamma Q)$, quadratic	$\sigma_\zeta^2 + \sigma_\mu^2 + \sigma_\eta^2$	0	0

D.8 Balancing discrepancy

Per quarter, the balancing residual collects whatever no auction carried and no agent could re-schedule:

$$\Delta_t \;=\; \big(\text{within-hour demand objects unpriced by any auction}\big) + \big(\text{unhedged production error}\big) + \varepsilon_t. \tag{19}$$

State $(1,1)$*.* Both auctions hourly: the spot trades only the hour-level update $\bar{\omega}$, so the ramp, the redistributive news, and the aggregator's forecast error all land in balancing: $\Delta_t = \zeta_t + \mu_t + \eta_t + \varepsilon_t$.

State $(1,Q)$*.* The spot becomes per-period: it absorbs the full update $\omega_t = \zeta_t + \mu_t$, and the aggregator re-schedules η_t per quarter (§D.2): $\Delta_t = \varepsilon_t$.

State (Q,Q)*.* The per-quarter forward carries the ramp ζ_t, the spot carries the news μ_t, the aggregator hedges η_t: $\Delta_t = \varepsilon_t$.

State	ramp ζ_t	redistributive μ_t	forecast error η_t	residual Δ_t
$(1,1)$	balancing	balancing	balancing	$\zeta_t+\mu_t+\eta_t+\varepsilon_t$
$(1,Q)$	spot	spot	aggregator (hedged)	ε_t
(Q,Q)	forward	spot	aggregator (hedged)	ε_t

The rows for $(1,Q)$ and (Q,Q) are identical in the residual column: the forward reform relocates rows between the two auctions but produces no further imbalance gain — the body's §3.5(v). These reductions are mechanical in the contracting grid and hold under any arbitrage regime.

D.9 Welfare derivations

We collect closed-form welfare components by state, consistent with the balancing ledger of §D.8 and with the channels listed in the body §3.6.

D.9.1 Components

Per clock-hour:

$$W \;=\; CS \;+\; \pi^{SB} \;+\; \pi^{Agg,\text{net}} \;+\; \pi^{Arb} \;-\; C^{I}_{\text{sys}}. \tag{20}$$

For linear residual demand $q=\bar{A}-bp$, consumer surplus at p^* is $CS=(\bar{A}-bp^*)^2/(2b)$. Gross surplus from consuming q at state A (willingness to pay minus production cost at c; balancing premia accounted separately) is

$$S(q;\,A) \;=\; \int_0^q \frac{A-x}{b}\,dx \;-\; c\,q \;=\; \frac{q\,(A-bc)}{b} \;-\; \frac{q^2}{2b}.$$

D.9.2 Imbalance cost decomposition (closed form, quadratic penalty)

Under the quadratic penalty $\gamma\,\mathbb{E}[\Delta_t^2]$ per quarter and the ledger of §D.8, with ζ_t deterministic ($Q^{-1}\sum_t \zeta_t^2=\sigma_\zeta^2$) and μ_t, η_t mean-zero and independent:

$$\begin{array}{lccc} & (1,1) & (1,Q) & (Q,Q) \\ \hline \gamma\,\mathbb{E}[(\mathcal{I}^{Agg})^2] \text{ (in } \pi^{Agg,\text{net}}) & \gamma Q\,\sigma_\eta^2 & 0 & 0 \\ C^{I}_{\text{sys}} \text{ (system side: } \zeta+\mu) & \gamma Q\,(\sigma_\zeta^2+\sigma_\mu^2) & 0 & 0 \\ \text{Total imbalance cost} & \gamma Q\,(\sigma_\zeta^2+\sigma_\mu^2+\sigma_\eta^2) & 0 & 0 \end{array} \tag{21}$$

omitting the unpriceable $\gamma Q\sigma_\varepsilon^2$, common to all three states. The two channels are disjoint (η_t enters only $\pi^{Agg,\text{net}}$; ζ_t and μ_t only C^I_{sys}). For uniform shocks, $\sigma_\mu^2=\Delta_\mu^2/3$ and $\sigma_\eta^2=\Delta_\eta^2/3$. The whole imbalance saving arrives at the spot reform; the (Q,Q) column equals the $(1,Q)$ column, the model's DA15 imbalance null.

D.9.3 Allocative cost of pricing the state per quarter (discrimination term)

Pricing the within-hour state per quarter is not free: it is third-degree price discrimination by a residual monopolist. Fix one clock-hour, the within-hour object ω_t with $\sum_t\omega_t=0$, and the margin b at which it is priced; write $\bar{q}=q^m(\bar{A})$ and $A_t=\bar{A}+\omega_t$, so $A_t-bc=2\bar{q}+\omega_t$.

Regime U (one price for the hour). The monopolist sets $\bar{p}=p^m(\bar{A})$; quarter-t consumption realises on the demand curve, $q_t^U=A_t-b\bar{p}=\bar{q}+\omega_t$; the contracted profile is flat at $\bar{q}$ and the balancing layer supplies the residual ω_t. Substituting into S:

$$S(q_t^U;\,A_t) \;=\; \frac{(\bar{q}+\omega_t)(3\bar{q}+\omega_t)}{2b}.$$

Regime D (per-quarter prices). The monopolist prices each quarter at $p^m(A_t)$; consumption is $q_t^D = q^m(A_t) = \bar{q} + \omega_t/2$ — monopoly pass-through one half — and there is no balancing residual:

$$S(q_t^D;\, A_t) \;=\; \frac{3\,(\bar{q}+\omega_t/2)^2}{2b}.$$

Difference. Summing over t and using $\sum_t \omega_t = 0$,

$$\sum_t S^D - \sum_t S^U \;=\; \frac{1}{2b}\sum_t \Big[3\bar{q}^2 + 3\bar{q}\,\omega_t + \tfrac{3}{4}\omega_t^2 \;-\; 3\bar{q}^2 - 4\bar{q}\,\omega_t - \omega_t^2\Big] \;=\; -\,\frac{\sum_t \omega_t^2}{8\,b}\;:$$

total output is unchanged and output is misallocated across quarters, at expected cost $Q\,\mathbb{E}[\omega^2]/(8b)$. Against it stands the avoided quadratic balancing penalty $\gamma\,Q\,\mathbb{E}[\omega^2]$, so the net system-side gain from pricing ω at margin b is

$$Q\,\mathbb{E}[\omega^2]\left(\gamma - \frac{1}{8b}\right), \qquad \text{positive iff } \gamma > \frac{1}{8b},$$

a threshold independent of the distribution of ω. At the spot reform the relevant margin is b_2; balancing premia in Spanish electricity markets exceed wholesale margins by a factor of 2–10, far above $1/(8b_2)$.

D.9.4 Markup channel

From the Cournot identity (6), $\text{markup}_m = q_m/b_{mt}$: at given cleared quantity, $\partial\,\text{markup}/\partial b_{mt} = -\,q_m/b_{mt}^2$, so per-firm residual-demand thickening compresses the markup — and, by level neutrality (Remark 1), this is the *only* channel through which the reform moves mean prices; ladder geometry contributes nothing to the mean.

D.9.5 Discreteness channel

The optimal-ladder expected profit shortfall relative to the continuous benchmark is $\Delta^2/(12\,b\,(2n^*+1)^2)$ (Proposition 1); the deeper endogenous depth at amplified screens (Corollary 2(ii)) reduces it. By Remark 1 the mean clearing price is unchanged by the ladder, so within a product this channel is mostly a CS$\leftrightarrow \pi^{SB}$ transfer, net of tranche costs $\kappa\, n^*$; the efficiency content is the discreteness loss itself.

D.9.6 Less strategic withholding under limited per-product arbitrage

The spot deadweight-loss triangle at markup $M = p_2 - c$ and slope b_2 is $b_2 M^2/2$. Under unlimited strategic arbitrage, $p_2 - c = 3(p_1 - c)/4$ (§D.4, R4): $\text{DWL} = 9\,b_2(p_1-c)^2/32$. Under binding per-product capacity, the wedge is $(p_1 - c)/2$, so $p_2 - c = (p_1 - c)/2$ and $\text{DWL} = b_2(p_1-c)^2/8 = 4\,b_2(p_1-c)^2/32$. The per-product saving from the regime change is

$$\frac{5\,b_2\,(p_1-c)^2}{32},$$

per Ito and Reguant Result 2 logic — a welfare gain that accompanies the wedge-SD non-collapse of §3.5(ii), not a cost of it.

D.9.7 Net welfare deltas (closed form)

Combining the contractibility channels (writing $a_1 \equiv \bar{A} - b_1 c$, aggregator and arbitrage legs netted into $\bar{A}$):

$$\Delta W^{\text{spot}} = \gamma\, Q\, \sigma_\eta^2 \;+\; Q\left(\sigma_\zeta^2 + \sigma_\mu^2\right)\Big(\gamma - \frac{1}{8\, b_2}\Big), \tag{22}$$

$$\Delta W^{\text{forward}} = \frac{Q\, \sigma_\zeta^2}{8}\Big(\frac{1}{b_2} - \frac{1}{b_1}\Big) \;+\; \frac{c\, a_1}{4\, b_1}\Big(Qb_1 - \sum_t b_{1t}\Big). \tag{23}$$

The spot-reform delta is the aggregator's hedging saving plus the net value of pricing the update at the spot margin: positive whenever $\gamma > 1/(8b_2)$, and dominated by the imbalance saving at observed balancing premia. The forward-reform delta is a *relocation*: the ramp's discrimination cost moves from the spot margin b_2 to the (thicker) forward margin b_1 — the first term, positive when $b_1 > b_2$ and zero when the margins coincide — plus the thinness scale-up of (18) valued at the pre-existing forward markup, carrying the sign of $Qb_1 - \sum_t b_{1t}$. Both forward-reform terms are second-order relative to (22): no imbalance term appears at the forward reform (the (Q, Q) column of (21) equals the $(1, Q)$ column), which is the model's account of the small DA15 aggregate footprint. The markup-compression and withholding channels add on top wherever b_f thickens or the arbitrage regime tightens, with the signs derived above; the bid-ladder rent channel is mostly distributional.

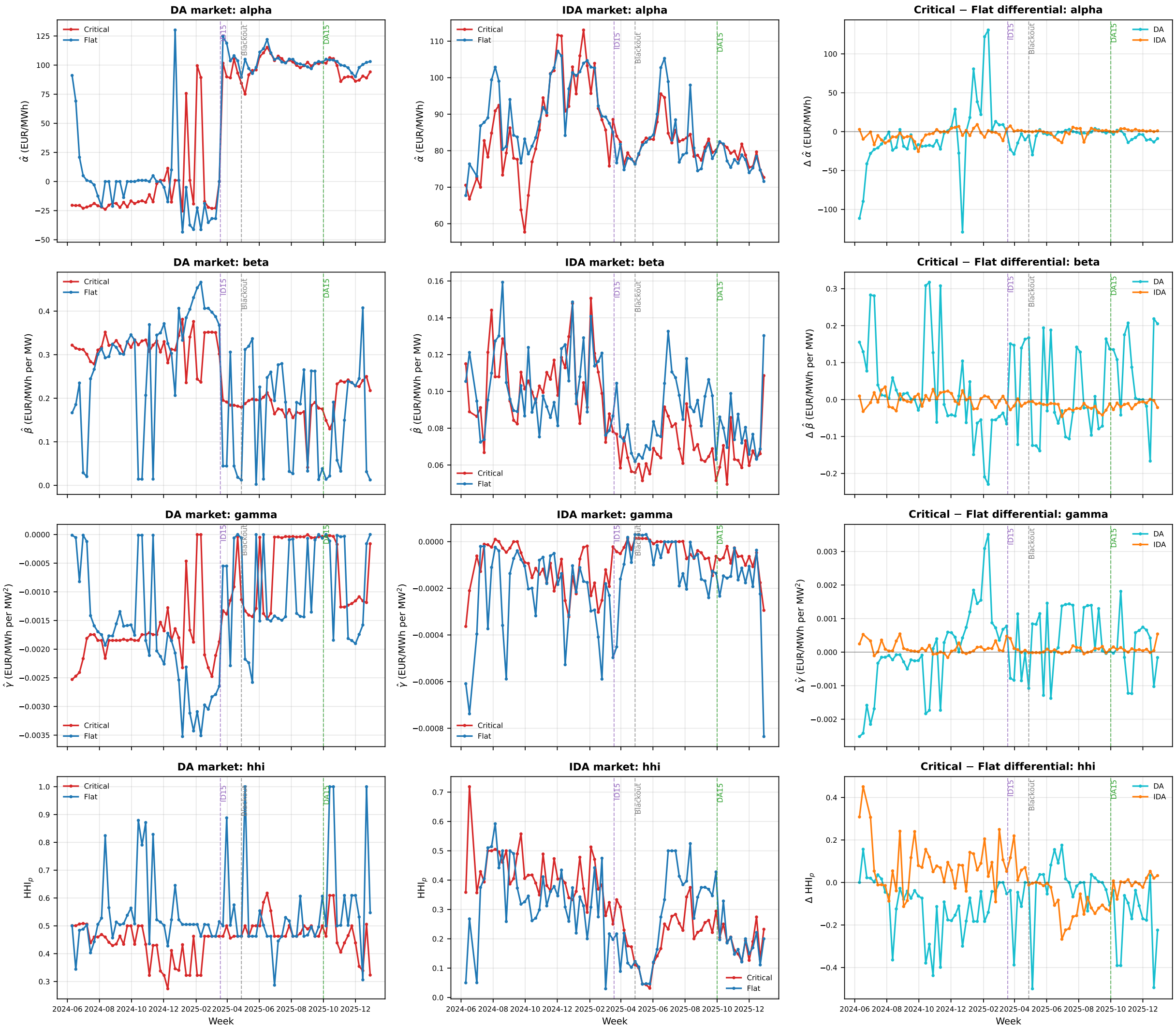

Figure 7: Structural decomposition: weekly CCGT per-curve $\hat{\alpha}, \hat{\beta}, \hat{\gamma}, \mathbf{HHI}_p$ by hour-class, 2024-06 to 2025-12. *Left column*: DA market levels by hour-class. *Middle column*: IDA market levels. *Right column*: critical − flat differential by market (cyan DA, orange IDA), the visual analogue of the bid-shape DiD coefficient. In-band per-curve OLS quadratic fit at $\delta = 150$ EUR/MWh; per-cell aggregation uses the median across curves with $n_{\text{tranche}} \geq 4$ and $\Delta q \geq 20$ MW for the $\hat{\beta}$ and $\hat{\gamma}$ rows (joint p–q identification floor); $\hat{\alpha}$ and HHI_p have no such floor and use all curves. Vertical dashes mark the three reform dates. The HHI_p row carries the cleanest visual signal (DA critical sits sustainedly below flat post-DA15, the visual signature of bid-stack deconcentration); $\hat{\alpha}$ is dominated by the spring-2025 shared negative-price shock; $\hat{\beta}$ and $\hat{\gamma}$ rows are noisier per the identification argument in the paragraph above and should be read in conjunction with the σ_p panel of Figure 6.

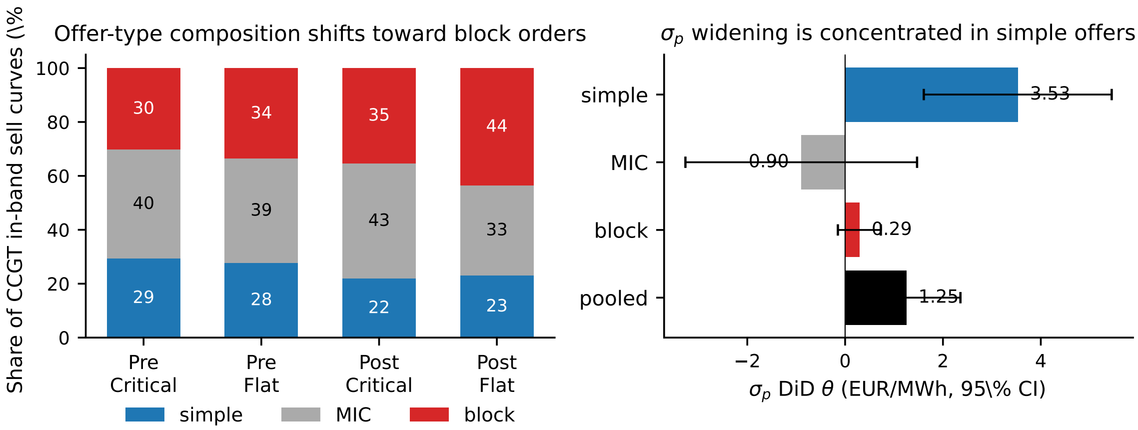

Figure 8: CCGT offer-type composition shift and within-type bid-shape DiD. DA15 windows: pre 2025-07-01 to 2025-09-30; post 2025-10-01 to 2025-12-31. *Left*: composition of CCGT in-band sell curves, pre vs post DA15, critical vs flat. *Right*: bid-shape DiD by offer type with 95% CI; the panel is plotted on the σ_p outcome, and the HHI_p DiD by offer type carries the same sign pattern with negative coefficients in every type (simple -0.007 n.s., MIC $-\mathbf{0.046}^{***}$, block $-\mathbf{0.043}^{***}$, pooled $-\mathbf{0.059}^{***}$).

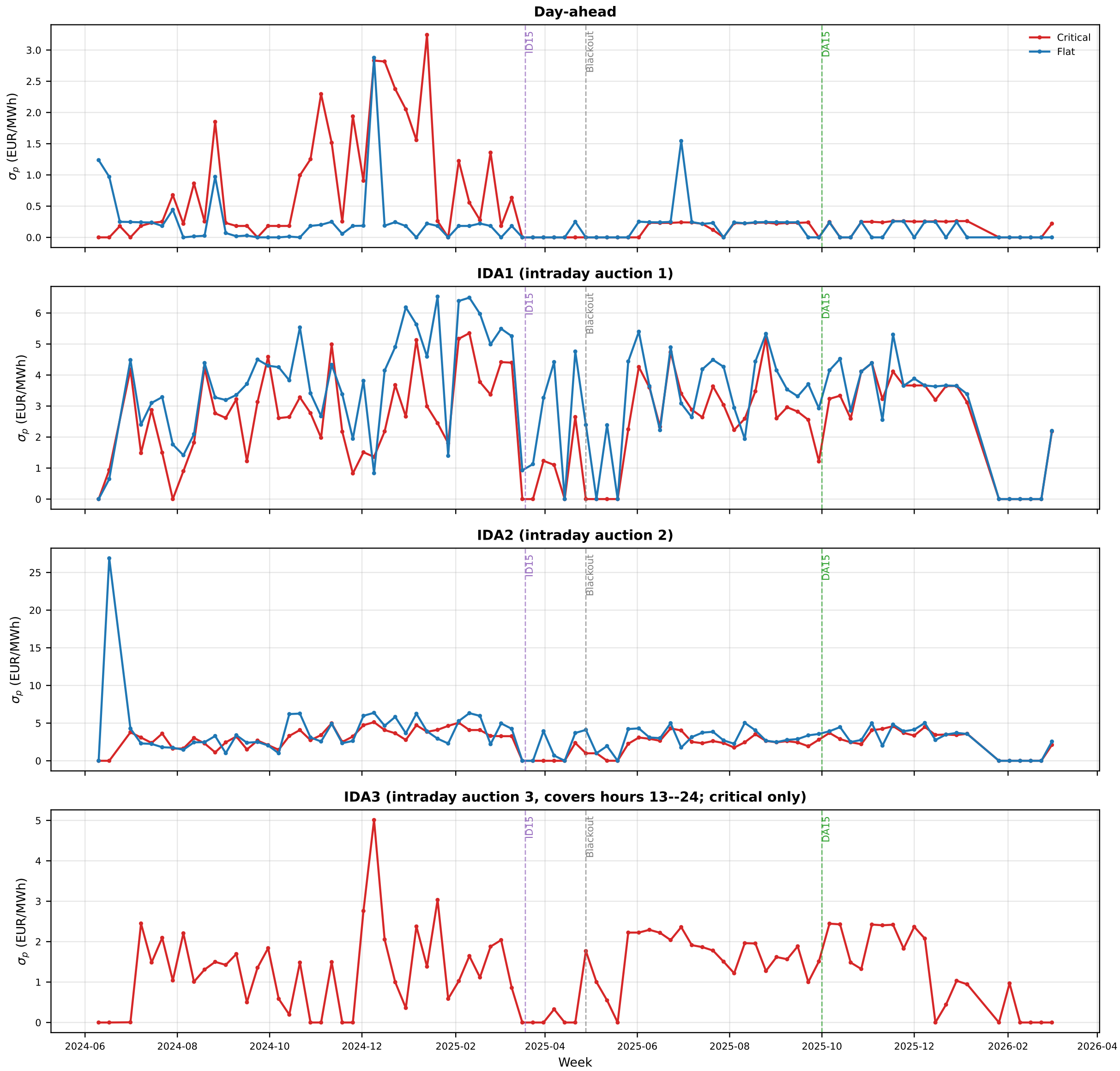

Figure 9: Per-session parallel-trends diagnostic: weekly CCGT σ_p by hour-class, 2024-06 to 2026-05. Stacked panels: day-ahead, then each of the three IDA sessions. Red: critical hours (5–8, 16–22). Blue: flat hours (1–3). IDA3 covers only hours 13–24, so its panel reports critical only. Bandwidth $|p_k - \text{MCP}| \leq 50$ EUR/MWh. Vertical dashes mark ID15 (2025-03-19), blackout (2025-04-28), and DA15 (2025-10-01). The IDA2 panel carries the most-active CCGT bid-shape pre-MTU15-IDA, and the three IDA panels visibly converge in level post-reform — the per-session story behind the pooled bid-shape DiD.

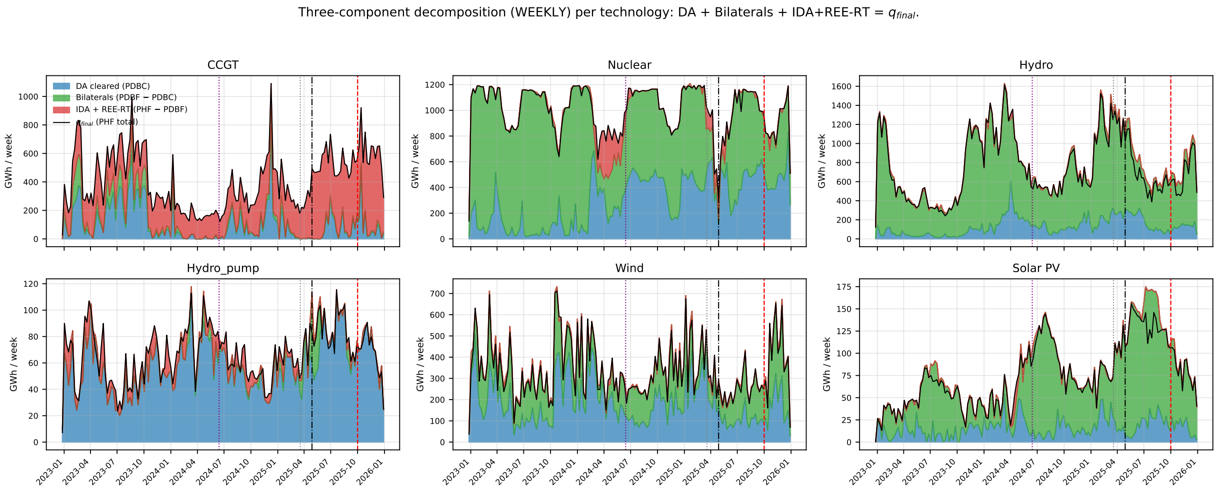

Figure 10: Weekly three-component decomposition of dispatch, by technology. PDBC (blue, day-ahead-cleared) + bilateral executions (PDBF − PDBC, green) + post-DA REE intervention layer (PHF − PDBF, red) = final post-IDA dispatch (black, equivalent to PHF). The red layer aggregates everything that happens to the schedule between the day-ahead programme (PDBF) and the post-IDA programme (PHF): *Fase I* pre-IDA up-redispatch and *Fase II* rebalance under PO 3.2 (paid pay-as-bid via the restriction-offer channel of PO 14.4 ap. 20.1, distinct from the day-ahead market offer), plus the net of all IDA clearings and REE's post-IDA real-time technical-restrictions step. Within the red layer, Fase I dominates the post-blackout growth (Figure 4 and §6.3). Vertical guides: MTU15-IDA, April 2025 blackout, MTU15-DA.

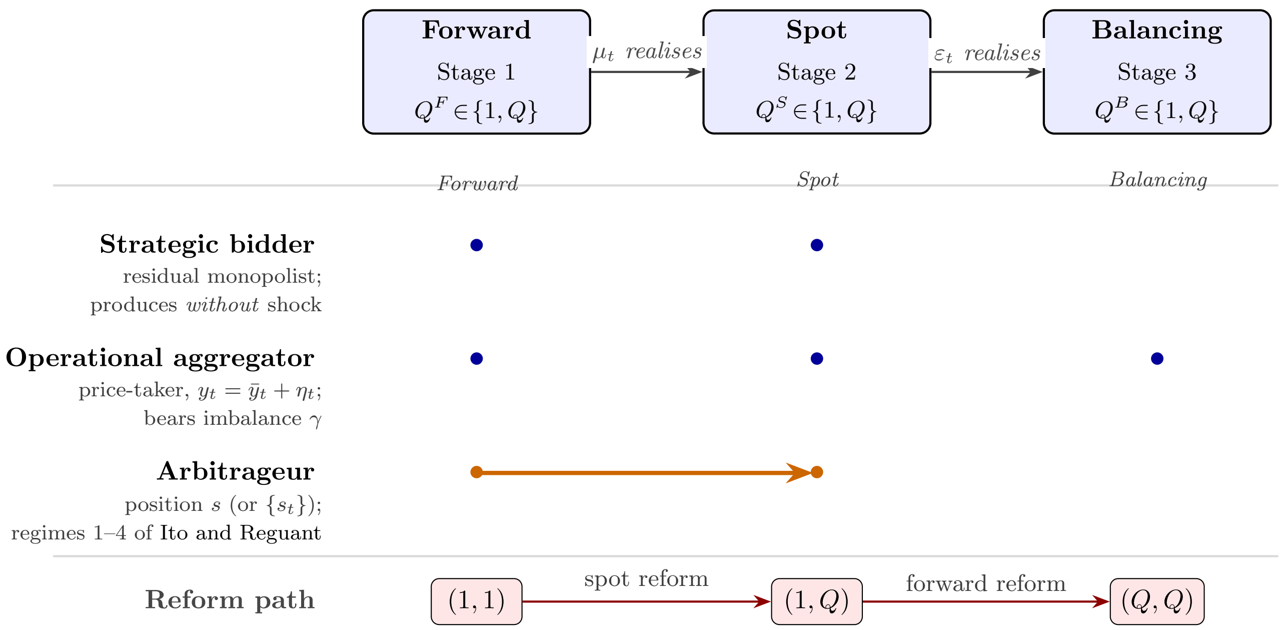

Figure 14: Sequential markets, granularity, and agents. *Top:* a forward market, a spot market, and a balancing settlement on a timeline; each has its own granularity selector $Q^m \in \{1, Q\}$ and information resolves between them. The forward market maps empirically to the electricity day-ahead auction, the spot market to the intraday auction. *Middle:* which agents act at which stages (blue dots), and the arbitrageur position s linking forward to spot (orange arrow). The strategic bidder produces without uncertainty and does not internalise the imbalance penalty; only the operational aggregator does, since it bears η_t. *Bottom:* the auction-side reform path $(Q^F, Q^S) = (1, 1) \to (1, Q) \to (Q, Q)$ traversed in the main analysis.

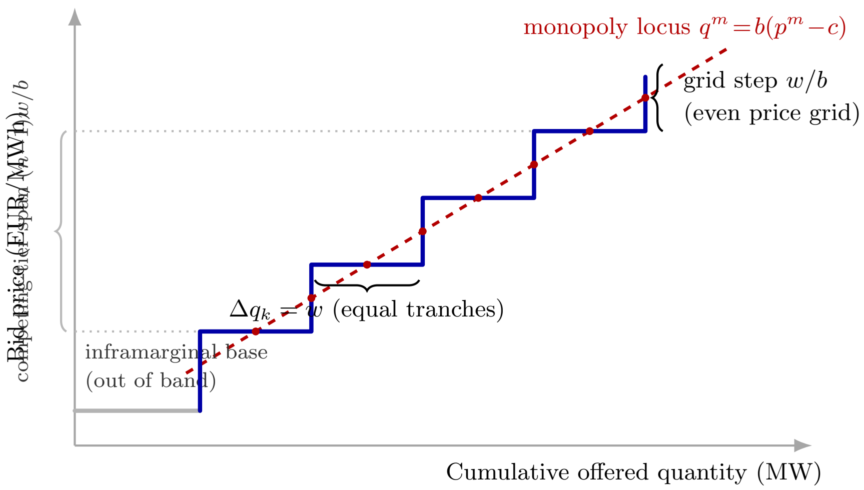

Figure 15: The optimal ladder and the monopoly locus ($n = 4$ competing tranches). The blue step offer weaves across the dashed monopoly locus: every horizontal segment crosses it at its midpoint (states where the tranche price is exactly the monopoly price) and every vertical riser crosses it at its midpoint (states where the cumulative quantity is exactly the monopoly quantity); red dots mark the implemented points. Competing tranches carry equal MW $w = 2\Delta/(2n+1)$ on an even price grid of step w/b; the inframarginal base sits below the contested range and outside the empirical band of §4.3.